When Does Trade Protection Help? Evidence from the Rise of American Manufacturing

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Abstract

When does trade protection reduce growth and when does it boost growth? I study this question in the context of US industrialization in the 19th century. I develop a novel empirical design leveraging differential county exposure to nationwide tariffs based on distance to international ports, using newly digitized tariff, import price and railway freight data. Exploiting a nationwide fiscal shock that suddenly raised tariffs for reasons unrelated to local industry conditions, I find protection reduces output growth for products close to the technological frontier but increases growth for products far behind it. I also find protection shifts patent activity towards far-from-frontier products, suggesting a reshaping of comparative advantage. A model of trade and innovation with variable markups generates these results: protection enables a far-from-frontier producer to capture the rents from innovating, while a close-to-frontier producer's high market share and low residual demand elasticity mean protection reduces the gain from innovating. This model provides an imperfect competition rationale for infant industry protection distinct from the usual learning-by-doing justification. I conclude by studying the welfare effects of trade, which can be negative unlike in workhorse trade models.

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1 Introduction

One of the oldest ideas in all of economics is the infant industry argument: trade protection insulates new domestic producers from foreign competition, thereby allowing these producers to grow (Hamilton 1791, List 1841, Mill 1848). This claim has proven to be remarkably durable, yet empirical evidence on its effects remains mixed. In the 19th century US context, Head (1994), Baack and Ray (1974), Sundararajan (1970) and Harley (1992) argue trade protection successfully led to the development of the steel rail and cotton textile industries. Others such as Irwin (2000a, 2000b) and Irwin and Temin (2001) argue tariffs had no major impact on the development of the tinplate, pig iron and cotton textile sectors. Klein and Meissner (2024) and Taussig (1915, 1931) argue positive effects for some sectors but null or negative effects for others. The broader industrial policy literature (Lashkaripour and Wu 2025) and trade-and-innovation literature (Shu and Steinwender 2019) also find mixed effects depending on policy, context and sector.

I revisit the 19th century US setting with newly digitized historical data and a novel empirical design, addressing endogeneity concerns associated with policymakers' targeting of sectors for protection (Juhasz, Lane and Rodrik 2024). In contrast to past studies, which tend to focus on a single product or sector, I estimate the effect of protection across multiple products, allowing me to identify a product characteristic that conditions whether protection helps or harms. I find protection reduces growth for products close to the technological frontier but raises it for those behind — rationalizing some of the disagreement in the literature. I develop a model of trade and innovation with variable markups that generates this pattern, and study its welfare and policy implications.

My empirical design is based on the insight that, much like a tariff discourages imports by raising their prices, high transportation costs to international ports also serve as protection for interior counties. When nationwide tariffs go up, areas near international ports see larger proportional increases in protection than interior areas, which are already shielded from imports by high transportation costs.[1] Methodologically, this approach isolates an across-county, within-product comparison. For example, when nationwide steel tariffs rise, Philadelphia steel producers — located at a major international port — see a larger proportional increase in protection than Chicago steel producers, located in the interior. Comparing their growth rates isolates the effect of protection from any nationwide trends in the steel industry, such as technological developments.[2]

[1] This spatial variation in tariff pass-through holds with additive domestic freight costs, a specification supported by tariff valuation rules and freight-rate estimates in this period; see Section 5.1.

[2] For example, the Bessemer process, invented in 1856, enabled the mass production of steel for the first time in human history.

By contrast, a large empirical trade literature exploits cross-product variation by comparing products facing larger versus smaller tariff changes following national policy changes (e.g. Pavcnik 2002, Topalova and Khandelwal 2011, De Loecker et al. 2016, Fajgelbaum et al. 2020). However, national product trends correlated with tariffs — whether coincidental or reflecting targeting of fast-growing sectors — can confound this comparison. My within-product design absorbs these trends via product fixed effects.

Local labor market approaches (e.g. Autor, Dorn and Hanson 2013) are also potentially confounded by national product trends, as Borusyak, Hull and Jaravel (2022) show.[3] These approaches compare the aggregate growth rates of counties with high versus low baseline employment shares in products facing larger national shocks, which may correlate with trends such as technological change. My design instead varies protection across counties within the same product, so product fixed effects absorb such trends; Appendix B.6 compares the two designs in more detail.

A second benefit of the within-product, across-space design is that I can identify product-specific treatment effects, in contrast to both the shift-share and cross-product approaches. As policymakers set tariffs by product, product-specific treatment effects are arguably more useful than an average across products. For example, a negative average treatment effect can be consistent with positive effects for a subset of products. In addition, the industrial policy literature stresses that the effects of trade protection vary across products or sectors (e.g. Lashkaripour and Wu 2025, Bartelme et al. 2025).

The 1860s provide a useful natural experiment: With tariffs accounting for 90% of Federal revenues (Irwin 2020), Civil War fiscal pressures drove rapid tariff increases, with Congress raising tariffs five times between 1861 and 1865. In addition, US trading partners did not retaliate, allowing me to isolate the import protection channel. Identification requires that nationwide tariff changes are uncorrelated with idiosyncratic county-product shocks. For this condition to be violated, policymakers would need to identify and target idiosyncratic county-product trends across roughly 1,500 counties and hundreds of tariff line items—an informational and cognitive burden far beyond what 19th-century wartime legislators could satisfy, let alone build Congressional consensus around. I validate this narrative by showing that county-product pairs receiving more protection are balanced on baseline observable characteristics relative to those receiving less. Finally, I employ a battery of robustness checks to control for wartime destruction, Emancipation,

[3]This is true for designs based on the "exogenous shocks" view of Borusyak, Hull and Jaravel (2022). The "exogenous shares" view of Goldsmith-Pinkham, Sorkin and Swift (2020) requires that industrial composition is uncorrelated with future trends. This is unlikely to hold in my context, as locations specializing in different sectors are likely to be exposed to different shocks such as immigrant flows, technological changes and unionization drives.

Reconstruction, military procurement, lobbying power and input disruptions.

Implementing this design requires combining historical data sources, several of which I newly digitize. I digitize product-level import prices and tariffs from the Foreign Commerce and Navigation Tables of the United States for 1859 and 1869. I digitize commodity-specific internal freight rates from the Aldrich Report (1893), combining this information with Donaldson and Hornbeck's (2016) transportation network to construct county-product-specific freight costs to port. To control for military procurement shocks, I digitize new data from the Quartermaster General's reports. To control for changes in the cost of imported inputs, I construct an input-output table using Atack, Bateman and Weiss' (2006) state-level census of manufactures. Finally, I harmonize British and American patent records from the 1850s and use UK-US patent ratios to measure product-level distance to the technological frontier — Britain being the industrial leader of this period.[4]

I find heterogeneous effects of protection depending on a product's distance to the technological frontier. A 1% increase in protection lowers output growth by 1.65 percentage points at the 25th percentile of distance to frontier but raises it by 1.01 percentage points at the 75th percentile. Employment and input use exhibit the same pattern, providing evidence the output effects are not driven solely by price changes. Turning to innovation, I find that protection shifts patent activity towards laggard products, suggesting a reshaping of comparative advantage.

I develop a model of trade and innovation with variable markups. Protection has two effects: a positive substitution effect that shifts demand toward domestic varieties, and an ambiguous innovation effect operating through its impact on the returns to innovation. For close-to-frontier products, the domestic producer already commands a large market share and faces a sufficiently low residual demand elasticity that protection reduces the incremental revenue gain from innovation. In some scenarios, it is profit-maximizing to cut costs associated with innovation and cede market share. If this innovation force is large enough, it can overtake the positive substitution effect and reduce output on net. For products sufficiently behind the frontier, the innovation effect instead amplifies the positive substitution effect. Protection expands sales while residual demand remains highly elastic, raising the payoff to innovation; thus, unlike close-to-frontier products, innovation and protection are complements, not substitutes. The model thus admits a threshold in distance to the frontier above which protection raises output and below which it may reduce it — a possibility I confirm in the data.

This model provides an imperfect competition rationale for infant-industry protection,

[4] A position it occupied since at least the First Industrial Revolution in the 1760s. See Section 2 for further context.

distinct from the usual learning-by-doing arguments.[5] Evidence of widespread imperfect competition in developing and developed economies (e.g. De Loecker and Eeckhout 2025, Díez et al. 2021) suggests this mechanism may have broad relevance. Additionally, this model allows a unilateral increase in protection to *reduce* output in partial equilibrium. By contrast, standard gravity frameworks predict that unilateral increases in output protection — absent retaliatory responses by partners — weakly increase output holding expenditures fixed (see Appendix B.5).

Under a general equilibrium version of the model, the welfare gains from trade are ambiguous, whereas standard gravity frameworks predict positive gains. The model's welfare formula nests the classic ACR formula (Arkolakis, Costinot and Rodríguez-Clare 2012), with additional ex-post statistics, one of which is the change in innovation, governing the sign and magnitude of the welfare effect.

Literature Review: My paper is connected to four different literatures. First, this paper contributes to the literature analyzing the role of trade protection in the development of American manufacturing (Taussig 1915, Sundararajan 1970, Baack and Ray 1974, Head 1994, Irwin 2000a, Irwin and Temin 2001, Greenland and Lopresti 2024, Klein and Meissner 2024), which I revisit with newly digitized data and a novel empirical design. The evidence in this literature is mixed: some papers find a positive effect, while others find a null or negative effect. My framework can explain some of these results based on a product's distance to the technological frontier. Irwin and Temin (2001) find cotton textiles did not benefit from protection — a near-frontier product in my data. Klein and Meissner (2024) find negative effects for textiles and metals, both near the frontier, and a positive effect for glass, one of the furthest behind. Taussig (1931) reaches the same conclusion for glass. In addition, the negative estimates are difficult to reconcile with standard trade frameworks, in which unilateral protection weakly raises domestic output, but can be explained in my model via reduced innovation incentives.

Second, this paper contributes to the empirical trade literature by offering a new within-product design to study the impact of trade protection within a country. This contrasts with a large empirical literature relying on cross-sectoral variation in tariff changes — comparing sectors receiving large versus small tariff cuts after a nationwide trade liberalization episode — such as Pavcnik (2002), Topalova and Khandelwal (2011), De Loecker et al. (2016), and Fajgelbaum et al. (2020). Shift-share approaches based on the

[5] In his famous critique, Baldwin (1969) argues that learning-by-doing justifies protection only if some additional distortion prevents firms from financing or appropriating future learning gains. If future productivity gains are privately appropriable, firms should be able to borrow against them in complete capital markets; if they are not, the appropriability problem would not necessarily be resolved with protection. By contrast, the mechanism in my model only requires imperfect competition in output markets.

"exogenous shocks" view also implicitly rely on cross-product comparisons, as noted by Borusyak, Hull and Jaravel (2022). These approaches may be susceptible to product-level trends, such as technological shocks, being correlated with tariffs, whether incidental or targeted by policymakers.[6] In addition, my design can identify product-specific treatment effects, rather than an average across products. Juhasz (2018) also uses a within-country, within-product comparison: she uses the rerouting of international trade routes during the Napoleonic Wars as exogenous variation in protection across French regions. My design has the advantage of not relying on transportation network changes, which may affect output and innovation through channels such as market access or information flows.

Third, this paper contributes to the trade-and-innovation literature (as reviewed by Shu and Steinwender 2019 and Akcigit and Melitz 2022). I show how protection's effects depend on the pre-existing competitive environment, which in turn depends on a product's distance to the frontier. My paper is related to Aghion et al. (2005), who show that theoretically and empirically the relationship between innovation and competition is an inverted U. They model competition as the share of profits that neck-and-neck firms can extract through collusion; I instead model protection as tariffs and transportation costs applied to imports, which is more specific to the trade setting. In addition, they study how the level of innovation varies with the level of competition, whereas I study how changes in innovation in response to changes in protection vary with distance to frontier. Aghion et al. (2009) study how the effect of entry on incumbent innovation is mediated by distance to frontier, whereas I study the effect of trade protection. Amiti and Khandelwal (2013) study how the effect of trade protection on quality upgrading is affected by distance to frontier. In addition, I connect these innovation responses to output growth.

Fourth, this paper contributes to the industrial policy literature, specifically the ex-post empirical analysis of such policies (as reviewed by Lashkaripour and Wu 2025). My paper addresses the identification challenges discussed by Juhasz, Lane and Rodrik (2024): protection can be directed toward products policymakers expect to grow, so cross-product comparisons risk confounding the effect of protection with the reasons it was granted. In addition, in contrast to other ex-post empirical analyses in this space (e.g. Lane 2025, Juhasz 2018, Choi and Levchenko 2025, Hanlon 2020), my paper focuses on multiple products rather than an individual sector. This clarifies a key condition determining the effect of protection, and hence which products a policymaker should expect protection to benefit or harm.

Roadmap: Section 2 describes the historical context. Section 3 presents a partial equilibrium model in which protection's effects depend on distance to frontier. Section 4

[6]See Appendix B.6 for a formal comparison between the shift-share and the design in this paper.

describes the data. Section 5 develops the empirical framework and identification strategy. Section 6 presents the results. Section 7 presents a general equilibrium model studying protection's welfare effects and how distance to frontier mediates these gains or losses.

2 Historical Context

In the 1850s, the United States was an agrarian economy lagging behind Western Europe's industrial powers — the United Kingdom, France, Belgium, and the German states (which later unified to form Germany). The UK stood furthest ahead, having led the world technologically since the First Industrial Revolution in the 1760s. Like many developing countries today, the US was a net exporter of primary products and a net importer of manufactures. In 1850, 53% of steel used domestically was imported despite a large domestic iron ore endowment, while chemicals such as soda ash and bleaching powder were entirely imported with no domestic production of any kind. By the 1890s, the US catches up and becomes a net exporter of manufactures, with GDP per capita 50% higher than at midcentury.

This transformation was accompanied by high average tariffs, ranging from 16% to 45% over the second half of the century as Figure 1 shows. This aggregate correlation between tariffs and growth is often cited to justify current protectionist trade policy in the US.[7] My analysis centers on the 1860s, which offer a useful natural experiment: with tariffs supplying 90% of Federal revenues (Irwin 2020), Civil War fiscal pressures drove average rates to their highest level since the Tariff of Abominations in 1828. Thus, this large and sudden tariff increase was plausibly driven by revenue needs rather than local industrial policy goals. Section 5.5 develops this argument in detail.

This period is often called the First Globalization, an era of unprecedented international economic and financial integration. It was driven by the rise of the steamship and telegraph, adoption of the gold standard, and unilateral and bilateral reduction of protectionist measures. Within the US, exposure to the First Globalization varied greatly depending on geographic distance and the transportation network.[8] As Special Commissioner David A. Wells (1869) put it in a report to Congress,

> "If the duty on pig iron were entirely removed, the American producer in

[7] See Donald Trump (2025): "We were at our richest from 1870 to 1913. That's when we were a tariff country" and Oren Cass (2024): "Behind some of the world's highest tariff barriers, the United States transformed from colonial backwater to continent-spanning industrial colossus".

[8] In the modern day, other papers make a similar case. Atkin and Donaldson (2015) show that the probability of finding foreign imports falls the further one goes into the interior. Bosker and Haasbroek (2025) and Cosar and Fajgelbaum (2016) find that access to ports mediates the impact of globalization.

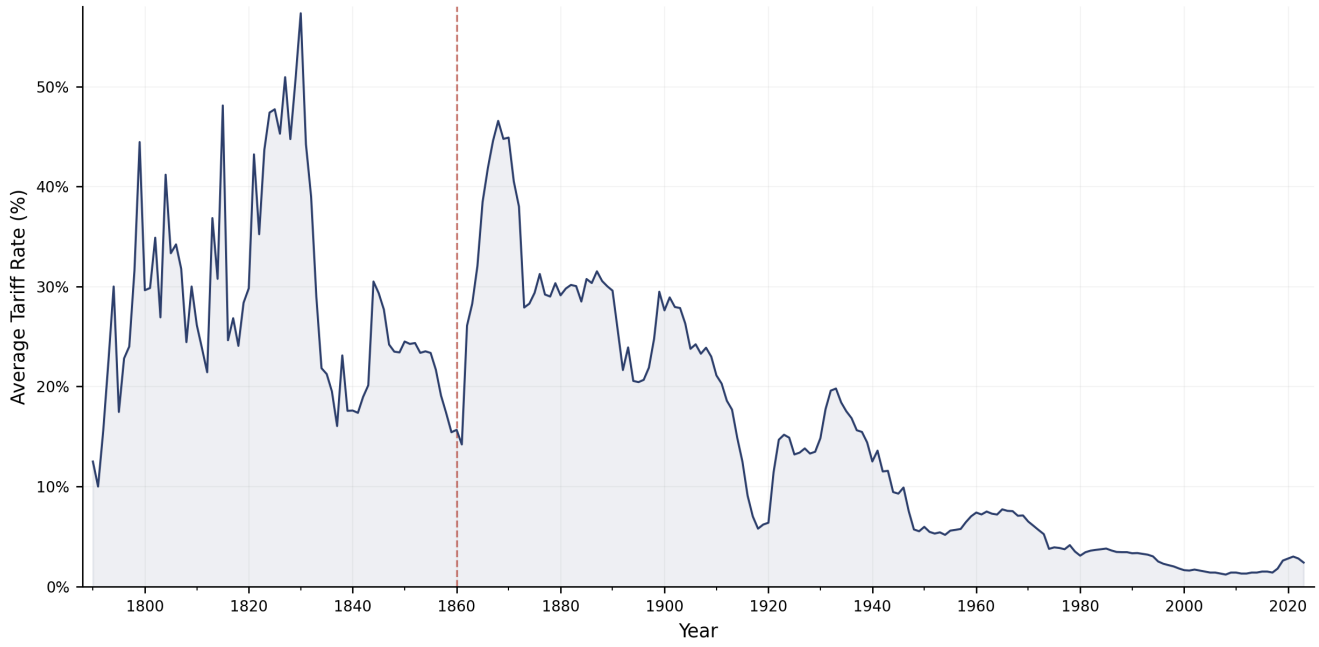


Figure 1: Average US Tariff Rate, 1790–2023

Notes: Duties collected as a percentage of total import value. Data for 1790–2000 from Table Ee424-430, Irwin (2006); 2001–2023 from the U.S. International Trade Commission (2024). The dashed vertical line marks 1860, the start of the tariff increases studied in this paper.

> the interior would still enjoy a protection in the cost of transportation . . . that intervene between the place of production and the port of entry . . . which circumstances renders the transport of a single pound of foreign pig iron to any considerable distance into the interior a matter of ordinary commercial impossibility."

Trade economist Frank Taussig (1915) shared this view: "The cost of transportation from the sea-board to the interior is such that, even in the absence of the duty, steel rails would be imported only to supply railways near tide-water."[9] High transportation costs thus protected interior producers against imports. Because initial levels of protection varied across space, a national tariff shock raises protection proportionally more in coastal counties than in the interior in the presence of additive transport costs — a feature I exploit in my empirical design in Section 5.1.[10]

[9]See also Sundararajan (1970): ". . . in the vast hinterland of the United States foreign competition is checked due to the existence of transportation costs. Whatever handicaps to American producers may be imposed by those costs, the influence of transport costs on competition from foreigners is felt only at the seacoast and adjacent areas."

[10]Gravity models in trade also imply the output response to a nationwide tariff shock is larger for coastal than interior areas. See Appendix B.5 for a formal analysis.

An attractive feature of this period is the absence of retaliation by trading partners, which isolates the import protection channel from retaliatory tariffs and lost export markets. The United Kingdom had committed to free trade following the repeal of the Corn Laws in 1846 and was phasing out tariffs on all goods except alcohol and sugar. The other European powers appeared unconcerned with the United States and were more focused on trade with their colonial empires and other European states. Appendix A.1 goes into further historical detail about the lack of retaliatory response by European powers.

Another important dimension of this period is the wide variation in American patenting activity across products, which I use to proxy for technological capabilities. As Table 1 shows, in some industries, such as cotton textiles, American producers were innovating at rates similar to Britain's — the result of decades of transatlantic technology transfer dating back to the early republic (Jeremy 1981). In glass, technological development lagged further behind Europe, with few innovations of domestic origin (Scoville 1944). In others, such as paints, the US lagged furthest behind: Sherwin-Williams was the first US paint firm to employ a chemist in 1884, over a century after British rivals such as Berger Paints (1760) and Winsor & Newton (1832) were founded by chemists. Appendix A.2 provides more examples and validation of my distance to frontier measure. As I show in this paper, this heterogeneity in technological proximity has sharp implications for how protection affects innovation incentives and output growth.

The US and British patent systems differed markedly in this period. The US system was substantially cheaper, operated a substantive novelty examination from 1836 onward, and was deliberately designed to be accessible to inventors of modest means. The British system was far more costly, with total fees amounting to roughly half a year's wages for a working man, and relied on a registration system with no examination for novelty until 1902 (Khan and Sokoloff 2001; Khan 2005). British patent counts therefore likely understate inventive activity relative to American counts, so my measure understates the true distance to frontier. However, to the extent these institutional differences generate a uniform proportional gap in the propensity to patent across industries, the measure is scaled by a common factor, leaving the ranking of products unchanged.

Local markets in this period were highly concentrated according to Bateman, Foust and Weiss (1971). I confirm this finding using recently digitized establishment-level data from the 1860 Census of Manufactures (Hornbeck et al. 2026). I compute Herfindahl-Hirschman indices (HHI) for each county-product market in my sample. The median county-product HHI in my sample is 0.56, and 82% of markets exceed 0.25, the Department of Justice's 2010 threshold for "highly concentrated" markets.[11] These concentrated local markets

[11]When aggregating state-wide, the median state-product market in my sample has an HHI of 0.20,

Table 1: Distance to Technological Frontier: British and US Patents, 1855–1859

Product	Distance to Frontier	British Patents	US Patents
Cotton Textiles	1.31	464	353
Worsted Goods	1.31	464	353
Boots	1.71	181	106
Hats	1.86	169	91
Iron Bar	2.53	374	148
Iron Pig	2.53	374	148
Steel	2.53	374	148
Paper	2.79	301	108
Beer	2.87	149	52
Whiskey	2.87	149	52
Glass	3.39	95	28
Gloves	3.64	222	61
Rope	3.88	128	33
Nails	4.98	214	43
Paints	7.82	297	38

Notes: Distance to frontier is the ratio of British to American patents in 1855–1859. A value of one indicates technological parity; higher values indicate the US lags further behind. Products sharing a patent count are grouped under a common class in the underlying patent data. British patent data from Hanlon (2016); US patent data from Shaw (undated, accessed 2025). See Section 4.1 for construction details and Appendix A.2 for historical validation.

motivate the Bertrand oligopoly framework I develop in Section 3.

3 A Theory of Protection and Distance to Frontier

To explain why protection can raise output for some products but reduce it for others, I develop a model in which innovation incentives depend on distance to the technological frontier. Motivated by the evidence on concentrated local markets, the model combines Bertrand oligopoly and CES demand with an innovation decision, generating variable markups.[12] This makes the innovation response to protection depend on the firm's initial market share, which in turn is determined by initial technology levels (i.e. distance to frontier). This yields novel results: protection can reduce output and innovation in partial equilibrium, and the model provides an imperfect competition rationale for infant

exceeding the DOJ's "moderately concentrated" threshold of 0.15; 43% of markets exceed the highly concentrated threshold.

[12]Variable markups can instead arise from the demand system rather than the market structure. In Appendix D.6, I show a model of monopolistic competition with Kimball demand delivers the same results.

industry protection. These results do not obtain under constant markup gravity models (such as monopolistic competition with CES demand), where unilateral protection always weakly raises output and never reduces the incentive to innovate.

Sections 3.1 and 3.2 set up the model. Sections 3.3 and 3.4 derive comparative statics and provide the economic mechanism. Later in Section 7, I study the welfare implications of trade in general equilibrium.

3.1 Setup

In county i's market for product z, a local producer competes a la Bertrand against a UK exporter. Local expenditure on product z is E_{iz}. I assume preferences across products are Cobb-Douglas with weights χ_{iz}, thus $E_{iz} = \chi_{iz} E_i$. In this section, total expenditures E_i are held fixed in partial equilibrium, and will be absorbed by the county fixed effect in the empirical section. In Section 7, I allow expenditures E_i to adjust in general equilibrium.

Demand is CES with elasticity $\sigma > 1$. If the local and UK prices are p_{iz}^{US} and p_{iz}^{UK} respectively, demand for the local variety is:

$$q_{iz}^{US} = (p_{iz}^{US})^{-\sigma} P_{iz}^{\sigma-1} E_{iz},$$

where the price index is $P_{iz} = \left[(p_{iz}^{US})^{1-\sigma} + (p_{iz}^{UK})^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$. The quality-adjusted productivity of the UK is T_z^{UK}. I assume all counties in the US share the same initial technology T_z^{US}. Define distance to the frontier as:

$$D_z := \frac{T_z^{UK}}{T_z^{US}}.$$

Thus D_z close to one means the US is close to the British frontier, while larger D_z means the US is further behind. I assume both countries have constant returns to scale production, as well as common factor prices normalized to one, relaxing the latter assumption in Section 7. Unit costs for the UK are thus:

$$c_z^{UK} = \frac{1}{T_z^{UK}}.$$

Let t_z denote the ad valorem tariff on product z, and let τ_{iz} be the freight cost from the US port to county i. Since tariffs were not assessed on domestic freight costs, the foreign firm's delivered unit cost is $c_{iz}^{UK} = c_z^{UK}(1 + t_z) + \tau_{iz}$. Total protection, expressed in ad valorem terms relative to foreign marginal cost, is thus:

$$TP_{iz} := \frac{c_{iz}^{UK} - c_z^{UK}}{c_z^{UK}} = t_z + \frac{\tau_{iz}}{c_z^{UK}}.$$

Interior regions have higher τ_{iz} and thus higher total protection TP_{iz}. The local firm can pay a fixed cost F to upgrade its technology from T_z^{US} to αT_z^{US}, with step size $\alpha > 1$.[13] In this section, I am agnostic to whether this upgrade reflects domestic innovative activity or adoption of existing technology, but I refer to this upgrading as innovation throughout as a shorthand. Let $I_{iz} \in \{0, 1\}$ denote the local firm's decision to innovate. Thus, the local firm's unit cost is:

$$c_{iz}^{US}(I_{iz}) = \frac{1}{T_z^{US}\alpha^{I_{iz}}}.$$

3.2 Pricing and Innovation

Under CES demand with Bertrand oligopoly, the local firm's residual demand elasticity—its own-price elasticity holding the UK price fixed—is

$$\varepsilon_{iz}^{US}(I_{iz}) := -\left.\frac{\partial \log q_{iz}^{US}}{\partial \log p_{iz}^{US}}\right|_{p_{iz}^{UK}} = \sigma - (\sigma - 1)\lambda_{iz}(I_{iz}).$$

I write the local firm's market share $\lambda_{iz}(I_{iz})$ and residual demand elasticity $\varepsilon_{iz}^{US}(I_{iz})$ as functions of I_{iz} to make their dependence on innovation explicit. A larger market share $\lambda_{iz}(I_{iz})$ implies a lower residual demand elasticity $\varepsilon_{iz}^{US}(I_{iz})$ and a higher markup $\mu_{iz}^{US}(I_{iz})$. Bertrand first-order conditions imply

$$p_{iz}^{US}(I_{iz}) = \frac{\varepsilon_{iz}^{US}(I_{iz})}{\varepsilon_{iz}^{US}(I_{iz}) - 1} c_{iz}^{US}(I_{iz}) = \frac{\sigma - (\sigma - 1)\lambda_{iz}(I_{iz})}{(\sigma - 1)\big(1 - \lambda_{iz}(I_{iz})\big)} c_{iz}^{US}(I_{iz}),$$

and

$$p_{iz}^{UK}(I_{iz}) = \frac{1 + (\sigma - 1)\lambda_{iz}(I_{iz})}{(\sigma - 1)\lambda_{iz}(I_{iz})}(1 + TP_{iz})c_z^{UK}.$$

The local firm only sells its products locally in this section, so output is equal to the domestic trade share times expenditures: $Y_{iz}(I_{iz}) = \lambda_{iz}(I_{iz})E_{iz}$. Operating profit is sales divided by the perceived elasticity:

$$\Pi_{iz}(I_{iz}) = E_{iz}\lambda_{iz}(I_{iz})\frac{1}{\varepsilon_{iz}^{US}(I_{iz})}.$$

[13] In Appendix D.7, I study a continuous-innovation version of this model, which delivers additional comparative statics results and maps more closely to the empirical specification for patents.

Innovation increases profits through two channels: It increases market share, and it increases the markup (captured by an increasing $\frac{1}{\varepsilon_{iz}^{US}(I_{iz})}$). The gain from innovation is the resulting profit differential:

$$G_{iz}(I_{iz}) := \Pi_{iz}(1) - \Pi_{iz}(0)$$

The firm innovates if this gain exceeds the fixed cost:

$$I_{iz}^* = 1\{G_{iz}(D_z, TP_{iz}) \geq F\}.$$

The gain from innovation G_{iz} is always positive. However, G_{iz} is not monotonic in total protection TP_{iz} under Bertrand oligopoly — protection can shrink the gap between post- and pre-innovation profits, potentially pushing G_{iz} below F and causing the firm to forego innovation. This is in contrast to CES demand with monopolistic competition, where G_{iz} is monotonic increasing in TP_{iz}.

Figure 2 illustrates this non-monotonicity numerically. Panel A plots G_{iz} against distance to the frontier before and after protection rises from $TP_{iz}^0 = 0.30$ (blue) to $TP_{iz}^1 = 0.59$ (orange). For distances below 1.29, the blue line lies above the orange line, so protection reduces G_{iz}; above this threshold, the ordering reverses and protection raises G_{iz}. In the red region (D_z between 0.90 and 1.10), G_{iz} exceeds F before the protection increase but falls below it afterward, so the firm stops innovating. In the green region (D_z between 1.48 and 1.81), G_{iz} lies below F before the protection increase but rises above it afterward, so the firm innovates only under more protection.

Panel B plots the elasticity $\partial \log G_{iz} / \partial \log TP_{iz}$ evaluated at $TP_{iz} = 0.30$. Below a distance to the frontier of 1.17, the elasticity is negative: protection reduces the innovation gain. Above 1.17 it is positive and rises with distance to the frontier over the range shown.

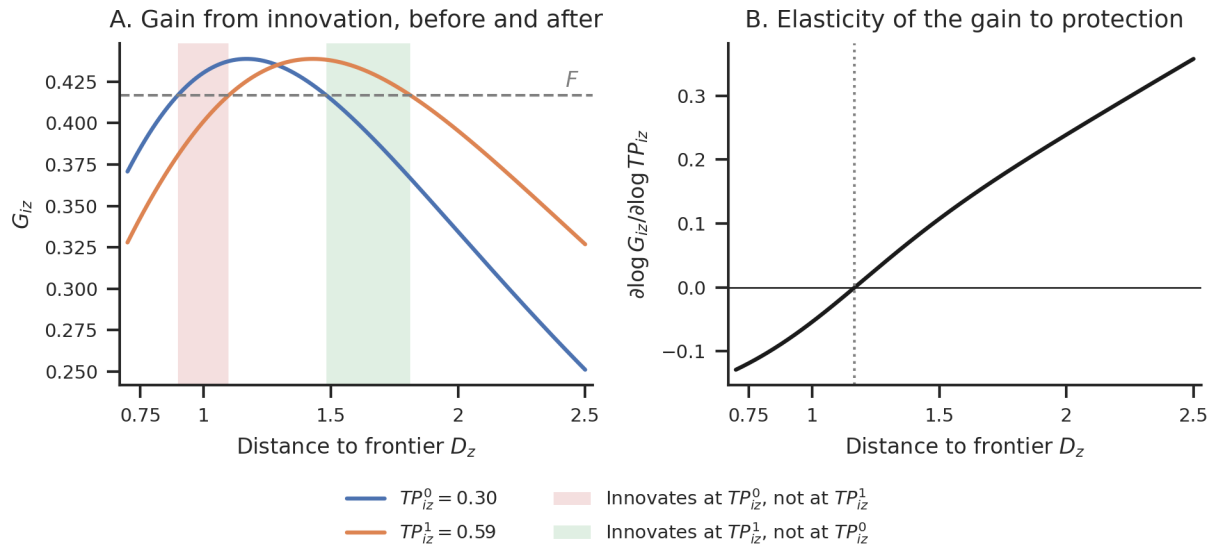
3.3 Comparative Statics

Let TP_{iz}^0 and $TP_{iz}^1 > TP_{iz}^0$ denote levels of total protection before and after an increase. We can decompose the change in output into two components:

$$\begin{aligned} \Delta \log Y_{iz} = & \underbrace{\log Y_{iz}(I_{iz}^*(TP_{iz}^0), TP_{iz}^1) - \log Y_{iz}(I_{iz}^*(TP_{iz}^0), TP_{iz}^0)}_{\text{substitution effect} > 0} \\ & + \underbrace{\log Y_{iz}(I_{iz}^*(TP_{iz}^1), TP_{iz}^1) - \log Y_{iz}(I_{iz}^*(TP_{iz}^0), TP_{iz}^1)}_{\text{innovation effect}}. \end{aligned} \tag{1}$$

The first term is the substitution effect. It holds the optimal innovation decision fixed at its pre-protection value, $I_{iz}^*(TP_{iz}^0)$, and measures how output changes when protection

Figure 2: Gain from Innovation and Protection



Notes: Panel A plots the level gain from innovation $G_{iz} = \Pi_{iz}(1) - \Pi_{iz}(0)$ against distance to the frontier D_z at two levels of total protection, $TP^0_{iz} = 0.3$ and $TP^1_{iz} = 0.59$. Shaded bands mark where the innovation decision differs between the two levels: the firm innovates at TP^0_{iz} but not at TP^1_{iz} (red) or at TP^1_{iz} but not at TP^0_{iz} (green). Panel B plots $\partial \log G_{iz} / \partial \log TP_{iz}$ evaluated at $TP_{iz} = 0.30$; the dotted line marks where it changes sign. Parameters: $\sigma = 8$, $\alpha = 3$, $F = 0.417$ and $E_{iz} = 1$. $D_z = T_z^{UK}/T_z^{US}$ is the productivity ratio in the model.

rises from TP^0_{iz} to TP^1_{iz}. The second term is the innovation effect. It holds protection fixed at its post-change level, TP^1_{iz}, and measures the additional change in output caused by the induced change in the innovation decision from $I^*_{iz}(TP^0_{iz})$ to $I^*_{iz}(TP^1_{iz})$.[14]

The substitution effect is always positive: an increase in protection raises the relative price of the foreign variety and induces local consumers to substitute toward the domestic variety in partial equilibrium, i.e. holding fixed total expenditures.

The sign of the innovation effect hinges on whether protection raises or reduces innovation, i.e. the sign of $\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})}$. Since the firm innovates if and only if $G_{iz} \geq F$, this reduces to whether protection raises or lowers the gain from innovation G_{iz}. Under Bertrand oligopoly, $\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})}$ can be positive or negative. For a near-frontier firm that already commands a large market share, the firm is on a sufficiently low-elasticity portion of its residual demand curve that the incremental gain from innovation can fall, pushing G_{iz} below F. Thus, in this case, protection makes it profit-maximizing to save on innovation costs, so $\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})} < 0$. If this innovation loss is large enough, it can overtake the substitution effect, resulting in a net negative effect of protection on output.

[14]The decomposition can instead be written by first allowing innovation to adjust and then changing protection, rather than first changing protection then allowing innovation to adjust. The main insights are unchanged.

This possibility of a negative innovation reaction relies crucially on variable markups.[15] Under perfect competition, there are no economic profits, so there is no profit-based motive to innovate: $\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})} = 0$. The innovation channel is absent entirely. Under monopolistic competition with CES demand, each firm is atomistic and faces a constant elasticity of demand, implying a constant markup of $\frac{\sigma}{\sigma-1}$. Therefore, the gain from innovation is a fixed fraction of pre-innovation profits, so protection always raises the incentive to innovate, i.e. $\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})} \geq 0$.[16] Far-from-frontier firms in my model approximate this benchmark: their low market share implies a residual demand elasticity close to σ, so protection raises G_{iz} and can push it above F.

3.4 Main Propositions

I now state the main comparative statics results formally. Propositions 1 and 2 characterize the effect of protection on innovation and output respectively.

Proposition 1: There always exists a threshold distance to frontier $D^{I,*}_{iz}$ such that for all $D_z \geq D^{I,*}_{iz}$, protection weakly increases innovation:

$$\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})} \geq 0 \quad \text{for } D_z \geq D^{I,*}_{iz}.$$

Below this threshold, protection can reduce innovation.

Corollary 1: Under additional conditions,[17] protection strictly reduces innovation for $1 \leq D_z < D^{I,*}_{iz}$.

$$\frac{\Delta I^*_{iz}}{\Delta \log(TP_{iz})} < 0 \quad \text{for } D_z < D^{I,*}_{iz}.$$

Proof. See Appendix D.3.1 and Appendix D.3.3.

Proposition 2: There always exists a threshold $D^{Y,*}_{iz}$ such that for all $D_z \geq D^{Y,*}_{iz}$, protection increases output:

$$\frac{\Delta \log Y_{iz}}{\Delta \log(TP_{iz})} > 0 \quad \text{for } D_z \geq D^{Y,*}_{iz}.$$

[15] These results are not specific to Bertrand competition. Declining residual elasticities can also arise from the demand system rather than the market structure. In Appendix D.6, I show that a model of Kimball demand with monopolistic competition also produces the same patterns.

[16] This is akin to the "market size effect" in the innovation literature (Akcigit and Melitz 2022, Schmookler 1966, Acemoglu and Linn 2004). Protection essentially increases the "market size" for domestic firms by raising the share of local consumption directed towards local producers.

[17] Namely, the gain from innovation is single-peaked in D_z and, at $D_z = 1$, $[\Pi_{iz}(1) - \Pi_{iz}(0)]_{TP_{iz}=TP^0_{iz}} \geq F > [\Pi_{iz}(1) - \Pi_{iz}(0)]_{TP_{iz}=TP^1_{iz}}$.

Below this threshold, protection can reduce output.

Corollary 2: If in addition the conditions of Corollary 1 hold and $\alpha > (1 + TP_{iz}^1)/(1 + TP_{iz}^0)$, then on the domain $D_z \geq 1$ protection strictly reduces output:

$$\frac{\Delta \log Y_{iz}}{\Delta \log(TP_{iz})} < 0 \quad \text{for } D_z < D_{iz}^{Y,*}.$$

Proof. See Appendix D.3.2 and Appendix D.3.4.

Taken together, Propositions 1 and 2 say that protection weakly increases output and innovation for products sufficiently far from the frontier. The novel result is that protection can potentially reduce output and innovation for products close to the frontier. Appendix D.1 illustrates the possibility of protection causing innovation and output to fall with a numerical example, with $\sigma = 8$, $\alpha = 3$, $D_z = 1.05$, and a 10 percentage point increase in protection. By contrast, gravity models predict that unilateral protection — absent any retaliatory response — always increases output in partial equilibrium (see Appendix B.5).[18] Under stronger assumptions, Corollaries 1 and 2 guarantee single crossing: protection strictly reduces output and innovation below their respective thresholds.

These theoretical results motivate the main regression specification, which interacts distance to frontier with trade protection so that the effect of protection is allowed to change sign across products (Equation (4) in Section 5). I also structurally interpret the regression coefficients through the lens of this model in Section 6.4.

Note that the heterogeneity in responses across products arises purely from variable markups interacting with differences in initial technology levels. Additional heterogeneity on the cost side, such as larger step sizes for more technologically backward products, i.e. $\frac{\partial \log \alpha_z(D_z)}{\partial D_z} > 0$ — the "advantage of backwardness" of Gerschenkron (1962) — would further amplify these forces.[19]

Additionally, with endogenous innovation, trade protection can actually lead to markups falling.[20] While the relationship between markups and innovation is theoretically ambiguous, the empirical results below imply that protection raises markups, innovation and growth jointly far from the frontier but lowers all three close to it. Thus, markups

[18] Thus standard frameworks cannot rationalize the negative effects of protection on output for some sectors as reported by Klein and Meissner (2024), but my model can.

[19] Advantage of backwardness reinforces the positive effects of protection among backward firms that still require protection to innovate: less protection is needed to induce upgrading, and the induced upgrade delivers larger output gains.

[20] Without endogenous innovation, protection always raises the local firm's market share and therefore its markup. With endogenous innovation, however, protection can induce the firm to forgo upgrading. If the resulting productivity loss outweighs the positive substitution effect— as the empirics suggest near the frontier—its market share and markup fall. Its price rises, but marginal cost rises proportionately more.

covary positively with innovation and growth in this setting, contrasting with accounts linking rising markups to declining innovation in the modern US (e.g. Akcigit and Ates 2021, Aghion et al. 2023).

4 Data

4.1 Data Sources and Construction

I digitize US tariffs and foreign import prices entering US ports from the Foreign Commerce and Navigation Tables of the United States for 1859 and 1869, and the Report of the Secretary of the Treasury on the State of the Finances for 1859. These tables record total import quantities and customs values by product, from which I infer per-unit import prices (see Figure 3 for an example). They also list the applicable tariff — whether ad valorem, specific, or composite — for each tariff line item. I convert all physical units into pounds, so prices are expressed as dollars per pound.

As there is no input-output table available in this period, I construct one using Atack, Bateman and Weiss' (2006) digitized state-level census of manufactures samples, which records input-specific usage.[21] I deflate all nominal values to 1870 dollars using the Historical Statistics of the United States (2006). County-product output, labor employment, total materials usage and capital are taken from the recently digitized Census of Manufactures for 1860 and 1870 (Rotemberg and Hornbeck 2024).

British patent data comes from Hanlon (2016). Table 2 lists the basket of products in my sample alongside British patent counts. As expected, products associated with the two Industrial Revolutions saw more patenting activity, with textiles — the quintessential First Industrial Revolution sector — at the top. Iron and steel, already the second most patented category, foreshadow their dominant role in the coming Second Industrial Revolution. US patent data comes from Shaw (undated, accessed 2025).

$$Dist_Front_z = \frac{\text{\# British Patents}_{1855-1859,z}}{\text{\# US Patents}_{1855-1859,z}}.$$

I harmonize product categories across the two countries' patent data and measure

[21] I limit the sample to single product establishments, as, like in the vast majority of manufacturing censuses even today, input usage is not recorded by product. See Orr (2022) and De Loecker et al. (2016) for further discussion on inferring input allocations in multi-product establishments. I first compute firm-specific Leontief shares, then aggregate across firms to obtain product shares using Domar weights (each firm's share of total revenue). Mathematically: Let $\zeta_{z' \to z}$ be the Leontief share of input z' used for z. Let f be an establishment. $\zeta_{z' \to z, f} = \frac{\text{expenditure on input } z' \text{ by } f}{\text{total input expenditure of } f}$, and $\zeta_{z' \to z} = \sum_{f \in z} w_f \zeta_{z' \to z, f}$, where $w_f = \frac{y_f}{\sum_{f' \in z} y_{f'}}$ and y_f is the revenue of establishment f.

No 9.—*General statement of foreign imports*—Continued.

	WHENCE IMPORTED.	MERCHANDISE PAYING SPECIFIC DUTIES.									
		IRON, AND MANUFACTURES OF IRON.									
		Bar iron, rolled or hammered.		Railroad iron, not above six inches high.		Boiler plate.		Band iron.		Hoop iron.	
		$15 per ton.		$12 per ton.		$20 per ton.		$20 per ton.		$20 per ton.	
		Cwt.	Dollars.	Cwt.	Dollars.	Cwt.	Dollars.	Cwt.	Dollars.	Cwt.	Dollars.
1	Sweden and Norway	8,105	30,686								
2	Swedish West Indies										
3	Denmark										
4	Danish West Indies										
5	Hamburg										
6	Bremen	5,505	14,069								
7	Holland										
8	Dutch West Indies										
9	Dutch Guiana										
10	Dutch East Indies										
11	Belgium										
12	England	33,174	63,243	48,744	62,226	1,284	2,973	1,915	3,751	5,770	12,942
13	Scotland	30	36								
14	Ireland										
15	Gibraltar										
16	Malta										
17	Canada	4	12			2	8				
18	Other British North American Possessions	5	12								
19	British West Indies										
20	British Honduras										
21	British Guiana										
22	British Possessions in Africa										
23	British East Indies										
24	France on the Atlantic										
25	France on the Mediterranean										
26	French West Indies										
27	French Guiana										
28	Spain on the Atlantic										

Figure 3: Excerpt from the 1859 Foreign Commerce and Navigation Tables

Notes: Below each product name the applicable tariff is listed (e.g. $15 per ton for bar iron, rolled or hammered). Port prices are inferred by dividing the "Dollars" column by the "Cwt." column, where Cwt. denotes hundredweight, equal to 100 pounds.

distance to the technological frontier as the ratio of British to American patents in the pre-shock period (1855-1859). This patent-based measure is similar in spirit to the one used by Akcigit, Ates and Impullitti (2024).[22] A value of one means the US has as many patents as the UK (thus it's "at the frontier"), while a large value signifies the US significantly lags behind the UK in that product category. As discussed in Section 2, the US patent system was cheaper and examined patents for novelty, while the British system was costlier and relied on registration (Khan and Sokoloff 2001; Khan 2005). My measure therefore likely understates true distance to frontier, though product rankings and empirical conclusions

[22]Alternative measures face their own challenges. R&D or innovative expenditures are not available for this period. Total factor productivity requires production function estimation, which in turn requires establishment-level data that is scarce in this period. Moreover, classical control function approaches (Olley and Pakes 1996, Levinsohn and Petrin 2003) rely critically on either perfect or monopolistic competition, as noted by Doraszelski and Jaumandreu (2021) and Ackerberg and De Loecker (2024), assumptions unlikely to hold across all of the markets I study and in direct contradiction to the model in Section 3. Finally, my data does not separate prices from quantities, measured productivity could reflect markups rather than quality or productivity differences.

Table 2: British Patents by Product Class

Product	British Patents	Category
Cotton Textiles	464	First IR
Worsted Goods	464	First IR
Iron Bar	374	First/Second IR
Iron Pig	374	First/Second IR
Steel	374	Second IR
Paper	301	First IR
Paints	297	Second IR
Glove	222	Neither
Nails	214	First/Second IR
Boots	181	Neither
Hats	169	Neither
Beer	149	Neither
Whiskey	149	Neither
Rope	128	Neither
Glass	95	Neither

Notes: British patent counts by product class, 1855–1859, from Hanlon (2016). IR = Industrial Revolution; products are classified by whether their crucial innovations occurred during the First Industrial Revolution (c. 1760–1840), the Second (c. 1870–1914), or neither. Products sharing a patent count are grouped under a common category in the source data. Patenting activity is concentrated in the Industrial Revolution sectors, with textiles at the top and iron and steel close behind.

remain unaffected provided the gap in patenting propensity is uniform across industries.

Table 1 displays the rankings of the products by distance to frontier, and Appendix A.2 presents historical context and validation for these rankings as a measure of technological gaps. For example, cotton textiles' near-frontier status reflects decades of transatlantic technology transfer dating to Samuel Slater and Francis Cabot Lowell (Jeremy 1981), while paints' position as the most distant product is consistent with virtually all major innovations in paint originating in Europe during the 19th century (Barnett 2004).

Finally, I extract transportation cost data from Donaldson and Hornbeck (2016) (DH hereafter).[23] DH's network contains information on whether travel is possible by waterway (fastest), railway (close to fastest) or wagon (very slow). Assigning values to the speed of travel per mile-ton and assigning a cost to transitioning between different modes of transportation, DH compute the shortest path between any two counties in the US and assign this value as the transportation cost.

[23]This was built using Atack's shapefiles (2015, 2016, 2017).

I build on DH's network by 1) adding in access to international ports and 2) estimating the functional form of transportation costs in the next subsection rather than assuming iceberg. I assume that trade operates according to a "hub-and-spoke" system: US port $\rightarrow$ US internal counties. I first compute the distance from the US county to one of five major US ports[24] using DH's map. I then use the minimum of these five distances as the county's transportation cost to its nearest major port. See Figure 4 for a depiction of the transport network in the East Coast and Midwest.

My final estimation sample accounts for 27.9% of output, 30.8% of value added, 36.1% of workers, and 32.4% of capital of the total 1860 manufacturing data. The sampled industries are somewhat more capital intensive and produced in larger establishments than manufacturing as a whole, consistent with the fact that products appearing in trade statistics — necessary to construct tariffs and port prices — tend to be tradable and produced at a larger scale. See Appendix Table A1 for further descriptive comparisons.

4.2 The Structure of Transportation Costs

Correctly measuring protection requires accurately specifying transportation costs. For this purpose, I digitize commodity-specific freight rates from the Aldrich Report (1893). Let $\tau_{odzs} = p_{ozs}^{\omega}\zeta_{ods}$, where o is an origin, d is the destination, z is a product, s is year, τ_{odzs} is transportation cost, p_{ozs} is the origin price of product z and ζ_{ods} is the freight rate common to all products.

When $\omega = 1$, this specification nests the iceberg formulation widely used in the trade and spatial literatures: transportation costs scale proportionally with origin prices. This may reflect higher handling costs for expensive goods or markup pricing by the transportation sector.[25] At the other extreme, when $\omega = 0$, transportation costs are fully additive, and are the same for all products regardless of the product's origin price. Thus, only bilateral features between o and d matter, such as physical distance between the two locations and transportation infrastructure. ω can also be between 0 and 1. Taking logs, I estimate:

$$\log \tau_{odzs} = \omega \log p_{ozs} + \delta_{ods} \tag{2}$$

The route-year fixed effect δ_{ods} absorbs everything specific to shipping between o and d in year s (including geographic characteristics and transportation infrastructure), so ω

[24]These are ports that had more than $7.5 million worth of imports in 1860: Boston, Baltimore, New York City, New Orleans and Philadelphia (Table 9, Foreign Commerce and Navigation of the US 1860). This is roughly $310 million in today's dollars (MeasuringWorth 2026).

[25]Hummels and Skiba (2004) show that a monopolist in the transportation sector's markup is increasing in the origin's good price.

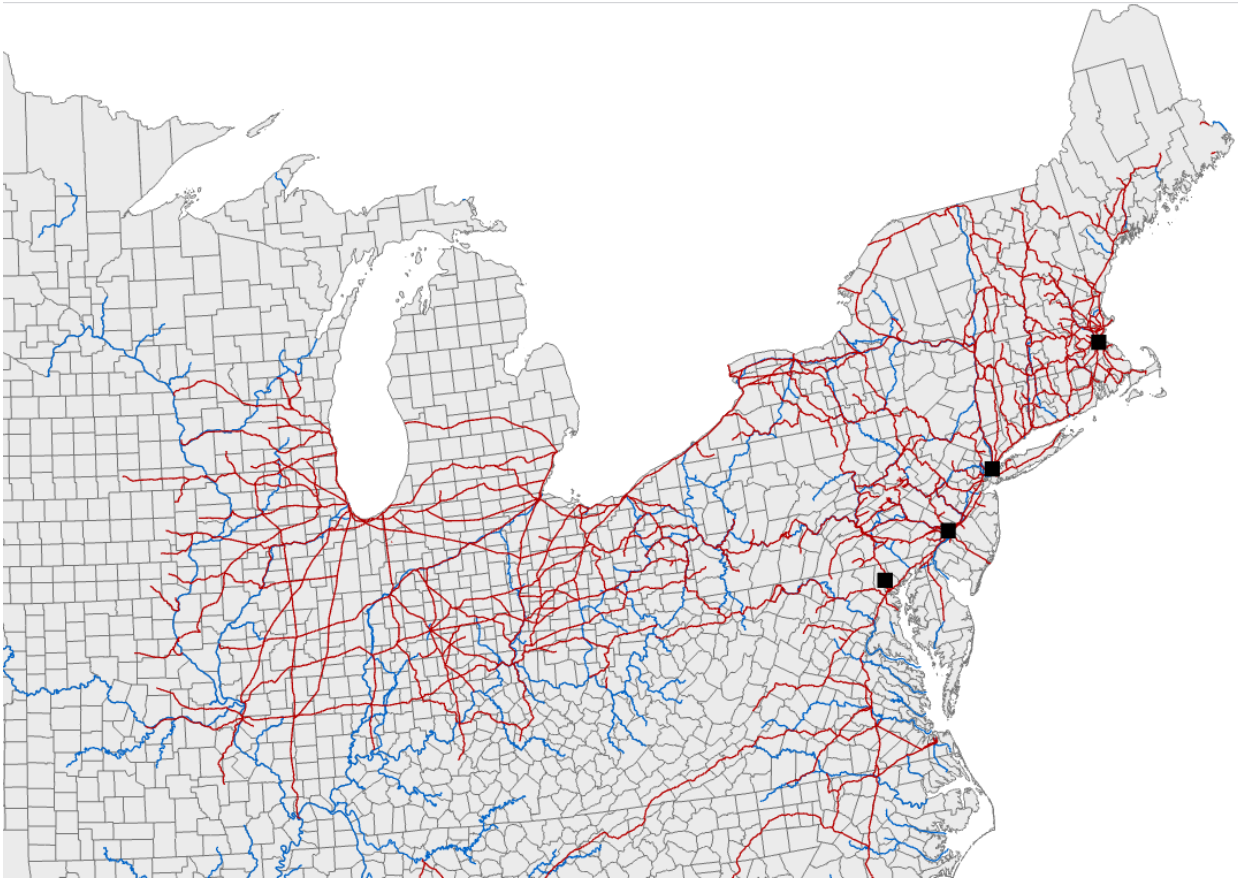


Figure 4: U.S. East Coast and Midwest Transportation Network, 1860

Notes: Railways in red, waterways in blue, with black squares marking major international ports (see footnote 24 for how these were selected). Source: Atack (2015, 2016, 2017).

is identified from variation in freight rates across products shipped along the same route in the same year. Critically — and unusually — the Aldrich Report contains information on transportation costs for a given route across different products. Thus τ_{odzs} is directly taken from the data (though only for a subset of routes and products). As for p_{ozs}, I set it to the price of imported products (inclusive of the tariff) if o is a port city.

As Table 3 shows, I find that $\hat{\omega} = 0.139$ with a 95% confidence interval of $[0.0959, 0.182]$, thus transportation costs are much closer to being additive than iceberg. I therefore set $\omega = 0.139$ in the baseline specification, but I examine robustness to $\omega = 0$ (fully additive case), $\omega = 0.0959$ (lower bound of 95% confidence interval), and $\omega = 0.182$ (upper bound of 95% CI) in Section 6.2.

Table 3: Transportation Cost Regression

	Log Transport Cost
Log Import Price ($\hat{\omega}$)	0.139***
	(0.022)
Observations	124
Route-Year FE	Yes

Notes: Estimates of Equation (2), regressing log commodity-specific transportation costs on log import price with route-year fixed effects, so that $\hat{\omega}$ is identified from variation in freight rates across products shipped along the same route in the same year. Transport costs digitized from the Aldrich Report (1893); port prices from the Foreign Commerce and Navigation Tables. The point estimate is 0.139, signifying transportation costs are close to additive. Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

5 Empirical Framework

This section develops the empirical framework. I first discuss the key measure of protection that incorporates both tariffs and transportation costs (Section 5.1). I then present the econometric specification (Section 5.2) and instrument (Section 5.3), followed by identification arguments (Section 5.4–5.6).

5.1 Treatment

Recall from Section 3 that total protection is the tariff t_z on product z and the freight cost τ_{iz} of delivering the foreign good to county i. Empirically, I measure it as

$$TP_{iz} := t_z + \frac{\tau_{iz}}{p_z}, \tag{3}$$

where p_z is the price of the British good at a US port, the empirical counterpart to unobserved British marginal costs c_z^{UK} in the theory section. Freight costs enter additively as domestic transport costs were not subject to tariffs. To economize on notation, I write tariffs in ad valorem form, but the empirics take into account whether a tariff line is specific, ad valorem or composite. Appendix B.4 provides the general formulas.

If $\tau_{iz} = 0$, (i.e. no transportation costs), then TP_{iz} simply reduces to the tariff t_z, which is the notion of protection used in the majority of empirical trade papers (e.g. Pavcnik 2002, Bustos 2011, Topalova and Khandelwal 2011, Fajgelbaum et al. 2020).

When $\tau_{iz} > 0$, transportation costs provide additional protection by raising import prices much like tariffs, as in Juhasz (2018) and Pascali (2017). See Figure 5 for a visual

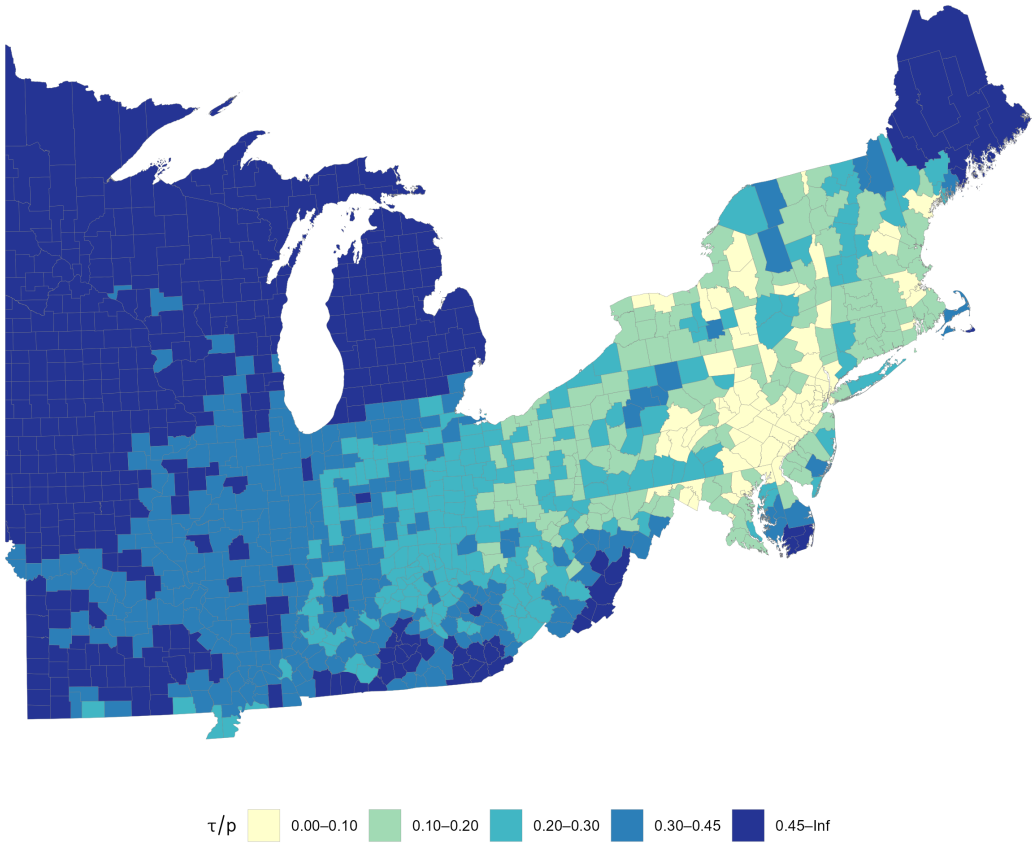


Figure 5: Natural Protection from Transportation Costs: Pig Iron, 1860

Notes: Transportation costs as a fraction of import price at port (τ_{iz}/p_z) for pig iron, 1860. Darker shading indicates higher transportation costs relative to port price, implying greater natural protection from foreign competition.

depiction of transportation costs across space for the product pig iron. Coastal counties near major ports such as New York and Philadelphia face negligible transportation costs relative to import prices. By contrast, for interior Midwest counties, transportation costs alone exceed 45% of the import price — roughly twice the protection provided by the 24% pig iron tariff in place in 1860.

Estimates in Section 4.2 using newly digitized commodity-specific freight rates show that transportation costs are mostly additive. This contrasts with the iceberg assumption commonly made in the trade and spatial literatures. Freight costs therefore provide greater protection for cheaper products, making product-specific port prices essential for measuring protection. I digitize these prices alongside tariffs, as described in Section 4.1.

Protection, in addition to making competitor import products more expensive, also makes foreign inputs used by domestic firms more expensive. Thus, I define an input protection measure in the following way:

$$\log(Input_TP_{iz}) = \sum_{z'} \zeta_{z' \to z} \log(TP_{iz'}),$$

where $\zeta_{z' \to z}$ is the Leontief share of input z' used in the production of z, using the state-level census of manufactures digitized by Atack, Bateman and Weiss (2006). See Section 4.1 for details on the construction of $\zeta_{z' \to z}$. I include this variable in all specifications.

5.2 Econometric Specification

I estimate the following regression, expressed in growth rates:

$$\Delta \log y_{iz} = \beta[\Delta \log(TP_{iz}) \times Dist_Front_z] + \theta \Delta \log(TP_{iz}) + \gamma X_{iz} + \Delta\delta_i + \Delta\delta_z + \Delta e_{iz} \quad (4)$$

where Δ is the difference operator (i.e. $\Delta \log y_{iz} = \log y_{iz,1870} - \log y_{iz,1860}$), i, z index county and product respectively. $\Delta\delta_i$ is a county fixed effect, $\Delta\delta_z$ is a product fixed effect and X_{iz} is a vector of controls, including the input protection measure defined in Section 5.1. θ is the baseline effect of protection for a product with distance to frontier of zero, and β captures how the effect of protection varies with distance to the frontier. The interaction of distance to frontier with protection is motivated by the model in Section 3, allowing for a potential sign flip in the effect of protection on output.

First, note that taking the difference controls for any persistent county-product factors δ_{iz}. For example, this would control for Pittsburgh being better at producing steel than Boston. Second, the county fixed effect $\Delta\delta_i$ controls for local equilibrium effects such as agglomeration, labor migration and local consumer price indices.[26] This would, for example, control for Chicago growing rapidly in all product categories. Third, unlike other designs, I am also able to include product fixed effects $\Delta\delta_z$, controlling for any nationwide trends in products such as technological developments. Thus, this design isolates within-product, across-county changes in trade protection.

In words, this design is comparing the growth rate of Philadelphia steel to Chicago steel when the change in protection is higher for Philadelphia steel than Chicago steel (while controlling for the average growth rate in Philadelphia and Chicago across all products). If

[26]The county fixed effect absorbs any change common to all products within a county, making identification more credible. As a result, β is identified from variation across products within a county and measures the effect of protection on output holding local real income fixed. This excludes the general equilibrium channel through which protection alters local real incomes — falling with higher import prices, rising with increased demand — and thereby demand for all products. Therefore, β should be interpreted as a partial equilibrium elasticity as opposed to a general equilibrium one. See also discussion in Chodorow-Reich (2019, 2020) and Guren et al. (2021).

Philadelphia steel grows faster than Chicago steel, we infer that output protection benefits growth for steel. One notable feature of this design is that identification operates entirely within-product — comparing counties producing the same product — rather than across products, as is common in the empirical trade literature.

However, the error term Δe_{iz} may still be correlated with $\Delta \log(TP_{iz})$. For example, newly built railroads may facilitate access to domestic markets and inputs, in addition to affecting protection from foreign imports. This would lead to biased estimates for β and θ. Thus I turn to an instrument to deal with this concern.

5.3 Instrument

Because changes in transportation costs may affect output through other channels — via access to inputs or markets for example — I instrument for $\Delta \log(TP_{iz})$ using only the tariff-driven component of protection changes. To do so, I use a first order approximation (see Appendix B.1 for derivation):

$$\Delta \log(TP_{iz}) \approx \underbrace{STS^t_{iz} \Delta \log t_z}_{IV} + STS^\tau_{iz} \Delta \log \tau_{iz} - STS^\tau_{iz} \Delta \log p_z, \tag{5}$$

where $STS^t_{iz} = \frac{t_z p_z}{t_z p_z + \tau_{iz}}$ and $STS^\tau_{iz} = \frac{\tau_{iz}}{t_z p_z + \tau_{iz}}$.

STS^t_{iz} captures the tariff's share in total protection for a given county-product iz pair. This expression has a "shift-exposure" structure as in Borusyak and Hull (2023), with cross-sectional exposures to aggregate shifters.[27]

I use the first component $STS^t_{iz} \Delta \log t_z$ as an instrument. Intuitively, counties where the tariff forms a larger fraction of local protection (i.e. those with high STS^t_{iz}) see larger percentage increases in protection for the same nationwide tariff increase. As coastal areas will generally have larger values of STS^t_{iz} than interior regions, this formalizes the intuition from Section 2 that tariffs have larger effects for coastal areas than interior areas.

One way to think about this is to note that interior regions have two forms of protection once a product arrives at a US port — tariffs and internal transportation costs — so a given tariff increase raises their protection by a smaller proportion than in coastal areas, which rely on the tariff alone. See Figure 6 for a visual depiction of the tariff-driven component of protection changes across space for pig iron. Coastal counties near major ports such as New York and Philadelphia see the largest percentage increases in protection, while

[27] This is technically not a shift-share according to the definition in Borusyak, Hull and Jaravel (2025), though in spirit it resembles the same structure of a local exposure to an aggregate shifter. Borusyak and Hull (2023) show that shift-shares are a special case of the "shift-exposure" framework.

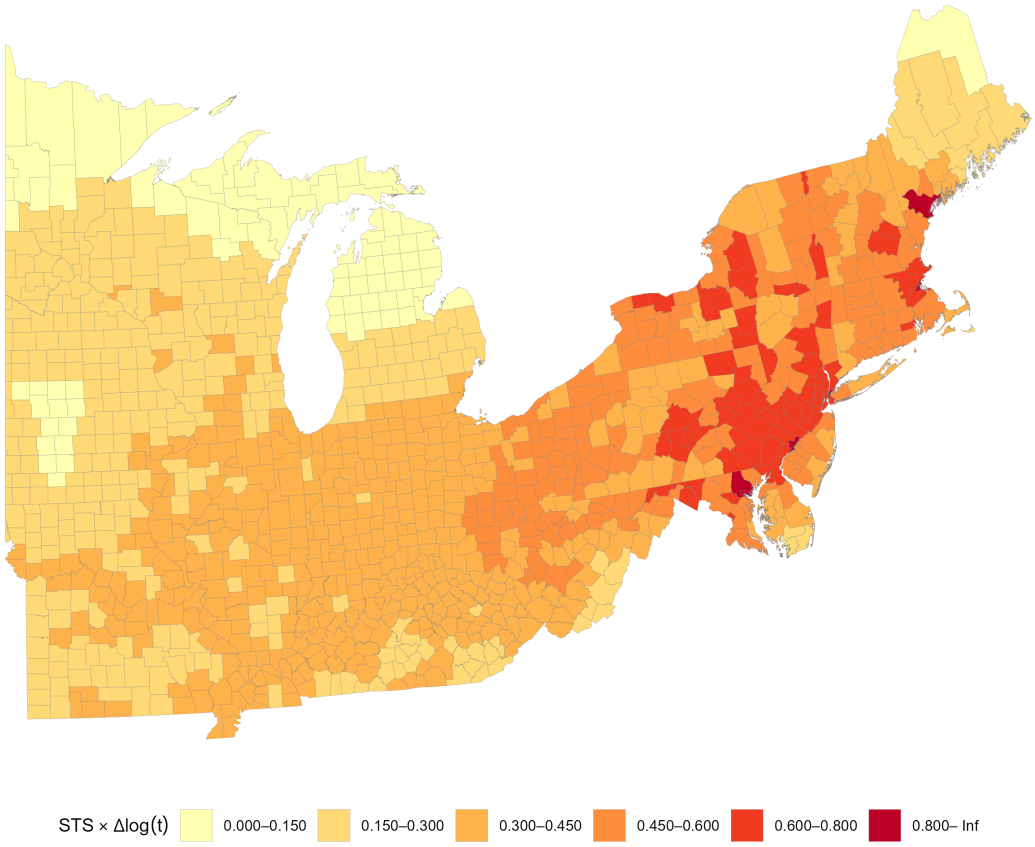


Figure 6: Instrument, Pig Iron, 1860–1870

Notes: The instrument $STS_{iz}^{t} \times \Delta \log t_z$ for pig iron, as defined in Equation (5). Darker shading indicates greater percentage increases in protection arising from the nationwide tariff shock. Coastal counties near major ports see the largest percentage increases in protection, while interior counties see smaller changes despite facing the same tariff increase.

interior counties — already shielded by high freight costs — see much smaller changes despite facing the same nationwide tariff increase.

The instrument developed in this section isolates the change in protection arising only from the change in tariffs.[28] The next three subsections will then argue that this component is exogenous to county-product shocks.

5.4 Identification

Formally, we need the following IV orthogonality condition to hold for identification of β and θ:

$$\mathbb{E}(STS_{iz}^{t} \Delta \log t_z \cdot \Delta e_{iz} | \Delta\delta_i, \Delta\delta_z) = 0 \tag{6}$$

[28] Another way to think about this is that this is the change in protection if the US had no new railways built in the decade 1860-1870 and port prices did not change.

A sufficient condition for Equation (6) is:

$$\mathbb{E}(\Delta \log t_z \cdot \Delta e_{iz} | \Delta\delta_i, \Delta\delta_z, STS_{iz}^t) = 0 \tag{7}$$

See Appendix B.2 for the proof. Condition (7) requires **nationwide** tariff changes to be uncorrelated with **local** product shocks, conditional on product and county fixed effects.

It is instructive to compare this condition with conditions required by other designs. One canonical approach is to take a nationwide trade liberalization episode and compare the change in outcomes for sectors that received larger tariff cuts to sectors that received smaller cuts (such as in Pavcnik (2002), Bustos (2011), De Loecker et al. (2016) and Flaaen and Pierce (2024)). For these cross-product comparisons, the equivalent condition would be: $\mathbb{E}(\Delta \log t_z \cdot \Delta e_z | \Delta\delta) = 0$, which requires **nationwide** tariff shocks to be uncorrelated with **nationwide** product shocks — as opposed to **local** product shocks. Thus a cross-product design is confounded if nationwide tariff changes are correlated with nationwide product trends, such as technological breakthroughs. For example, the Bessemer process, invented in 1856, enabled the mass production of steel for the first time in human history and transformed the industry over exactly the decade I study. A design comparing steel to other products might attribute its growth purely to the tariff. My design is robust to this, since the product fixed effect $\Delta\delta_z$ absorbs any nationwide product trend.

In addition, Borusyak, Hull and Jaravel (2022) show that shift-share designs under the exogenous shocks view are equivalent to a shock-level regression, so identification rests on the same cross-product comparison. Thus shift-shares are also potentially confounded if products receiving different tariffs are on differential trends, while my approach is robust to this concern. The other path to identification for shift-shares, based on the exogeneity of shares (Goldsmith-Pinkham et al. 2020), is unlikely to hold in my context, or in any setting where baseline industrial composition correlates with subsequent local trends. Locations specializing in different sectors are likely to be exposed to a variety of different shocks such as immigrant flows, technological changes and unionization drives. See Appendix B.6 for further comparison of this design with the shift-share design.

There are two major threats to Condition (7) holding: 1) endogenous tariff setting by policymakers, and 2) other shocks that are coincidentally correlated with tariffs. The rest of the section will discuss the trade policy environment, provide a narrative for why neither endogeneity concern is likely to hold in my context, and present a balance test showing that county-product pairs receiving more protection are comparable on observable characteristics to those receiving less.

5.5 Policy Environment and Narrative

In this period, 90% of Federal revenues came from tariffs (Irwin 2020), and federal budgets and surpluses played a pivotal role in tariff policy. In November 1860, Abraham Lincoln is elected President. Out of fears that Lincoln would restrict slavery, Southern states begin seceding from the Union and the Civil War begins two months after Lincoln's inauguration. While many assumed the War would end by Christmas 1861, it drags on and turns out to be an unprecedented fiscal burden, an order of magnitude larger than any other fiscal shock in US history up until that point. Figure 7 shows the dramatic spike in federal expenditures and deficits during the war years.

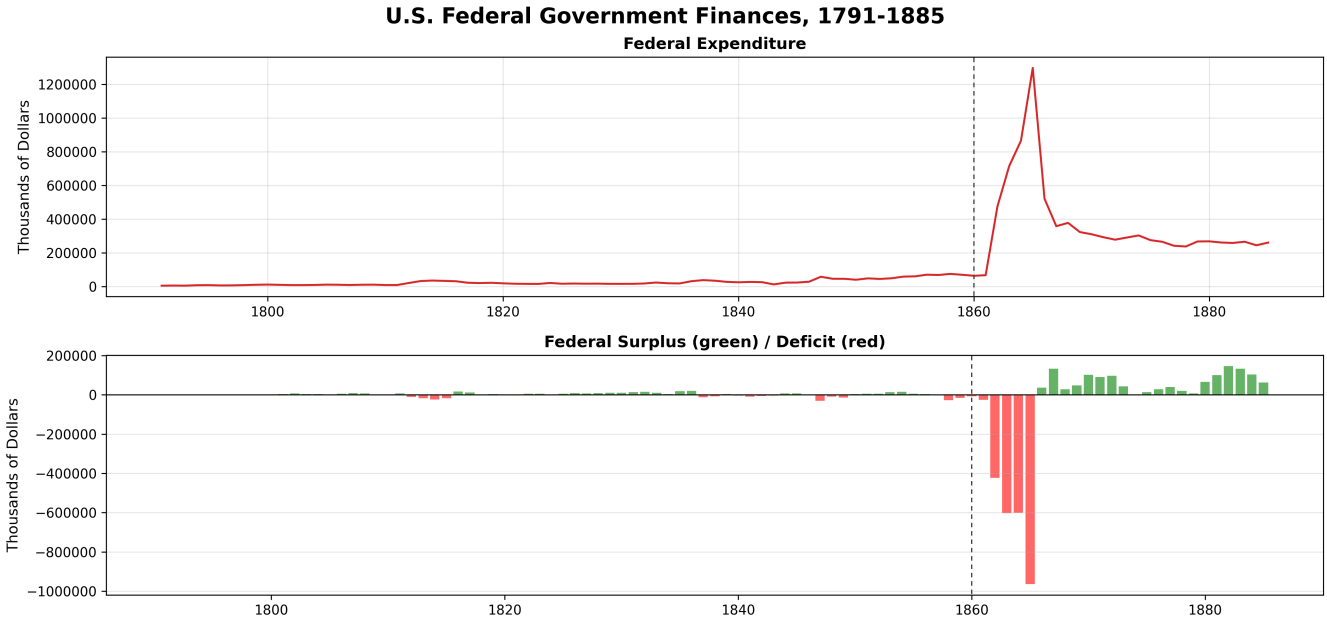


Figure 7: U.S. Federal Government Finances, 1791–1885

Notes: Top panel reports federal expenditure in thousands of 1870 dollars; bottom panel reports the federal surplus (green) and deficit (red) on the same basis. Dashed line indicates 1860, reflecting Civil War fiscal pressures, an order of magnitude larger than any prior fiscal shock. Data from Carter et al. (2006), Historical Statistics of the United States, Millennial Edition.

To finance the war effort, Congress passes five major tariff increases (and many more minor revisions) in 1861-1865. For comparison, Congress only passed one major tariff bill in the 1850s. Average tariffs went from 16% in 1860 to 45% by 1870. The unprecedented fiscal demands of the Civil War suggest that revenue generation was an overriding priority for federal policymakers. Nonetheless, other goals — such as industrial development or protecting a lobby — could have been pursued in tandem. A key advantage of my study is that it is robust to any product-level trend correlated with tariff changes, whether targeted

by policymakers or merely coincidental. This is not true of other approaches, such as shift-shares or cross-sector analyses.

In order for endogenous tariff setting to pose a threat to Condition (7), Federal policymakers would need to react to idiosyncratic county-product shocks in a way that systematically deviates from the county-time, county-product and product-time average. First, policymakers would need to identify county-product trends that deviate from the product-time, county-time and county-product averages — across roughly 1,500 counties and hundreds of tariff line items. Then they would need to set tariff levels to systematically exploit these residual trends. Finally, they would need to build majority consensus on this targeting in both the House and Senate. This is already a high cognitive and informational burden on policymakers — one only exacerbated by the wartime fiscal pressures of this period. In contrast, targeting national product-level trends is more tractable and easier to build Congressional consensus around. As a further check, I examine robustness to dropping the five largest cities in 1860 from the sample in Section 6.2, as these counties may have had particularly large influence over trade policy due to lobbying power or visibility.

This assuages concerns about endogenous tariff setting, but other Civil War shocks — such as military procurement, Emancipation and the effect of battles — loom large. I address these concerns in Section 6.2.

5.6 Balance Tests

While Condition (7) cannot be directly tested, balance tests provide suggestive evidence. If county-product pairs receiving more protection differ systematically on observable characteristics from those receiving less, it would suggest the threat of endogenous tariff setting or confounding shocks looms large—either of which would violate Condition (7). For example, systematic imbalance might indicate that policymakers were able to overcome the informational constraints discussed above and selectively target county-product pairs with certain characteristics. This would then raise concerns they could also identify and target idiosyncratic county-product trends, violating Condition (7).

I regress a number of initial period variables on the change in input and output protection, with product and county fixed effects.[29] These characteristics include total labor

[29]Mathematically:

$$\log X_{iz,1860} = \phi \left(STS_{iz}^{t} \times \Delta \log t_z \right) + \psi \sum_{z'} \zeta_{z' \to z} \left(STS_{iz'}^{t} \times \Delta \log t_{z'} \right) + \delta_z + \delta_i + \epsilon_{iz}$$

(county-product pairs with larger labor forces may have more electoral and lobbying power), firm size (county-product pairs with larger firms may be able to overcome collective action problems and lobby as in Olson (1965)) and output (implying more resources for lobbying).

Table 4: Balance Test

	IV Output		IV Input	
Dependent Variable	Coef.	SE	Coef.	SE
Log Total Labor Initial	−1.921	(2.854)	−3.299	(2.051)
Log Establishments Initial	−2.015	(1.498)	−0.739	(1.098)
Log Capital Initial	−0.286	(2.369)	−2.545	(2.418)
Log Labor Productivity Initial	−0.694	(1.489)	0.539	(0.540)
Log Output Initial	−2.194	(1.908)	−3.190	(1.832)
Log Output per Estab. Initial	−0.183	(1.929)	−2.405	(1.496)
Log Average Wage Initial	−0.635	(0.443)	0.242	(0.357)
Log Capital-Labor Ratio Initial	1.635	(2.696)	0.754	(1.094)

Notes: Each row reports a separate regression of a baseline (1860) county-product characteristic on the output and input protection instruments, including county and product fixed effects, as in footnote 29. The output instrument is $STS^t_{iz} \times \Delta \log t_z$; the input instrument is its Leontief-weighted counterpart across supplying products. Standard errors two-way clustered by county and product in parentheses. None of the sixteen coefficients is significant at the 10% level: county-product pairs receiving larger increases in protection are statistically balanced on baseline characteristics against those receiving smaller increases. Table 5 reports the same test omitting product fixed effects. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 4 presents the results. None of the sixteen coefficients is significant at the 10% level: county-product pairs receiving larger increases in protection are balanced on initial period observable characteristics against those receiving smaller increases, assuaging concerns of selective targeting by policymakers.[30]

The value of the within-product comparison becomes clear when I run the same balance test, but omit the product fixed effect δ_z, which controls for any nationwide product trend common to all units, such as technological breakthroughs. As we see in Table 5, the number of significant correlations goes from 0/16 to 6/16 (at the 10% level), raising concerns that Condition (7) does not hold when we also exploit cross-product

[30]While statistically insignificant, the magnitudes of the output IV coefficients for labor, establishments and output are economically meaningful. Results are robust to including these variables as controls in the main specification, see Appendix C.4. In addition, the sign runs opposite to what a lobbying story would predict: county-product pairs with higher initial output have more resources with which to lobby the central government, and so would be expected to receive more protection, not less.

variation. It appears output protection is skewed towards county-product pairs with higher initial output, higher wages and higher labor productivity. Isolating the within-product comparison is thus crucial for the exogeneity narrative — and is infeasible in canonical approaches, including both shift-share and nationwide liberalization designs.

Table 5: Imbalance Test: Without Product Fixed Effects

	IV Output		IV Input	
Dependent Variable	Coef.	SE	Coef.	SE
Log Total Labor Initial	−0.143	(0.151)	−0.885*	(0.444)
Log Establishments Initial	0.018	(0.056)	−0.232	(0.170)
Log Capital Initial	0.032	(0.128)	−0.877*	(0.437)
Log Labor Productivity Initial	0.264***	(0.089)	0.331	(0.197)
Log Output Initial	0.208*	(0.107)	−0.430	(0.355)
Log Output per Estab. Initial	0.189	(0.131)	−0.192	(0.397)
Log Average Wage Initial	0.062**	(0.024)	0.192**	(0.081)
Log Capital-Labor Ratio Initial	0.175	(0.124)	0.008	(0.272)

Notes: Each row reports a separate regression of a baseline (1860) county-product characteristic on the output and input protection instruments, including county fixed effects but omitting product fixed effects. The specification is otherwise identical to Table 4. Standard errors two-way clustered by county and product in parentheses. Six of sixteen coefficients are significant at the 10% level, compared with zero of sixteen when product fixed effects are included (Table 4), indicating that the within-product comparison is what delivers balance. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

6 Results and Interpretation

6.1 Output and Patent Results

I now run the specification in equation (4), reproduced here:

$$\Delta \log y_{iz} = \beta[\Delta \log(TP_{iz}) \times Dist_Front_z] + \theta \Delta \log(TP_{iz}) + \gamma X_{iz} + \Delta\delta_i + \Delta\delta_z + \Delta e_{iz}$$

where i is county, z is product and Δ represents a decade difference, which in this context is a ten year difference from 1860 to 1870. TP_{iz} is the measure of output protection, and X_{iz} is a vector of controls, including the measure of input protection as discussed in Section 5.1. $\Delta\delta_i$ is a county fixed effect and $\Delta\delta_z$ is a product fixed effect. $STS^t_{iz}\Delta \log t_z$

is the IV for $\Delta \log(TP_{iz})$, where $STS_{iz}^{t} = \frac{t_z p_z}{t_z p_z + \tau_{iz}}$ is the county-product exposure to the nationwide tariff shock, as discussed in Section 5.3.

Table 6: Effect of Trade Protection on Output Growth

	Δ Log Output	
	OLS	IV
Δlog(TP) $\times$ Dist_Front	1.359^{***}	1.493^{***}
	(0.411)	(0.398)
Δlog(TP)	-5.144^{**}	-4.424^{**}
	(1.834)	(1.997)
Observations	833	833
Input TP Control	Yes	Yes
County FE	Yes	Yes
Product FE	Yes	Yes
First-stage F-stat (KP)	—	15.5

Notes: Estimates of Equation (4), relating decadal changes in log output (1860–1870) to changes in log total protection TP_{iz} and its interaction with distance to the British technological frontier. $\Delta \log(TP_{iz})$ is instrumented with $STS_{iz}^{t} \times \Delta \log t_z$, the tariff's share in total protection interacted with the nationwide tariff change. Dist_Front is the ratio of British to US patents in 1855–1859, so higher values indicate products further behind the frontier. The IV estimates imply the effect of protection turns from negative to positive at a British-to-US patent ratio of 2.96 which is simply the ratio of the estimated protection coefficient -4.424 to the interaction coefficient 1.493. All specifications control for the change in input protection and include county and product fixed effects. Standard errors two-way clustered by county and product in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

In Table 6, $\hat{\beta} > 0$ and $\hat{\theta} < 0$. At baseline (i.e. product with distance to frontier of 0), a 1% increase in protection leads to a 4.4% fall in output growth. Each unit increase in distance-to-frontier raises this elasticity by 1.5, implying the effect of protection flips from negative to positive at a British-to-US patent ratio of roughly 2.96.[31] To put this in context, protection harms growth for near-frontier products such as cotton textiles (1.31) and iron and steel (2.53), but boosts growth for laggard products such as glass (3.39) and paints (7.82).[32]

[31] 2.96 is obtained by dividing the coefficient for total protection by the coefficient for the interaction term, i.e. $\frac{4.424}{1.493}$.

[32] The first-stage Kleibergen-Paap F-statistic of 15.5 exceeds the conventional threshold of 10 from Stock

These results can rationalize the mixed findings in the infant industry literature. Taussig (1931) argues that glass benefited from protection — consistent with my framework, as glass (3.39) exceeds the 2.96 threshold. Irwin (2000a) argues that a large share of the pig iron sector could have survived without protection — again consistent, as pig iron (2.53) falls below the threshold. Irwin and Temin (2001) argue that cotton textiles did not benefit from protection, which my results corroborate given cotton textiles' near-frontier status (1.31).

I examine the effect of protection on other manufacturing outcomes. As we see in Table 7, protection exhibits the same frontier-dependent pattern for firm size, value added, employment and materials usage, with the only exception being capital.[33] These additional results indicate that the effects are not purely operating via changes in prices and markups, but that physical quantities are also changing. Additionally, in a class of models that includes mine, the change in firm size is a sufficient statistic for the change in markups. The firm size results thus suggest that protection lowered markups near the frontier and raised them far behind it, as the model in Section 3 predicts.

I also examine whether protection exhibits the same frontier gradient for innovative activity. In this period, there were no formal R&D laboratories; invention was conducted by independent inventors, firm owners and workers (Lamoreaux and Sokoloff 2005). Consistent with the demand-pull view of innovation (Schmookler 1966), patenting rates responded systematically to local market conditions and expected profits (Sokoloff 1988, Khan and Sokoloff 1993). This is consistent with the model in Section 3, where innovation responds to the profit gain from upgrading — a margin that protection shifts by changing the competitive environment the firm faces. Changes in county-level patent counts therefore capture how local innovators adjust their activity in response to changes in protection. Using patent-level data from Shaw (undated, accessed 2025), geocoded to the county level using Berkes, Karger and Nencka (2023), I run specification (4) with patent issuances as the dependent variable.

As Column 2 of Table 8 shows, protection exhibits the same frontier-dependent pattern for innovation as for output. At baseline, a 1% increase in protection results in a 7.1% fall in patent issuances. Each unit increase in distance to frontier increases this patent elasticity by 1.3. Protection thus shifts patent activity toward laggard products, suggesting a reshaping

and Yogo (2005), though recent work has shown these critical values can be unreliable for valid inference (Montiel Olea and Pflueger, 2013; Lee et al., 2022; Andrews, Stock and Sun, 2019). As a more robust check, I construct weak-instrument-robust confidence sets based on Andrews (2018), implemented using Sun (2018). Both $\hat{\beta}$ and $\hat{\theta}$ remain significant at the 5% level under these tests, which are valid regardless of instrument strength.

[33] One explanation is that the massive increase in federal borrowing to finance the Civil War crowded out private credit needed for capital expenditures.

Table 7: Trade Protection and Firm Size, Value Added and Factor Usage

	Δ Log Output per Estab.	Δ Log Value Added	Δ Log Labor	Δ Log Materials	Δ Log Capital
Δlog(TP) $\times$ Dist_Front	1.818***	1.015**	1.383**	1.905***	-0.675
	(0.482)	(0.413)	(0.530)	(0.475)	(0.498)
Δlog(TP)	-6.049***	-3.588	-6.896**	-6.076**	-0.969
	(1.976)	(2.061)	(2.560)	(2.235)	(2.712)
Observations	833	832	833	833	833
Input TP Control	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes	Yes
First-stage F-stat (KP)	15.5	15.6	15.5	15.5	15.5

Notes: Estimates of Equation (4), relating decadal changes (1860–1870) in each outcome to changes in log total protection TP_{iz} and its interaction with distance to the British technological frontier. $\Delta \log(TP_{iz})$ is instrumented with $STS_{iz}^{t} \times \Delta \log t_z$, the tariff's share in total protection interacted with the nationwide tariff change. Dist_Front is the ratio of British to US patents in 1855–1859, so higher values indicate products further behind the frontier. Output per establishment, value added, labor and materials all display the same frontier gradient as output in Table 6. All specifications control for the change in input protection and include county and product fixed effects. Standard errors two-way clustered by county and product in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

of comparative advantage.

6.2 Civil War Confounds and Robustness

The Civil War was not merely a fiscal shock — it had wide-ranging impacts across a variety of social and economic domains, making War-related omitted variables loom large. I address this with a battery of robustness checks. My baseline specification excludes the Confederate South due to the potential effects of battles, Confederate governance, Emancipation, and Reconstruction. Results are robust to including the Confederate South in the sample (Column 1 of Table 9). I also examine sensitivity to excluding the four slave states that fought on the side of the Union — Kentucky, Maryland, Missouri, and Delaware; results are robust to their exclusion (Column 2 of Table 9). Finally, I restrict the sample to counties — whether North or South — without a documented Civil War battle from the Civil War Sites Advisory Commission (CWSAC, 1993), thereby removing any county that may have experienced direct physical destruction of productive capacity. Results are once again robust (Column 3 of Table 9).

Table 8: IV Estimates: Distance to Frontier and Protection Effects on Patent Growth

	Δ Log Patents	
	OLS	IV
Δlog(TP) $\times$ Dist_Front	1.175*	1.312**
	(0.596)	(0.572)
Δlog(TP)	-7.459^{*}	-7.138^{*}
	(3.872)	(3.537)
Observations	281	281
Input TP Control	Yes	Yes
County FE	Yes	Yes
Product FE	Yes	Yes
First-stage F-stat (KP)	—	45.2

Notes: Estimates of Equation (4) with the decadal change in log patent issuances (1860–1870) as the dependent variable. $\Delta \log(TP_{iz})$ is instrumented with $STS^{t}_{iz} \times \Delta \log t_z$, the tariff's share in total protection interacted with the nationwide tariff change. Dist_Front is the ratio of British to US patents in 1855–1859, so higher values indicate products further behind the frontier. Protection therefore shifts patent activity toward laggard products, the same frontier gradient found for output in Table 6. Patent data from Shaw (undated, accessed 2025), geocoded to counties following Berkes, Karger and Nencka (2023). All specifications control for the change in input protection and include county and product fixed effects. Standard errors two-way clustered by county and product in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Beyond direct conflict, the Civil War also generated economic shocks, as military procurement created geographically concentrated demand boosts. I digitize the report of the Quartermaster General documenting Civil War purchases by county; Column 4 of Table 9 shows that results are robust to dropping counties with any documented military purchases.

The Civil War also severely disrupted the supply and transportation of cotton, the vast majority of which was produced in the South. If this disruption differentially affected cotton textile producers on the coast relative to interior producers, it could confound the identifying variation. Column 1 of Table 10 shows results are robust to dropping cotton textiles. Appendix Table C3 examines robustness to dropping other products one-by-one. Reassuringly, the interaction coefficient remains positive and significant and the baseline effect remains negative in all cases when removing products one-by-one.

In addition, I drop the counties containing the five largest cities in 1860 — New

Table 9: Robustness: Civil War Sample Restrictions

	Δ Log Output			
	With South	Excl. Border	Excl. Battles	Excl. Procurement
Δlog(TP) $\times$ Dist_Front	1.508***	1.203***	1.568***	1.695***
	(0.406)	(0.267)	(0.406)	(0.549)
Δlog(TP)	-4.159^{*}	-3.634^{*}	-4.691^{**}	-5.073^{**}
	(2.109)	(1.995)	(2.037)	(2.339)
Observations	883	771	853	800
Input TP Control	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes
First-stage F-stat (KP)	16.6	19.2	16.2	13.6

Notes: IV estimates of the main specification (Equation (4)) under alternative sample restrictions addressing Civil War confounds. The baseline sample excludes the Confederate South; column 1 adds it back. Column 2 further excludes the four Union slave states (Kentucky, Maryland, Missouri, Delaware). Column 3 excludes counties with a documented Civil War battle (CWSAC, 1993). Column 4 excludes counties with documented military purchases from the Quartermaster General's reports. The implied sign-flip threshold ranges from 2.76 to 3.02 across columns, against 2.96 in the baseline. Standard errors two-way clustered by county and product in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

York City, Philadelphia, Baltimore, Boston and New Orleans — which may have had particularly large influence over trade policy due to lobbying power or visibility, potentially representing a threat to exogeneity. Columns 2 and 3 of Table 10 respectively show results are robust to dropping these counties, and to dropping their neighboring counties as well.

Beyond Civil War-related concerns, I also examine robustness to alternative specifications of transportation costs. In Section 4.2, I estimated $\hat{\omega} = 0.139$ and used the following specification for transport costs: $\tau_{iz} = p_z^{0.139}\zeta_i$. In Table 11, results are robust to setting $\omega = 0$ (corresponding to fully additive transport costs), $\omega = 0.096$ (lower bound of the 95% confidence interval for ω estimated in Section 4.2) and $\omega = 0.182$ (upper bound).

I also examine robustness to specifying distance to frontier in logarithms rather than levels. The log transformation compresses the distribution, reducing the leverage of the most distant products. As Table 12 shows, the interaction remains positive and significant. The implied threshold is $\exp(-\hat{\theta}/\hat{\beta}) = 2.85$, close to the 2.96 implied by the linear specification, indicating that the location of the sign flip is not sensitive to the functional form of the interaction.

Table 10: Robustness: Excluding Cotton Textiles and Top 5 Cities

	Δ Log Output		
	Excl. Cotton Textiles	Excl. Top 5 Cities	Excl. Cities & Neighbors
Δlog(TP) $\times$ Dist_Front	1.663***	1.690***	1.527***
	(0.469)	(0.544)	(0.485)
Δlog(TP)	−4.833**	−4.395*	−4.250**
	(2.146)	(2.169)	(1.966)
Observations	743	801	739
Input TP Control	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Product FE	Yes	Yes	Yes
First-stage F-stat (KP)	15.9	15.7	18.1

Notes: IV estimates of the main specification (Equation (4)) under alternative sample exclusions. Column 1 drops cotton textiles, whose supply and transport were severely disrupted by the war. Columns 2 and 3 drop the counties containing the five largest cities in 1860 (New York City, Philadelphia, Baltimore, Boston and New Orleans), and those counties together with their neighbors, addressing the concern that these locations had outsized influence over trade policy. The implied sign-flip threshold ranges from 2.60 to 2.91, against 2.96 in the baseline. Standard errors two-way clustered by county and product in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

6.3 Mechanisms and Alternative Interpretations

The model presented in Section 3 generates the patterns seen in Tables 6 and 8. The key insight is that protection substitutes for innovation as a source of market power for close-to-frontier products. These producers already command a large market share and face a sufficiently low residual demand elasticity that protection reduces the incremental gain from innovating. For products further behind the frontier, residual demand is more elastic and the negative innovation effect is weaker, allowing the positive demand substitution effect — the shift in demand toward domestic varieties from higher import prices — to dominate. In this section I discuss whether alternative models can generate the same empirical results.

Heterogeneous Trade Elasticities

First, I discuss an alternative interpretation based on trade elasticities. Standard trade theory predicts that more substitutable products see a larger output response to a given increase in protection. If products far from the frontier also happen to have higher elasticities of substitution, this could rationalize the positive interaction coefficient without

Table 11: Robustness to Transportation Cost Specification (ω)

	Δ Log Output		
	$\omega = 0$	$\omega = 0.096$	$\omega = 0.182$
Δlog(TP) $\times$ Dist_Front	1.066***	1.340***	1.666***
	(0.288)	(0.354)	(0.452)
Δlog(TP)	-3.289^{*}	-4.017^{**}	-4.885^{**}
	(1.557)	(1.820)	(2.225)
Observations	833	833	833
Input TP Control	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Product FE	Yes	Yes	Yes
First-stage F-stat (KP)	10.6	14.0	16.9

Notes: IV estimates of Equation (4) under alternative values of ω, the elasticity of transportation costs with respect to product price, which enters the protection measure through $\tau_{iz} = p_z^{\omega}\zeta_i$. Column 1 imposes fully additive transport costs; columns 2 and 3 use the lower and upper bounds of the 95% confidence interval around the estimate of $\hat{\omega} = 0.139$ from Table 3. The implied sign-flip threshold ranges from 2.93 to 3.09, against 2.96 in the baseline. Standard errors two-way clustered by county and product in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

any innovation channel as predicted by my model in Section 3.[34] I test this by correlating $Dist_Front_z$ with trade elasticities from three sources: Caliendo and Parro (2015), Shapiro (2016), and Giri, Yi, and Yilmazkuday (2021).

As Table 13 shows, the correlations are uniformly negative: products far from the frontier have *lower*, not higher, elasticities of substitution, ruling out this alternative interpretation. Results are also robust to directly controlling for $(\sigma_z - 1) \times \Delta \log(TP_{iz})$ in the main specification (Appendix Table C2).

Pricing-to-Market

The empirical framework defines total protection as tariffs plus transportation costs expressed in ad valorem form, which is theoretically consistent with the model in Section 3. However, the British exporter charges a destination-varying markup $\mu_{iz} = \frac{1+(\sigma-1)\lambda_{iz}}{(\sigma-1)\lambda_{iz}}$, which is decreasing in the domestic share λ_{iz} and hence in the level of protection in county i, representing incomplete passthrough.[35] Strictly speaking, this markup enters the error

[34]However, this would not be able to rationalize the sign flip or the negative effect of trade protection on output, without adding an additional channel such as innovation or X-inefficiencies.

[35]Equivalently, we can think of this as an American importer buying from the British exporter at the port, paying the tariff to the Federal government, then shipping the product to the interior while charging

Table 12: Effect of Trade Protection on Output Growth: Log Distance to Frontier

	Δ Log Output
Δlog(TP) $\times$ log(Dist_Front)	6.156**
	(2.331)
Δlog(TP)	$-$6.455*
	(3.055)
Observations	833
Input TP Control	Yes
County FE	Yes
Product FE	Yes
First-stage F-stat (KP)	18.0

Notes: IV estimates of Equation (4), replacing $Dist_Front_z$ with $\log(Dist_Front_z)$ in the interaction. $\Delta \log(TP_{iz})$ is instrumented with $STS^t_{iz} \times \Delta \log t_z$. The implied sign-flip threshold is $\exp(6.455/6.156) = 2.85$, close to the 2.96 from the linear specification in Table 6. All specifications control for input protection and include county and product fixed effects. Standard errors two-way clustered by county and product in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 13: Correlation between Distance to Frontier and Trade Elasticities

Elasticity Source	Correlation with $Dist_Front_z$
Caliendo and Parro (2015)	-0.229
Shapiro (2016)	-0.555
Giri, Yi and Yilmazkuday (2021)	-0.170
Median	-0.533

Notes: Each row reports the correlation across products between the distance-to-frontier measure $Dist_Front_z$ and the trade elasticity from the indicated source. All correlations are negative: products further from the British frontier are *less* substitutable, not more, thus heterogeneous trade elasticities cannot explain the results in Table 6.

term as an omitted variable in the main specification, Equation (4). Can such pricing-to-market behavior on its own account for the empirical results without an innovation channel?

Two arguments make this unlikely. First, pricing-to-market on its own cannot deliver a negative effect of protection on output. For output to fall, the exporter's markup would have to cut its markup by more than the tariff increase, representing *negative* passthrough,

a destination-specific markup. The analysis is the same.

which is very rarely observed empirically and difficult to motivate theoretically. My framework predicts positive and incomplete passthrough, i.e. the British firm cuts its markup when tariffs rise to dampen the loss in market share. The majority of other models also predict positive passthrough, whether complete or incomplete.

Second, and more decisively, a framework with no innovation channel makes no predictions for patenting. As Table 8 shows, protection shifts patent activity with the same frontier gradient as output. Thus, mechanically, pricing-to-market alone without an innovation channel cannot generate this pattern.

Quality Upgrading via Alchian-Allen Effects

Another interpretation arises from the Alchian-Allen effect (Alchian and Allen 1964, Hummels and Skiba 2004), where additive transport costs lower the relative cost of high-quality varieties.[36] In the baseline Alchian-Allen logic, high-τ_{iz} locations should receive relatively higher-quality imports. An ad valorem tariff applies proportionally to the product price, eroding the shipping advantage that additive transport costs give high-quality goods. Higher tariffs therefore push the foreign import mix toward lower-quality varieties, an effect strongest in high-τ_{iz} interior markets (Hummels and Skiba 2004).[37]

However, while the Alchian-Allen effect clearly predicts changes in the import quality mix after tariffs are applied, it does not make a prediction for the output response of interior firms. First, while some interior firms may upgrade quality to escape competition, others contract as competition at the low end intensifies, with the total effect on county output ambiguous. Second, even for firms that do move upmarket, this does not guarantee a higher output response. A firm that upgrades its quality sells higher priced units, but it may reduce total quantities produced, resulting in an ambiguous effect on output. In addition, there is no clear reason, regardless of its sign, why this effect should be different for far-from-frontier items relative to close-to-frontier items.

Factor Migration and Spatial Spillovers

The baseline specification treats county-product pairs as independent, but protection in one pair may affect outcomes in another. Such spillovers violate the stable unit treatment value assumption (SUTVA) and may bias $\hat{\beta}$ and $\hat{\theta}$. This issue is also related to the difference

[36] To see this in my notation, take low- and high-quality varieties of product z, with prices $p_z^L < p_z^H$. Total protection is $TP_{iz}(p) = t_z + \tau_{iz}(p)/p$. Since I estimate transportation costs as $\tau_{iz}(p) = p^{0.139}\zeta_i$ in Section 4.2, $\tau_{iz}(p_z^H)/p_z^H < \tau_{iz}(p_z^L)/p_z^L$, i.e. transportation costs are a larger fraction of the price for lower-quality varieties. Therefore, $TP_{iz}(p_z^H) < TP_{iz}(p_z^L)$.

[37] Let $a_H \equiv \tau_{iz}(p_z^H)/p_z^H$ and $a_L \equiv \tau_{iz}(p_z^L)/p_z^L$, with $a_H < a_L$. The relative delivered price of the high-quality variety is $(p_z^H/p_z^L)\{[(1+t_z)+a_H]/[(1+t_z)+a_L]\}$. Since $\partial\{[(1+t_z)+a_H]/[(1+t_z)+a_L]\}/\partial t_z = (a_L - a_H)/[(1+t_z)+a_L]^2 > 0$, an increase in the ad valorem tariff raises the relative delivered price of high-quality foreign varieties.

between local and aggregate estimates, a topic at the heart of the "missing-intercept" or micro-to-macro literature (Chodorow-Reich 2019, 2020).

First, we discuss factor migration. Suppose that more protected county-product pairs draw factors, workers for example, away from less protected ones. The county fixed effect $\Delta\delta_i$ absorbs any component of these flows that is common across products within a county. What remains is reallocation across county-product pairs. To ascertain the degree of this reallocation, I use the 1850, 1860 and 1870 full-count population censuses, which have since been declassified, permitting record linkage across these decades. I take these links from the IPUMS Multigenerational Longitudinal Panel (Ruggles et al. 2025), which matches individuals across censuses using a supervised machine-learning algorithm trained on hand-classified matches.

I derive the estimating equation from a discrete choice model of worker location, which relates bilateral flows into county-product pairs to protection at the destination. Appendix C.5 presents the full details. As Appendix Table C7 shows, I find that workers flow from less to more protected county-product pairs, but that there is no differential effect by distance to frontier. This analysis thus suggests that $\hat{\theta}$ is biased upwards, but suggests no bias in $\hat{\beta}$.

Second, we discuss demand spillovers. When a county becomes more protected, local consumers may substitute not only toward producers in their own county, but also toward producers in other counties. Let this omitted spillover term equal $S_{iz} = \sum_{j \neq i} \omega_{ijz} \Delta \log TP_{jz}$, where ω_{ijz} measures the bilateral spatial link between county i and j for product z. This term enters with a positive coefficient, since protection elsewhere raises demand for i's output. The sign of this bias is governed by $\text{Cov}(\Delta \log TP_{iz}, S_{iz})$. This covariance is likely positive: coastal counties i have stronger links to other coastal counties j, and those counties receive the largest protection shock, raising S_{iz}. Demand spillovers therefore bias $\hat{\theta}$ upward as well, with no obvious direction of bias for $\hat{\beta}$.

Under either channel the negative estimates for close-to-frontier products are conservative: the true θ is more negative still. Even accounting for these spillovers, a standard gravity model cannot rationalize a negative response of output to protection without adding channels such as the variable markups and innovation in this paper.

6.4 Structural Interpretation of Regression Coefficients

I now use the model in Section 3 to structurally interpret the estimated coefficients in Section 6.1. In Appendix D.2, I show the finite change in output with respect to a finite change in total protection can be written, using a first-order heuristic, as:

$$\frac{\Delta \log Y_{iz}}{\Delta \log(TP_{iz})} \approx \underbrace{(\sigma-1)(1-\lambda_{iz})}_{\text{Gravity benchmark}} \cdot \underbrace{\frac{\sigma+(\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}{\sigma^2-(\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}}_{\text{Bertrand dampening}\in(0,1)} \cdot \underbrace{\frac{TP_{iz}}{1+TP_{iz}}}_{\text{scaling}} \cdot \left(1+\log\alpha \cdot \frac{\Delta I_{iz}}{\Delta \log(1+TP_{iz})}\right) \tag{8}$$

The first term, $(\sigma-1)(1-\lambda_{iz})$, is the expenditure-switching elasticity implied by standard gravity models: output responds more to protection when import penetration $(1-\lambda_{iz})$ is high, as there is more expenditure to redirect toward the local variety,[38] and when products are more substitutable (higher σ). See Appendix B.5 for a derivation of this expression in a gravity setting. As low τ_{iz} locations (i.e. close to port locations) will tend to have larger baseline import penetration $1-\lambda_{iz}$ for a given product, they will see larger output increases via this channel.

The second term is a Bertrand dampening factor that is strictly between zero and one. This factor captures strategic markup adjustment: when protection raises the foreign firm's delivered cost, part of this increase is absorbed by endogenous markup adjustments rather than wholly translating into expenditure switching. In standard gravity frameworks, markups are constant, so this dampening term is equal to 1.

Finally, the term inside the parentheses governs whether the net effect is positive or negative: the total effect turns negative when $\frac{\Delta I_{iz}}{\Delta \log(1+TP_{iz})} < -\frac{1}{\log\alpha}$, that is, when protection reduces innovation sufficiently to overwhelm the mechanical substitution effect. Proposition 2 and Corollary 2 in Section 3 characterize when this occurs as a function of distance to the technological frontier.

As Appendix C.1 of Borusyak and Hull (2021) shows, in the presence of heterogeneous treatment effects, an IV regression identifies a convex weighted average of unit-level causal effects under a suitable monotonicity condition, which holds mechanically in my setting due to the definitions of $\Delta \log TP_{iz}$ and $STS^t_{iz}\Delta \log t_z$. Suppose we run the following regression:

$$\Delta \log y_{iz} = \sum_z \beta_z[\Delta \log(TP_{iz})] + \gamma X_{iz} + \Delta\delta_i + \Delta\delta_z + \Delta e_{iz}$$

with $STS^t_{iz}\Delta \log t_z$ as an IV for $\Delta \log(TP_{iz})$. According to Borusyak and Hull Appendix C.1 (2021), this $\hat{\beta}_z$ estimates a variance-weighted average of the county-level treatment effects:

[38] To see this in the extreme case, suppose a location has no foreign imports, i.e. $1-\lambda_{iz}=0$. In this case, an increase in protection is redundant under this framework, as there are no imports to substitute away from towards the local variety.

$$\hat{\beta}_z \xrightarrow{p} \frac{\sum_i \beta_{iz} w_{iz}^2}{\sum_i w_{iz}^2} \tag{9}$$

where $w_{iz}^2 = Var(STS_{iz}^t \Delta \log t_z \mid \Delta\delta_i, \Delta\delta_z)$ is the variance of the instrument conditional on fixed effects. Under the model in Section 3, $\beta_{iz} \approx \frac{\Delta \log Y_{iz}}{\Delta \log(TP_{iz})}$, which is unpacked in Equation (8) above in this Section. The w_{iz}^2 weights are larger for coastal counties, which have high STS_{iz}^t and therefore more instrument variation. Therefore, $\hat{\beta}_z$ recovers a convex weighted average of β_{iz}s, with the responses of coastal areas weighted more heavily (i.e. areas more affected by tariff increases).

The main empirical specification in Equation (4) then linearly projects β_z onto distance to the frontier: $\beta_z = \theta + \beta \cdot Dist_Front_z$. Therefore, $\hat{\beta}$ measures how a convex, coastal-weighted average of county-level treatment effects varies linearly with distance to the technological frontier.

7 Welfare and General Equilibrium

In Section 3, I presented a model in which the effect of trade protection on output and innovation depends on distance to the technological frontier. This was a partial equilibrium exercise: aggregate objects such as income and wages were held fixed. This partial equilibrium model maps naturally to the main empirical specification (Equation (4) in Section 5), where county fixed effects $\Delta\delta_i$ absorb changes in local income and wages; the estimated coefficients should therefore be interpreted as partial equilibrium elasticities.

However, trade protection can also affect aggregate income and wages, a point similar to the one made by the "missing intercept" or "micro-to-macro" literature (Chodorow-Reich (2019, 2020) and Guren et al. (2021)). For example, real incomes can fall due to higher import prices or rise due to increased demand from protection, affecting the output of all products (including those with no changes in protection). Thus, for a full accounting of the effects of trade protection, I incorporate general-equilibrium channels in this section and use the resulting framework to study the welfare implications of trade.

Novel policy and welfare implications emerge from this model relative to standard frameworks. First, unlike the class of trade models covered by Arkolakis, Costinot and Rodriguez-Clare (ACR 2012), the welfare effects of trade can be negative. Furthermore, the model provides a static imperfect competition rationale for infant industry protection, distinct from the usual learning-by-doing justification. Second, the domestic trade share is not a sufficient ex-post statistic to characterize the welfare effects as in ACR; we need an additional ex-post statistic, namely, the change in innovative activity. Third, from an

ex-ante perspective, my framework generally predicts lower gains from trade for high distance-to-frontier products.[39]

Section 7.1 presents the model. Section 7.2 presents market clearing conditions and the definition of equilibrium. Finally, Section 7.3 studies the welfare effects of trade under this model.

7.1 Model Setup

I extend the environment from Section 3 to general equilibrium, making wages and expenditures endogenous. In this setting, large oligopolistic firms internalize their impact on wages and expenditures, posing technical difficulties as noted by Neary (2016). I therefore assume a nested demand structure, as in Atkeson and Burstein (2008). For each product z, let submarkets be indexed by $\psi_z \in [0,1]$.[40] In each submarket ψ_z, the local firm in county i and a UK firm compete à la Bertrand in county i's market, and the local firm chooses whether to innovate before setting prices, as in Section 3. Thus, while firms are large and granular within their own submarket, each submarket is measure zero in the local economy. Therefore, each individual firm cannot affect local aggregates, though collectively their behavior shapes local incomes and wages.

To close the model under balanced trade, I also allow the local firm to export to the UK, so that export earnings finance imports. To avoid introducing additional notation, I relegate the export problem to Appendix D.4 and focus on the domestic problem.

We have three nests in this model: products, submarkets within products, and varieties within submarkets. I assume upper-tier preferences across products z are Cobb-Douglas with expenditure share χ_{iz}, so

$$E_{iz} = \chi_{iz} E_i, \qquad \sum_z \chi_{iz} = 1.$$

where E_i is total expenditure. Within product z, the product-level quantity aggregator is Cobb-Douglas across submarkets $\psi_z \in [0,1]$:

$$\log Q_{iz} = \int_0^1 \log Q_{iz}(\psi_z) d\psi_z.$$

where $Q_{iz}(\psi_z)$ is consumption in submarket ψ_z and Q_{iz} is aggregate consumption of

[39] For close-to-frontier products this is ambiguous.

[40] For example, within product category $z =$ cotton textiles, one submarket is white shirts while another is black dresses.

product z. The corresponding product-level price index is

$$\log P_{iz} = \int_0^1 \log P_{iz}(\psi_z) d\psi_z.$$

Within each submarket ψ_z, the local and UK varieties are CES substitutes:

$$Q_{iz}(\psi_z) = \left[q_{iz}^{US}(\psi_z)^{\frac{\sigma-1}{\sigma}} + q_{iz}^{UK}(\psi_z)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \qquad \sigma > 1.$$

The submarket price index is

$$P_{iz}(\psi_z) = \left[p_{iz}^{US}(\psi_z)^{1-\sigma} + p_{iz}^{UK}(\psi_z)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}.$$

7.1.1 Submarket Firm Problem

The domestic pricing game is identical to the setting in Section 3, with the only difference being that competition between the two firms occurs at the submarket level rather than at the product level. In addition to the domestic market, the local and British firms compete in the UK market, whose details are relegated to Appendix D.4.1 for ease of exposition.

Let E_{i,ψ_z} denote expenditure of county i in submarket ψ_z, which satisfies $E_{iz} = \int_0^1 E_{i,\psi_z} d\psi_z$. In each submarket ψ_z, the local firm chooses whether to innovate:

$$I_{iz}(\psi_z) \in \{0, 1\}$$

The marginal cost of the local firm under no innovation and innovation is, respectively:

$$c_{iz}^{US}(0) = \frac{w_i}{T_z^{US}}, \qquad c_{iz}^{US}(1) = \frac{w_i}{\alpha T_z^{US}}, \qquad \alpha > 1$$

The UK firm has delivered marginal cost

$$c_z^{UK} = \frac{w^{UK}}{T_z^{UK}}$$

Total protection $TP_{iz} = t_z + \frac{\tau_{iz}}{c_z^{UK}}$ is equal to the ad-valorem equivalent of tariffs and transportation costs, as in Section 3. Given innovation status $I_{iz}(\psi_z)$, firms compete in prices within the submarket. Let

$$\lambda_{iz}(\psi_z) = \left(\frac{p_{iz}^{US}(\psi_z)}{P_{iz}(\psi_z)} \right)^{1-\sigma}$$

denote the local firm's market share in submarket ψ_z in county i. The residual demand elasticity faced by the local firm is:

$$\varepsilon_{iz}(\psi_z) = \sigma - (\sigma - 1)\lambda_{iz}(\psi_z).$$

Thus the local firm's optimal price satisfies

$$p_{iz}^{US}(\psi_z) = \frac{\varepsilon_{iz}(\psi_z)}{\varepsilon_{iz}(\psi_z) - 1} c_{iz}^{US}(I_{iz}(\psi_z)) \tag{10}$$

The British firm's optimal price is, where ε_{iz}^{UK} is the residual demand elasticity:

$$p_{iz}^{UK}(\psi_z) = \frac{\varepsilon_{iz}^{UK}(\psi_z)}{\varepsilon_{iz}^{UK}(\psi_z) - 1}(1 + TP_{iz})c_z^{UK} = \frac{1 + (\sigma - 1)\lambda_{iz}(\psi_z)}{(\sigma - 1)\lambda_{iz}(\psi_z)}(1 + TP_{iz})c_z^{UK} \tag{11}$$

Submarket domestic sales are $E_{i,\psi_z}\lambda_{iz}(\psi_z)$. Analogous demand and pricing conditions hold in the UK market as detailed in Appendix D.4.1. Let $\Pi_{iz}^{X}(\psi_z)$ be the local firm's operating profits in the UK export market. Total operating profits are therefore:

$$\Pi_{iz}(\psi_z) = \frac{E_{i,\psi_z}\lambda_{iz}(\psi_z)}{\varepsilon_{iz}(\psi_z)} + \Pi_{iz}^{X}(\psi_z).$$

Let $G_{iz}(\psi_z)$ denote the gain from innovation, the difference in operating profit between innovating and not:

$$G_{iz}(\psi_z) := \Pi_{iz}\left(\psi_z;\, c_{iz}^{US}(1)\right) - \Pi_{iz}\left(\psi_z;\, c_{iz}^{US}(0)\right)$$

The submarket US firm innovates if and only if this gain exceeds the fixed cost:

$$I_{iz}(\psi_z) = \mathbf{1}\{G_{iz}(\psi_z) \geq F\} \tag{12}$$

7.1.2 Aggregation Across Submarkets

Within each county i and product z, all submarkets ψ_z share the same primitives: T_z^{US}, T_z^{UK}, τ_{iz}, t_z, w_i, w^{UK}, F, and α. The corresponding UK submarkets likewise share common primitives. I restrict attention to pure symmetric equilibria, in which all submarkets within county i and product z choose the same innovation status:

$$I_{iz}(\psi_z) = I_{iz} \qquad \forall \psi_z \in [0, 1]. \tag{13}$$

Let $G_{iz}^{s} \equiv G_{iz}(I_{iz,-\psi_z} = s)$ denote the gain from innovation when all other submarkets

choose $s \in \{0, 1\}$.[41] I impose a strategic substitutability condition in innovation incentives:

$$G_{iz}^0 > G_{iz}^1 \qquad \forall i, z.$$

Namely, the gain from innovating is strictly higher when other submarkets do not innovate than when they do. This assumption guarantees uniqueness of the pure symmetric equilibrium.

The unique innovation choice I_{iz}, along with common market conditions, implies $p_{iz}^{US}(\psi_z) = p_{iz}^{US}$, $p_{iz}^{UK}(\psi_z) = p_{iz}^{UK}$, $P_{iz}(\psi_z) = P_{iz}$, $\lambda_{iz}(\psi_z) = \lambda_{iz}$ and $\lambda_{UK,z}(\psi_z) = \lambda_{UK,z}$ for all i, z and ψ_z, where $\lambda_{UK,z}(\psi_z)$ denotes the local firm's market share in the UK in submarket ψ_z. Let Y_{iz} denote total sales by county i firms in product z. Sales aggregate as follows:

$$Y_{iz} = \int_0^1 [\lambda_{iz}(\psi_z) E_{i,\psi_z} + \lambda_{UK,z}(\psi_z) E_{UK,\psi_z}] d\psi_z = \int_0^1 [\lambda_{iz} E_{i,\psi_z} + \lambda_{UK,z} E_{UK,\psi_z}] d\psi_z$$

$$= \lambda_{iz} \int_0^1 E_{i,\psi_z} d\psi_z + \lambda_{UK,z} \int_0^1 E_{UK,\psi_z} d\psi_z = \lambda_{iz} E_{iz} + \lambda_{UK,z} E_{UK,z}$$

where E_{UK,ψ_z} denotes UK expenditure in submarket ψ_z.

7.2 General Equilibrium

This economy is closed by three clearing conditions. First, goods market clearing:

$$Y_{iz} = E_{iz} \lambda_{iz} + E_{UK,z} \lambda_{UK,z} \tag{14}$$

Second, aggregate income in county i equals labor income and profits net of innovation costs:

$$Y_i = w_i L_i + \Pi_i - F \sum_z I_{iz} \tag{15}$$

where w_i is the wage, $L_i = \sum_z L_{iz}$ is the labor endowment, and $\Pi_i = \sum_z \Pi_{iz}$ are operating profits earned across all products.

Third, aggregate expenditure satisfies:

$$E_i = Y_i + TR_i \tag{16}$$

[41] Existence of a pure symmetric equilibrium is guaranteed if for all i and z:

$$G_{iz}^0 \leq F \qquad \text{or} \qquad G_{iz}^1 \geq F.$$

where $TR_i = \rho_i TR$ is the tariff revenue rebated to county i by the Federal government, with ρ_i is the share of Federal tariff revenue rebated to county i, with $\sum_i \rho_i = 1$. Tariffs are levied nationally, so total revenue is collected across all counties:

$$TR = \sum_j \sum_z \frac{t_z}{1+t_z} (1 - \lambda_{jz}) E_{jz} \tag{17}$$

Analogous clearing conditions for the UK are provided in Appendix D.4.2. Given primitives, an equilibrium is a vector of prices, innovation choices, and wages such that firms set prices optimally, innovation choices satisfy Equation (12), and the clearing conditions in Equations (14), (15), and (16) and their UK counterparts hold.

7.3 Welfare

We now study the welfare implications of trade through the lens of this model. I define welfare as the ratio of total expenditure to the price index:

$$W_i = \frac{E_i}{P_i}.$$

where $P_i = \prod_z P_{iz}^{\chi_{iz}}$ and $E_i = \sum_z E_{iz}$. I begin with the one-product case, and thus omit the product index z. Unlike in the rest of the paper, whether protection arises from tariffs or from transportation costs matters, as tariffs generate revenues which may affect real incomes. I first abstract from revenue effects, and study shocks to transportation costs $\Delta \log \tau_i$. In Appendix D.5.4, I reintroduce revenue effects, which add an additional term to the expression below. Note λ_i is the share of expenditure spent on the domestic variety. The welfare change is thus:

$$\Delta \log W_i = \underbrace{\frac{1}{1-\sigma} \Delta \log \lambda_i}_{\text{ACR term}} + \underbrace{\Delta I_i \log \alpha}_{\text{innovation effect}} + \underbrace{\Delta \log(L_i - FI_i + \Pi_i^X)}_{\text{resource effect}}.$$

See Appendix D.5.2 for the derivation of this expression. The first term is the classic ACR domestic-share term (Arkolakis, Costinot, and Rodriguez-Clare, 2012), which fully summarizes welfare changes in standard gravity frameworks. A larger fall in the domestic share leads to larger gains in welfare, with the exact mechanism depending on the particular microfoundation.[42]

My framework has an additional term. Unlike in the class of models covered by ACR, the change in domestic expenditure share is no longer a sufficient ex-post statistic for

[42] In Krugman (1980), for instance, the gains arise from love of variety, while in a Ricardian setting (Eaton and Kortum 2002) they arise from specialization according to comparative advantage.

the welfare gains from trade. It also matters whether the trade shock changes the firm's innovation decision, beyond its effect on the domestic expenditure share.

This additional term allows for the possibility of a welfare loss from trade liberalization. Trade can reduce the incentive to innovate, i.e. $\Delta I_i < 0$. If in addition, the step size is larger than the innovation resource cost (i.e. $\log \alpha > \Delta \log[L_i - FI_i + \Pi_i^X]$), signifying that innovation has net positive social value, the innovation effect is negative and may dominate the ACR gains from trade term, resulting in a net welfare loss from trade. Appendix D.5.1 provides a numerical example of a welfare loss from moving toward trade.

How does distance to frontier affect the welfare gains from trade? A high distance to frontier has two effects pulling in opposite directions: On the one hand, when the foreign producer is far ahead, the domestic share will decline more dramatically after a move towards free trade, increasing the ACR gains from trade term. On the other hand, as Proposition 1 in Section 3 suggests, the possibility of a decrease in innovation is higher for far from frontier products relative to close to frontier products when trade is liberalized.

This formula provides a basis for quantifying welfare changes. Given domestic market shares, innovation choices, and export profits before and after the trade shock, together with σ, α, F, and L_i, the welfare effect can be computed. Unfortunately, this quantification is not feasible in my setting because county-level trade shares, export profits, and fixed-cost data are unavailable.

Relative to the gravity models covered in Arkolakis et al. (2012), this model tends to predict lower welfare gains for high distance-to-frontier products after a shock to trade costs (assuming innovation has net positive social value). Two channels push in this direction. First, endogenous markup adjustments dampen the change in the domestic share relative to a constant-markup model, attenuating the ACR term (see Equation (8)). Second, a move toward trade can trigger a cut in innovation, subtracting further from the gains and potentially reversing their sign. However, for close-to-frontier products the comparison is ambiguous: the first channel still attenuates the ACR term, but liberalization raises the return to innovation for these products, so the innovation effect works in the opposite direction.

Next, we turn to a multiproduct version of the gains from trade formula, presented below:

$$\Delta \log W_i = \underbrace{\sum_z \chi_{iz} \frac{1}{1-\sigma} \Delta \log \lambda_{iz}}_{\text{ACR term}} + \underbrace{\sum_z \chi_{iz} \Delta I_{iz} \log \alpha}_{\text{innovation effect}} + \underbrace{\Delta \log \left(L_i - F \sum_z I_{iz} + \sum_z \Pi_{iz}^X \right)}_{\text{resource cost}}$$
$$\underbrace{- \sum_z \chi_{iz} \Delta \log \mu_{iz}^{US}}_{\text{procompetitive effect}} + \underbrace{\Delta \log \left(1 + \frac{\sum_z (\Pi_{iz} - \Pi_{iz}^X)}{L_i - F \sum_z I_{iz} + \sum_z \Pi_{iz}^X} \right)}_{\text{rent effect}}.$$

where $\mu_{iz}^{US} = \frac{\varepsilon_{iz}^{US}}{\varepsilon_{iz}^{US}-1}$ is the markup. See Appendix D.5.3 for a proof of this expression. The first two terms are the multiproduct analogues of the corresponding single-product terms, weighted by Cobb-Douglas expenditure shares χ_{iz}. Two additional terms arise, capturing reallocation across products. The procompetitive effect captures changes in markups: if trade lowers markups, i.e. $\Delta \log \mu_{iz} < 0$, it raises welfare by lowering consumer prices. This effect is larger when the product is a larger share of consumption (high χ_{iz}). The rent effect captures changes in domestic profits, which under my assumptions are rebated to the representative consumer in county i and therefore raise welfare when profits rise relative to the county's other resources, $L_i - F \sum_z I_{iz} + \sum_z \Pi_{iz}^X$. This effect is larger when the product accounts for a larger share of domestic profits. In the one-product case, these two effects exactly cancel: an increase in markups raises consumer prices but is offset exactly by the increase in domestic profits rebated to the consumer. In the multiproduct case, this cancellation no longer holds product by product because markup changes are weighted by expenditure shares, while profit changes are weighted by each product's contribution to aggregate profits. Thus markup increases are welfare reducing when they occur in products that are important for consumption, but welfare enhancing when they occur in products that are important sources of domestic profits. Because of these additional terms, we do not have a simple sufficient condition for welfare gains from trade as in the single product case. The formula with tariff-revenue effects can be found in Appendix D.5.4.

How does distance to frontier affect these two terms? Both terms are larger in magnitude (but with opposite signs) for close-to-frontier products than far-from-frontier products: close-to-frontier firms see larger cuts in markups but also larger reductions in rents, as they command larger market shares and face a lower residual demand elasticity, giving them both more market power to lose and more rents to lose. Therefore, if close-to-frontier products tend to have high profit intensity relative to their expenditure shares, the rent losses from liberalization dominate the procompetitive gains, reducing the net welfare

benefit of trade for those products. Whether the procompetitive or rent effect dominates in the aggregate therefore depends on the joint distribution of expenditure shares and profit intensities across the frontier spectrum.

8 Conclusion

This paper establishes empirically in the 19th century US context that the effect of protection on output growth is negative for products near the technological frontier, yet positive for products further behind, using new data and a novel empirical design addressing various threats to exogeneity. It also provides a theoretical framework generating these results, with novel implications for welfare and policy analysis.

What do these results mean for the practice of trade and industrial policy today? First, an important caveat: as discussed in Section 2 and Appendix A.1, US trading partners did not retaliate against American tariffs in this period. Today, retaliatory tariffs are the norm rather than the exception, meaning that protection comes at the additional cost of reduced export market access — a cost absent from my estimates. For example, Behnam (2026) finds that while the 2018 US tariffs boosted domestic manufacturing output, retaliatory tariffs by trading partners, such as the European Union, Canada and China, more than offset these gains. I therefore interpret the results in this paper as an upper bound for the efficacy of trade protection in the modern era.

Second, the model suggests, if protectionist tariffs are to be used, that they be used in a carefully targeted manner. This framework suggests Liberation Day tariffs imposed by the second Trump administration — which applied uniform country-wide tariff rates based on bilateral trade deficits — are too blunt an instrument. For example, the 32% tariff applied to Taiwan made no distinction between advanced semiconductors, iron bolts and tapioca starch used to produce boba tea, products likely to be at varying distances to the technological frontier.

On the other hand, the Biden administration imposed very high 100% tariffs on Chinese electric vehicles (EVs) in 2024. Whether these tariffs are successful according to the framework in this paper is dependent on the relative quality and productivity of Chinese EVs relative to US EVs. Chinese battery cell prices are over 20% lower than American ones and EV production costs are over 30% lower (IEA 2025), and some analysts argue that Chinese EV manufacturers have equaled or surpassed their American counterparts in product quality (Ezell 2024). If these assessments are correct, my framework suggests such targeted tariffs could raise the return to innovation and boost growth for American EV producers.

While this paper focuses on technological gaps, other determinants of competitiveness also condition the innovation and output response to protection, such as wages and transport costs. For example, a developing country with sufficiently low wages or poor international port infrastructure may be insulated from foreign competition despite a large technological gap, so that liberalization triggers a domestic innovation gain. Conversely, if wages are not low enough to compensate for a large technological gap, liberalization poses greater risks to domestic output and innovation.

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Appendix A Additional Historical Context

Appendix A.1 Why Didn't Major Trade Partners Retaliate?

The three major European powers — the United Kingdom, France, and Prussia (which was to become Germany) — did not retaliate against the US for their high tariffs.

The United Kingdom, the industrial leader in this period, had committed to free trade in the mid 19th century, and begun eliminating tariffs on all goods except for sin goods such as alcohol. This had occurred after a bitter political battle over The Corn Laws[43], a set of protectionist measures designed to protect domestic British agrarian producers, which lead to food riots and arguably contributed to the Irish famine. In 1846, the British Prime Minister, Robert Peel, was able to overcome opposition and repealed the Corn Laws. This set into motion a process that ended up reducing tariffs on all goods other than sin goods by 1860. Furthermore, retaliatory measures to high American tariffs, though debated by the British especially during the Civil War period, did not end up passing.

In the mid-19th century, France begun a process of bilateral trade reductions with partner countries. There is no documented active bilateral treaty between France and the US in 1860-1870 (Pahre 2008). Furthermore, there is no evidence that the US was a major consideration of the general trade policy for France (i.e. tariffs applied to countries not subject to a specific bilateral treaty). Generally, it appears France, unlike the UK, was mostly concerned with trade with other European nations and its colonial empire, not the US.

Finally, the German states had yet to unify (which had occurred in 1871), inhibiting unified responses to tariff policy. For example, the two major German ports, Hamburg and Bremen, were not part of the Zollverein customs union, and thus not subject to the tariff policy of the rest of Germany. In addition, every member in the Zollverein had a veto till 1867. Finally, just like France, it appeared that the US was not a major consideration of the German states. Rather, it seems the focus of German trade policy was with other European states.

Appendix A.2 Validation of Distance to Frontier Measure

This appendix provides historical evidence that the patent-based distance to frontier measure captures meaningful technological differences across products.

Cotton textiles (1.31). Despite restrictions on technology transfers, British emigrants such as Samuel Slater — referred to as the "father of American manufactures" (White 1836) — alongside Americans like Francis Cabot Lowell, who memorized British power loom designs during a visit to England, established the New England textile sector. By the early 19th century, technology flowed in both directions across the Atlantic (Jeremy 1981), consistent with near-frontier status.

Iron and steel (2.53). Allen (1979): "In the middle of the nineteenth century Britain was the major supplier of iron and steel to the world market, while Germany and the

[43]The Economist magazine was founded as an anti-Corn Law journal. David Ricardo, in his capacity as a member of parliament, had also advocated against the Corn Laws.

United States were substantial importers... I shall demonstrate that Britain's mid-century export success was due to its superior technical efficiency." Temin (1964) adds that before the Civil War, American manufacturers produced only small quantities of steel, as they were "unable to master the demanding requirements to create steel through puddling," with imports from Sheffield dominating the domestic market.

Beer (2.87). Pattinson (2014): "While at the beginning of the century, British breweries had led the world in technology and scale of operation, by the end of it the Americans were the ones at the bleeding edge of innovation" — consistent with mid-range distance to frontier in the 1850s.

Glass (3.39). Scoville (1944): "The first glassmen in the United States naturally patterned their methods after those current in European countries, and they continued to do so throughout most of the nineteenth century. Few innovations were of American origin. Technological development lagged appreciably behind that in Belgium, England, France, and Germany."

Paints (7.82). Virtually all major pigment innovations of the nineteenth century originated in Europe (Barnett 2004). Leading British paint firms were founded by chemists—Berger Paints by a Frankfurt-trained chemist in 1760, Winsor & Newton by a chemist-artist partnership in 1832. By contrast, the first American paint firm to employ a chemist was Sherwin Williams in 1884.

Appendix A.3 Sample Characteristics

Table A1: Characteristics of the Matched Sample Relative to Manufacturing, 1860

Measure	All Manufacturing	Sample
Capital share	0.405	0.448
Labor share	0.147	0.160
Materials share	0.447	0.392
Output per establishment	14,427.48	21,560.35
Workers per establishment	9.66	18.52
Output per worker	1,493.37	1,164.28
Value added per establishment	6,256.82	10,319.28
Value added per worker	647.64	557.25
Capital per worker	766.22	694.16

Table A1 compares the characteristics of my sample to those of the broader manufacturing sector. The two are broadly similar, though the industries in my sample are somewhat more capital intensive and are produced in larger establishments on average. This is consistent with the fact that products appearing in trade statistics, which are necessary to construct tariffs and port prices, are more likely to be tradable and produced at larger scale than more locally oriented manufacturing activities.

Appendix B Treatment Construction and Identification Results

Appendix B.1 First Order Approximation of $\Delta \log(TP_{iz})$

$$\Delta \log(TP_{iz}) \approx d \log(TP_{iz}) = \frac{1}{TP_{iz}} d(TP_{iz}) = \frac{1}{TP_{iz}} d\left(t_z + \frac{\tau_{iz}}{p_z}\right)$$

Focusing on the last term:

$$d\left(\frac{\tau_{iz}}{p_z}\right) = \frac{1}{p_z^2}\left(p_z \, d\tau_{iz} - \tau_{iz} \, dp_z\right) = \frac{1}{p_z^2}\left(p_z \, \tau_{iz} \, d\log \tau_{iz} - \tau_{iz} \, p_z \, d \log p_z\right) =$$

$$\frac{\tau_{iz}}{p_z}\left(d\log \tau_{iz} - d \log p_z\right)$$

Plugging this back:

$$d \log(TP_{iz}) = \frac{1}{TP_{iz}}\left(d(t_z) + \frac{\tau_{iz}}{p_z}\left(d\log \tau_{iz} - d \log p_z\right)\right)$$

Note:

$$\frac{1}{TP_{iz}} d(t_z) = \frac{t_z}{t_z + \frac{\tau_{iz}}{p_z}} d \log t_z = \underbrace{\frac{p_z \, t_z}{p_z \, t_z + \tau_{iz}}}_{STS^t_{iz}} d \log t_z$$

Also note:

$$\frac{1}{TP_{iz}}\left(\frac{\tau_{iz}}{p_z}\left(d\log \tau_{iz} - d \log p_z\right)\right) = \underbrace{\frac{\tau_{iz}}{t_z \, p_z + \tau_{iz}}}_{STS^\tau_{iz}}\left(d\log \tau_{iz} - d \log p_z\right)$$

Putting these two together:

$$d \log(TP_{iz}) = STS^t_{iz} \, d \log t_z + STS^\tau_{iz}\left(d\log \tau_{iz} - d \log p_z\right)$$

Appendix B.2 Proof of Sufficiency of Identification Condition

The IV orthogonality condition is:

$$\mathbb{E}(STS^t_{iz} \Delta \log t_z \cdot \Delta e_{iz} | \Delta \delta_i, \Delta \delta_z) = 0$$

I show the following orthogonality condition implies the above:

$$\mathbb{E}(\Delta \log t_z \cdot \Delta e_{iz} | \Delta \delta_i, \Delta \delta_z, STS^t_{iz}) = 0$$

Proof (Borusyak and Hull 2023):

By Law of Iterated Expectations,

$$\mathbb{E}(STS_{iz}^t \Delta \log t_z \cdot \Delta e_{iz} | \Delta\delta_i, \Delta\delta_z) = \mathbb{E}(\mathbb{E}(STS_{iz}^t \cdot \Delta \log t_z \cdot \Delta e_{iz} | STS_{iz}^t, \Delta\delta_i, \Delta\delta_z))$$

$$= \mathbb{E}(STS_{iz}^t \cdot \mathbb{E}(\Delta \log t_z \cdot \Delta e_{iz} | STS_{iz}^t, \Delta\delta_i, \Delta\delta_z))$$

If $\mathbb{E}(\Delta \log t_z \cdot \Delta e_{iz} | STS_{iz}^t, \Delta\delta_i, \Delta\delta_z) = 0$, we yield the desired result

Appendix B.3 Elasticity of Revenue to the Tariff

Revenue $R = ptM$, where p is price of imports, t is tariff and M is quantity of imports.

$$\frac{\partial R}{\partial 1+t} = \underbrace{\frac{\partial t}{\partial 1+t}}_{1} pM + \frac{\partial M}{\partial 1+t} tp + tM \frac{\partial p}{\partial 1+t}$$

Elasticity of revenue to tariff is equal to:

$$\epsilon_t^R := \frac{\partial R}{\partial 1+t} \frac{1+t}{ptM} = \frac{pM(1+t)}{ptM} + \frac{\partial M}{\partial 1+t} \frac{tp(1+t)}{ptM} + \frac{\partial p}{\partial 1+t} \frac{tM(1+t)}{ptM}$$

Focusing on the middle term:

$$\frac{\partial M}{\partial 1+t} \frac{tp(1+t)}{ptM} = \frac{\partial M}{\partial 1+t} \frac{1+t}{M} = \frac{\partial \log M}{\partial \log 1+t} = \frac{\partial \log M}{\partial \log p(1+t)} \frac{\partial \log p(1+t)}{\partial \log(1+t)} = \epsilon_t^D \rho$$

where ϵ_d^M is the import elasticity of demand and ρ is the passthrough of tariffs to the (tariff-inclusive) price.

Focusing on the last term:

$$\frac{\partial p}{\partial 1+t} \frac{tM(1+t)}{ptM} = \frac{\partial p}{\partial 1+t} \frac{(1+t)}{p} = \frac{\partial \log p(1+t)}{\partial \log 1+t} - 1$$

where in the second equality we used the result that $\frac{\partial p}{\partial 1+t} = (\frac{\partial p(1+t)}{\partial 1+t} - p)\frac{1}{1+t}$ (Below is derivation of this)

$$\frac{\partial p(1+t)}{\partial (1+t)} = p + (1+t)\frac{\partial p}{\partial (1+t)} \iff \frac{\partial p}{\partial (1+t)} = \frac{1}{1+t}(\frac{\partial p(1+t)}{\partial (1+t)} - p)$$

Putting everything together, we yield:

$$\epsilon_t^R = \frac{1+t}{t} + \rho(\epsilon_t^D + 1) - 1$$

Appendix B.4 Formulas Including Specific Tariffs

In the main text, for brevity and ease of exposition I present tariffs as if they are all ad valorem. In the empirics, I take into account whether tariffs are ad valorem or specific. Here, I present the formulas taking into acccount specific tariffs.

The pricing equation with specific tariffs now is:

$$p_{iz} = p_z + t_z^{AV} p_z + t_z^{SP} + \tau_{iz}$$

where t_z^{AV} denotes ad-valorem tariff, and t_z^{SP} denotes specific tariffs.

The ad valorem equivalent of tariffs and tranpsortation cost is:

$$TP_{iz} := \frac{p_{iz} - p_z}{p_z} = t_z^{AV} + \frac{t_z^{SP}}{p_z} + \frac{\tau_{iz}}{p_z}$$

Finally, the first-order approximation in this case will be:

$$d\log(TP_{iz}) = STS_{iz}^{t^{AV}} d\log t_z^{AV} + STS_{iz}^{\tau}(d\log\tau_{iz} - d\log p_z) + STS_{iz}^{t^{SP}}(d\log t_z^{SP} - d\log p_z)$$

where $STS_{iz}^{t^{SP}} = \frac{t_z^{SP}}{t_z^{AV} p_z + t_z^{SP} + \tau_{iz}}$

Appendix B.5 Spatial Impact of Tariffs

This appendix seeks to show that the impact of a tariff is larger in coastal areas than interior areas via the lens of a gravity equation.

Appendix B.5.1 Gravity Equation

Consider the following gravity equation. The gravity equation can be microfounded by Anderson (1979), Krugman (1980), Eaton and Kortum (2002), and Melitz (2003):[44]

$$X_{iiz} = p_{iiz}^{1-\sigma} P_{iz}^{\sigma-1} E_{iz}$$

where i indexes county, z indexes product. X_{iiz} is sales of county i product z in county i, p_{iiz} is the local price, P_{iz} is the local price index, E_{iz} is total expenditure in county i, and $\sigma > 1$ is the elasticity of substitution.

Let preferences across products be Cobb-Douglas. Thus, $E_{iz} = \chi_{iz} E_i$, where E_i is total expenditure in county i and χ_{iz} is the Cobb-Douglas preference weight county i places on z goods. Note this allows different counties to have different preferences across products.

The local price index aggregates domestic and foreign (imported) varieties:

$$P_{iz} = (p_{iiz}^{1-\sigma} + p_{ifz}^{1-\sigma})^{\frac{1}{1-\sigma}}$$

[44] See Head and Mayer (2014) for discussion on these microfoundations and others.

where p_{ifz} is the delivered price of foreign goods to county i. Log-differentiating the gravity equation:

$$d\log X_{iiz} = (1-\sigma)d\log p_{iiz} + (\sigma-1)d\log P_{iz} + d\log E_{iz}$$

As gravity frameworks use CES demand and assume either perfect or monopolistic competition as the market structure, markups are constant, and domestic prices do not adjust to changes in foreign prices. Thus, a tariff shock, which directly affects the price of foreign products, does not affect domestic prices in the gravity framework: $d\log p_{iiz} = 0$. The change in price index is equal to:

$$d\log P_{iz} = \frac{p_{iiz}^{1-\sigma}}{P_{iz}^{1-\sigma}} d\log p_{iiz} + \frac{p_{ifz}^{1-\sigma}}{P_{iz}^{1-\sigma}} d\log p_{ifz}$$

Let λ_{ifz} equal county i's expenditure share on foreign imports:

$$\lambda_{ifz} = \frac{X_{ifz}}{E_{iz}} = \frac{p_{ifz}^{1-\sigma} P_{iz}^{\sigma-1} E_{iz}}{E_{iz}} = p_{ifz}^{1-\sigma} P_{iz}^{\sigma-1} = \frac{p_{ifz}^{1-\sigma}}{P_{iz}^{1-\sigma}}$$

Thus:

$$d\log P_{iz} = \lambda_{ifz} \cdot d\log p_{ifz}$$

Intuitively, the price index in county i rises more in counties with larger import shares λ_{ifz} and for areas where the foreign price shock $d\log p_{ifz}$ is larger. Both of these tend to be higher for low-τ_{iz} or coastal areas. Finally, I assume preferences do not change in this period, so $d\log \chi_{iz} = 0$. This leaves changes in total expenditures in county i, $d\log E_i$, which will be absorbed by the county fixed effect $\Delta\delta_i$, signifying the estimate captures partial equilibrium responses.

$$d\log E_{iz} = \underbrace{d\log \chi_{iz}}_{0} + \underbrace{d\log E_i}_{\Delta\delta_i}$$

Putting everything together, we have:

$$d\log X_{iiz} = (\sigma-1)\lambda_{ifz} \cdot d\log p_{ifz}$$

The delivered price of foreign goods incorporates both the factory-gate price, tariffs, and transport costs:

$$p_{ifz} = p_{fz}(1+t_z) + \tau_{iz} = p_{fz} \cdot (1+TP_{iz}), \qquad TP_{iz} \equiv t_z + \frac{\tau_{iz}}{p_{fz}}$$

where TP_{iz} is total protection, the ad valorem equivalent of the combined tariff and transport wedge. Taking the differential with $d\log p_{fz} = 0$:

$$d\log p_{ifz} = d\log(1+TP_{iz})$$

Finally for simplicity we assume the county only sells to itself (no exports). Thus, $Y_{iz} = X_{iiz}$. Therefore, the change in output can be written as:

$$d \log Y_{iz} = (\sigma - 1)\lambda_{ifz} \cdot d \log(1 + TP_{iz})$$

When nationwide tariffs go up ($d \log t_z > 0$), the change in protection, i.e. $d \log TP_{iz}$, is larger for low-τ_{iz} than high-τ_{iz} locations. In addition, the foreign import share λ_{ifz} tends to be higher for low-τ_{iz} areas relative to high-τ_{iz} areas, further amplifying the effect. Thus, the predicted change in output in a gravity framework will be larger for coastal i than for interior i. According to appendix C.1 of Borusyak and Hull (2021), under a monotonicity condition, the base specification estimates a convex weighted average, where weights are proportional to λ_{ifz} (locations with higher λ_{ifz}, higher import shares, are weighed more heavily in these calculations). In addition, this effect is always positive.

Appendix B.6 Comparison with Shift-Share Design

This appendix formally contrasts the shift-share design and the design in my paper. For ease of exposition, I omit controls other than fixed effects from this discussion.

A shift-share design is interested in an outcome at the regional level, e.g. county level. For example, a researcher may be interested in how output Y_i in county i is affected by its import exposure. A standard shift-share in this literature (i.e. the local labor market approach) is defined as:

$$Z_i^{SS} = \sum_z s_{iz} \Delta g_z,$$

where s_{iz} is the baseline share of product z employment in county i, and Δg_z is an aggregate shock to product z. For example, Δg_z could be changes in nationwide product tariffs t_z, as in this paper. Or Δg_z could be growth in Chinese imports in product z as in Autor, Dorn and Hanson (2013). The shift-share regression is:

$$\Delta Y_i = \beta Z_i^{SS} + \Delta e_i.$$

By contrast, the specification in my paper is:

$$\Delta Y_{iz} = \beta \Delta \log(TP_{iz}) + \Delta \delta_i + \Delta \delta_z + \Delta e_{iz},$$

where Y_{iz} is manufacturing output in county i and product z, $\Delta \log(TP_{iz})$ is the change in trade protection in county i and product z, and $\Delta \delta_i$ and $\Delta \delta_z$ are county and product fixed effects.

I instrument $\Delta \log(TP_{iz})$ with

$$Z_{iz} = STS_{iz}^t \Delta g_z,$$

where STS_{iz}^t is the share of the tariff in total protection as defined in Section 5.3. In this paper, $\Delta g_z = \Delta \log t_z$ where t_z is the nationwide tariff on product z.

Shift-share designs compare regions with different industrial compositions after a nationwide product-level shock, whereas my design compares the same product across counties with differential increases in protection due to differences in transportation costs to international ports. Put differently, the shift-share regression studies outcome changes

at the regional level i, whereas my regression studies outcome changes at the region-product level iz. This allows me to include product fixed effects and compare across counties within the same product.

These two approaches rely on different identification assumptions. As shown by Equation 7, the identification condition for my design is:

$$\mathbb{E}\left(\Delta g_z \cdot \Delta e_{iz} \mid STS_{iz}^t, \Delta\delta_i, \Delta\delta_z\right) = 0.$$

Exogenous Shocks. Shift-share identification arguments typically rely on one of two justifications. I first consider the exogenous-shocks view of Borusyak, Hull and Jaravel (2022), which requires:

$$\mathbb{E}\left(\Delta g_z \cdot \Delta \bar{e}_z \cdot s_z\right) = 0,$$

where $s_z = \sum_i s_{iz}$, and

$$\Delta \bar{e}_z = \frac{\sum_i s_{iz} \Delta e_i}{\sum_i s_{iz}}.$$

Shift-share designs require the national shock Δg_z to be uncorrelated with **national** product-level shocks $\Delta \bar{e}_z$; my design instead requires the national shock Δg_z to be uncorrelated with **local** county-product shocks Δe_{iz}. In this sense my identification condition is more granular.[45] In addition, my design allows for the inclusion of product fixed effects $\Delta\delta_z$, unlike shift-share designs. Therefore, any product-wide shock common to all regional units i, such as technological breakthroughs or federal policy changes, may confound shift-share designs, while being absorbed by product fixed effects in my design.

Exogenous Shares. Now I compare my design to the exogenous-shares view of Goldsmith-Pinkham, Sorkin and Swift (2020):

$$\mathbb{E}\left(s_{iz} \Delta e_i\right) = 0.$$

The identification condition here requires that initial industrial composition is uncorrelated with future location shocks. In this sense, within the shift-share family, the exogenous-shares view is more granular than the exogenous-shocks view: identification comes from the interaction of a region-product variable, s_{iz}, with a regional shock, Δe_i, rather than from the interaction of a product-level shock, Δg_z, with a product-level residual, $\Delta \bar{e}_z$. However, this shifts the identifying assumption onto the determinants of baseline industrial composition. In the trade context, this assumption is demanding: regions specializing in different products may also be differentially exposed to non-trade shocks, such as immigration flows, technological change, unionization drives, or region-specific industrial decline.

[45] In some contexts, where the nationwide shock Δg_z is a policy lever such as tariffs, this condition is arguably more plausible because violating it requires a more specific form of targeting. For example, policymakers targeting fast-growing products nationally would threaten shift-share designs, but would not threaten my design unless they targeted products based on differential trends across counties within the same product, which is a higher informational and cognitive burden on policymakers.

Appendix C Additional Empirical Results

Appendix C.1 Robustness to Trade Elasticity

Table C2: Robustness: Controlling for Trade Elasticity

	Δ Log Output			
	(1)	(2)	(3)	(4)
	Median	CP (2015)	Shapiro (2016)	GYY (2021)
$\Delta\log(\text{TP}) \times$ Distance to Frontier	2.650**	1.585***	3.586**	1.995***
	(1.027)	(0.433)	(1.370)	(0.695)
$(\sigma_z - 1) \times \Delta\log(\text{TP})$	0.851	1.426	0.993*	0.395
	(0.602)	(2.030)	(0.539)	(0.491)
$\Delta\log(\text{TP})$	−15.009*	−9.242	−23.539*	−9.253
	(7.914)	(6.883)	(11.410)	(6.052)
Observations	833	833	833	833
Input AVE Control	Yes	Yes	Yes	Yes
County-Year FE	Yes	Yes	Yes	Yes
Product-Year FE	Yes	Yes	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	7.59	3.55	25.62	7.71

Notes: Each column reports IV estimates of specification 4 of the change in log protection on log output, augmented with an interaction between the trade elasticity of substitution ($\sigma_z - 1$) and the change in log protection. Each column uses a different source for σ_z: Column 1 uses the median across sources, Column 2 uses Caliendo and Parro (2015), Column 3 uses Shapiro (2016), Column 4 uses Giri, Yi, and Yilmazkuday (2021). The distance-to-frontier interaction remains significant across all specifications. Standard errors clustered at the county and product level in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Appendix C.2 Dropping Products

Here we run our baseline specification, but we drop products one-by-one.

Table C3: Leave-One-Out Robustness Check

Product Excluded	$\Delta \log(\text{TP}) \times$ *Dist_to_Front*	$\Delta \log(\text{TP})$	KP F	N
Beer	1.823***	−5.839**	12.09	614
	(0.468)	(2.354)		
Boots	1.640***	−4.589*	16.00	592
	(0.451)	(2.315)		
Cotton Goods	1.659***	−4.821**	15.82	743
	(0.468)	(2.142)		
Glass	1.502***	−4.342*	21.42	797
	(0.428)	(2.060)		
Glove	1.443***	−4.359*	15.63	826
	(0.408)	(2.035)		
Hats	1.350***	−3.721*	16.40	794
	(0.387)	(2.052)		
Iron Bar	1.348**	−4.017*	13.51	766
	(0.471)	(2.092)		
Iron Pig	1.174**	−2.967**	12.20	774
	(0.516)	(1.321)		
Nails	1.394***	−4.511**	10.17	808
	(0.395)	(1.870)		
Paints	1.938**	−5.751*	13.83	823
	(0.881)	(2.833)		
Paper	1.528***	−4.287**	15.55	716
	(0.421)	(2.120)		
Rope	1.113***	−3.066*	15.44	798
	(0.289)	(1.525)		
Steel	1.503***	−4.465**	15.44	830
	(0.403)	(2.012)		
Whiskey	1.674***	−5.935**	15.65	729
	(0.503)	(2.355)		
Worsted Goods	1.498***	−4.430**	15.45	830
	(0.400)	(2.004)		

Notes: Standard errors in parentheses. Each row reports IV coefficient estimates from a regression excluding the indicated product. TP denotes total protection (tariff plus transportation cost). KP F is the Kleibergen-Paap rk Wald F-statistic. * p<0.10, ** p<0.05, *** p<0.01.

Appendix C.3 Extensive Margins

I run the following specification:

$$\mathbf{1}[y_{iz,1860} = 0,\ y_{iz,1870} > 0] = \beta[\Delta \log(TP_{iz}) \times Dist_Front_z] + \theta \Delta \log(TP_{iz}) +$$
$$\gamma X_{iz} + \Delta\delta_i + \Delta\delta_z + \Delta e_{iz}$$

The dependent variable is an indicator function taking the value of 1 if a county-product pair has zero production in 1860 but strictly positive production in 1870. The right hand side variables are the same as main text.

Table C4: Extensive Margin

	$1[y_{iz,1860} = 0,\ y_{iz,1870} > 0]$	
	OLS	IV
	(1)	(2)
Δlog(TP) $\times$ Dist_Front	-0.002	-0.007
	(0.013)	(0.022)
Δlog(TP)	0.033	0.082
	(0.067)	(0.126)
Observations	11,501	11,501
Input TP Control	Yes	Yes
County-Year FE	Yes	Yes
Product-Year FE	Yes	Yes
First-stage F-statistic (Kleibergen-Paap)	—	20.31

We find no statistically significant differential effect of trade protection on entry by distance to the frontier. This suggests that my model, which purely operates on the intensive margin, is the appropriate framework for this setting.

Appendix C.4 Regression with Initial Variables

Table 4 shows that the change in protection arising from the tariffs is statistically balanced on observable characteristics at a 10% level. However, some of the coefficients are economically significant even if they are not statistically significant. I thus include these initial variables in the main specification as a robustness check.

As Table C5 shows, the distance-to-frontier interaction remains positive and significant at the 1% level across all four specifications, and the baseline protection effect remains negative and significant.

Appendix C.5 Migration Theory and Analysis

This appendix develops the migration analysis referenced in Section 6.3. Section Appendix C.5.1 presents a discrete choice model of worker location. Section Appendix C.5.2 derives

Table C5: Robustness: Controlling for Initial Characteristics

	Δ Log Output			
	(1) Output	(2) Estab.	(3) Labor	(4) Cap./Labor
Δlog(TP) × Dist_Front	1.474***	1.707***	1.594***	1.977***
	(0.418)	(0.446)	(0.441)	(0.529)
Δlog(TP)	−5.231**	−5.962***	−5.499**	−4.993*
	(1.821)	(1.920)	(1.869)	(2.382)
Observations	833	833	833	833
Initial Char. Control	Yes	Yes	Yes	Yes
Input TP Control	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes
First-stage F-stat (KP)	15.57	15.12	15.72	16.23

Notes: IV estimates of the main specification (Equation (4)), augmented with a control for the baseline (1860) county-product characteristic indicated in each column header. Column 1 controls for initial output, Column 2 for initial number of establishments, Column 3 for initial labor, and Column 4 for initial capital-to-labor ratio. The distance-to-frontier interaction remains significant across all specifications. Standard errors two-way clustered by county and product in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

the estimating equation. Section Appendix C.5.3 describes the data and the assignment of workers to products. Section Appendix C.5.4 discusses the treatment of zero flows. Section Appendix C.5.5 presents results.

Appendix C.5.1 Model

Individual ι, initially residing in county o and working in product k, chooses a destination county i and destination product z. Utility from choosing (i, z) is

$$U_{\iota oikz} = \alpha \log w_{iz} - \alpha \log P_i + Z_i + Z_z - C_{oi} - C_{kz} + e_{\iota oikz},$$

where w_{iz} is the wage in county i and product z, P_i is the local price index, Z_i and Z_z are county and product amenities, C_{oi} is the cost of moving from o to i, and C_{kz} is the cost of switching from product k to product z.

Assuming $e_{\iota oikz}$ is distributed i.i.d. Type-I Extreme Value, the log choice probability is (McFadden 1978):

$$\log \frac{f_{oikz}}{L_{ok}} = \alpha \log w_{iz} - \alpha \log P_i + Z_i + Z_z - C_{oi} - C_{kz} - \Phi_{ok},$$

where f_{oikz} is the flow from origin county-product (o,k) to destination county-product (i,z), L_{ok} is employment in the origin county-product, and

$$\Phi_{ok} = \log\left[\sum_{i',z'} \exp\left(\alpha \log w_{i'z'} - \alpha \log P_{i'} + Z_{i'} + Z_{z'} - C_{oi'} - C_{kz'}\right)\right]$$

is the inclusive value for workers initially in (o,k), summarizing the attractiveness of all destination options available to them. Equivalently,

$$f_{oikz} = L_{ok} \exp\left(\alpha \log w_{iz} - \alpha \log P_i + Z_i + Z_z - C_{oi} - C_{kz} - \Phi_{ok}\right).$$

Appendix C.5.2 Estimating Equation

Aggregating over origin products k:

$$f_{oiz} \equiv \sum_k f_{oikz}.$$

Substituting the expression above and summing over k,

$$f_{oiz} = \exp\left(\alpha \log w_{iz} - \alpha \log P_i + Z_i + Z_z - C_{oi}\right) \Lambda_{oz}, \qquad \Lambda_{oz} \equiv \sum_k L_{ok} \exp\left(-C_{kz} - \Phi_{ok}\right).$$

Taking logs,

$$\log f_{oiz} = \alpha \log w_{iz} - \alpha \log P_i + Z_i + Z_z - C_{oi} + \log \Lambda_{oz}.$$

First-differencing between the 1850–1860 and 1860–1870 flows gives

$$\Delta \log f_{oiz} = \alpha \Delta \log w_{iz} + \underbrace{\left(-\alpha \Delta \log P_i + \Delta Z_i - \Delta C_{oi}\right)}_{\Delta\delta_{oi}} + \underbrace{\left(\Delta Z_z + \Delta \log \Lambda_{oz}\right)}_{\Delta\delta_{oz}}.$$

The fixed effect $\Delta\delta_{oi}$ absorbs changes in origin-destination migration costs and in destination county conditions such as amenities and cost-of-living; $\Delta\delta_{oz}$ absorbs changes in product amenities and the effective supply of workers from origin county o to product z, taking into account switching costs between different products. Differencing additionally removes any persistent component of corridor-specific flow δ_{oiz}.

I proxy destination wage growth with protection:

$$\Delta \log w_{iz} = \theta \Delta \log(TP_{iz}) + \beta\left[\Delta \log(TP_{iz}) \times Dist_Front_z\right] + \zeta_{oiz},$$

where ζ_{oiz} is the component of wage growth orthogonal to the tariff-induced labor demand shock. Substituting into the first-differenced migration equation gives

$$\Delta \log(f_{oiz}) = \beta\left[\Delta \log(TP_{iz}) \times Dist_Front_z\right] + \theta \Delta \log(TP_{iz}) + \gamma \Delta \log(Input_TP_{iz}) + \Delta\delta_{oi} + \Delta\delta_{oz} + \Delta\varepsilon_{oiz}, \tag{18}$$

As in the main specification, $\Delta \log(TP_{iz})$ and its interaction with $Dist_Front_z$ are instrumented with $STS^t_{iz}\Delta \log t_z$ and its interaction with $Dist_Front_z$. Standard errors are two-way clustered by destination county and product.

Appendix C.5.3 Data and Product Assignment

I use the 1850, 1860 and 1870 full-count population censuses, taking individual links from the IPUMS Multigenerational Longitudinal Panel (Ruggles et al. 2025). This yields two decade pairs, 1850–1860 and 1860–1870, and hence one difference in flows. I assign each linked individual to a product using their destination-year occupation, coded in the 1880 occupational classification that underlies the IPUMS OCC variable for all samples from 1850 to 1900. Table C6 reports the crosswalk.

Table C6: Mapping from Occupations to Products

Product	OCC	Occupation Title
Cotton Textiles	170	Cotton-mill operatives
Worsted Goods	264	Woolen mill operatives
Boots	142	Boot and shoemakers
Hats	189	Hat and cap makers
Iron Bar	191	Iron and steel works and shops operatives
Iron Pig	191	Iron and steel works and shops operatives
Steel	191	Iron and steel works and shops operatives
Paper	216	Paper mill operatives
Beer	146	Brewers and maltsters
Whiskey	171	Distillers and rectifiers
Glass	183	Glass-works operatives
Glove	184	Glove makers
Rope	232	Rope and cordage makers
Nails	209	Nail makers
Paints	—	No corresponding occupation

Notes: Occupation codes follow the 1880 classification underlying the IPUMS OCC variable, which applies to all full-count samples from 1850 to 1900. Iron Bar, Iron Pig and Steel share a single code; workers are split across the three in proportion to each county's 1860 output shares from the Census of Manufactures. The 1880 classification contains no occupation corresponding to paint manufacture, so Paints is absent from the migration sample. The broad definition additionally assigns codes 224 (print-works operatives) and 253 (thread makers) to Cotton Textiles, 190 (hosiery and knitting mill operatives) to Worsted Goods, 195 (leather curriers, dressers, finishers and tanners) to Boots, 250 (straw workers) and 179 (fur workers) to Hats, and 248, 249, 176 and 255 (boiler, stove, file, and tool and cutlery makers) to the iron products. Code 137 (bleachers, dyers and scourers) serves both textile products and is split by 1860 output shares.

I restrict the sample to men aged fifteen and over. The 1850 occupation universe covers

only males aged fifteen and over, and the linking procedure cannot match women who changed surnames upon marriage before 1880, so this restriction makes the two decade pairs comparable. I apply the same geographic exclusions as in the main sample and restrict the grid to the 833 county-product pairs that appear in the estimation sample of Table 6.

I report two alternative product definitions alongside the baseline. The broad occupation definition adds supporting occupations that plausibly work within the same establishment, such as print-works operatives and thread makers for cotton textiles, or boiler and stove makers for iron and steel. The industry definition uses the IPUMS IND1950 variable in place of occupation, which results in a coarser categorization: it merges cotton with worsted, beer with whiskey, hats with gloves, and the three iron products.

Appendix C.5.4 Dealing with Zero Flows

Most county-product pairs receive no linked workers from most origin counties in a given decade: 99.6% of cells in the balanced grid are zero. Taking logs of the dependent variable would therefore drop the vast majority of observations.

As a transparent way to handle these zeros without dropping them, I follow Chen and Roth (2024), who advocate calibrating explicitly how much one values the extensive margin relative to the intensive margin:

$$m(y) = \begin{cases} \log(y) & \text{if } y > 0 \\ -x & \text{if } y = 0. \end{cases}$$

Other approaches such as $\log(1+f_{oiz})$ or $\operatorname{arcsinh}(f_{oiz})$ transformations implicitly place a weight on the extensive margin relative to the intensive margin that depends on the units of measurement (Chen and Roth 2024). Poisson pseudo-maximum likelihood (Santos Silva and Tenreyro, 2006) offers another approach to handling zeros. However, it is inappropriate in my context because the dependent variable is a first difference of flows, which can be negative, and PPML requires non-negative outcomes.

The appropriate calibration depends on how large a new corridor is relative to a typical existing one. The median positive flow in my data is a single worker, so a corridor appearing at all is comparable in magnitude to a corridor that already exists. I therefore take $x = 1.0$ as the default, under which a new corridor is valued at one log point — equivalent to a 172% increase in an existing flow, or 1.7 workers at the median. I report $x = 0.1$ and $x = 0.5$ as well.

Appendix C.5.5 Results

Table C7 reports estimates of Equation (18). At the default calibration, the coefficient on protection is positive and significant: workers flow toward more protected county-product pairs. The interaction with distance to the frontier is small, negative and statistically insignificant in every specification. Protection therefore draws labor toward protected products, but not disproportionately toward the far-from-frontier products whose output expands most. Results are similar under the broad occupation definition.

Table C7: Effect of Trade Protection on Bilateral Worker Flows

	(1)	(2)	(3)
Δlog(TP) $\times$ Dist_Front	-0.0038	-0.0040	-0.0040
	(0.0024)	(0.0025)	(0.0025)
Δlog(TP)	0.0240^{**}	0.0247^{**}	0.0200^{*}
	(0.0100)	(0.0101)	(0.0108)
Product definition	OCC	OCC-B	IND
Observations	1,423,597	1,423,597	1,423,597
Input TP Control	Y	Y	Y
Origin $\times$ Dest. FE	Y	Y	Y
Origin $\times$ Product FE	Y	Y	Y
First-stage F-stat (KP)	15.6	15.6	15.6

Notes: IV estimates of Equation (18) at the default extensive-margin calibration $x = 1.0$. The dependent variable is the change in transformed bilateral flows, $\Delta m(f_{oiz})$, where $m(y) = \log(y)$ for $y > 0$ and $m(0) = -x$. OCC is the baseline occupation definition, OCC-B the broad occupation definition, and IND the industry definition, all described in Table C6. Origins are defined at the county level. $\Delta \log(TP_{iz})$ and its interaction with $Dist_Front_z$ are instrumented with $STS^t_{iz} \times \Delta \log t_z$ and its interaction with $Dist_Front_z$. Standard errors two-way clustered by destination county and product in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Finally, we examine alternative values of the extensive-margin weight x. As Table C8 shows, results at $x = 0.5$ are similar to the default: protection increases flows, and there is no statistically significant differential effect by distance to the frontier. At $x = 0.1$ the point estimates carry the same signs, though the effect of protection on flows is no longer statistically significant.

Table C8: Bilateral Worker Flows: Alternative Extensive-Margin Calibrations

	$x = 0.5$			$x = 0.1$		
	(1)	(2)	(3)	(4)	(5)	(6)
Δlog(TP) $\times$ Dist_Front	-0.0026	-0.0027	-0.0026	-0.0017	-0.0016	-0.0014
	(0.0019)	(0.0019)	(0.0019)	(0.0016)	(0.0015)	(0.0015)
Δlog(TP)	0.0176**	0.0176**	0.0148	0.0125	0.0120	0.0107
	(0.0081)	(0.0081)	(0.0085)	(0.0076)	(0.0073)	(0.0074)
Product definition	OCC	OCC-B	IND	OCC	OCC-B	IND
Observations	1,423,597	1,423,597	1,423,597	1,423,597	1,423,597	1,423,597
Input TP Control	Y	Y	Y	Y	Y	Y
Origin $\times$ Dest. FE	Y	Y	Y	Y	Y	Y
Origin $\times$ Product FE	Y	Y	Y	Y	Y	Y
First-stage F-stat (KP)	15.6	15.6	15.6	15.6	15.6	15.6

Notes: As in Table C7, but reporting alternative values of the extensive-margin calibration x, which sets the value assigned to a zero flow under the Chen and Roth (2024) transform. Lower values of x place less weight on corridors appearing or disappearing relative to changes in existing flows. Section Appendix C.5.4 discusses the choice of $x = 1.0$ as the default. Standard errors two-way clustered by destination county and product in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Appendix D Additional Theoretical Results

Appendix D.1 Numerical Example

Take

$$\sigma = 8, \qquad E_z = 1, \qquad D_z = 1.05, \qquad \alpha = 3, \qquad TP_z^0 = 0.25, \qquad TP_z^1 = 0.35, \qquad F = 0.4343$$

Solving the Bertrand-CES share equation gives

$$G_z(D_z, TP_z^0) = 0.4371,$$

and

$$G_z(D_z, TP_z^1) = 0.4316.$$

Then

$$G_z(D_z, TP_z^0) > F > G_z(D_z, TP_z^1),$$

so the firm innovates before protection but not after:

$$I_z^0 = 1, \qquad I_z^1 = 0.$$

Before the protection increase, the firm innovates and captures a 92.4% market share:

$$Y_z^0 = E_z \lambda_z \left(\frac{3 \times 1.25}{1.05} \right) = \lambda_z(3.57) = 0.9242.$$

After the protection increase, the firm no longer innovates and its market share falls to 66.1% despite higher protection:

$$Y_z^1 = E_z \lambda_z \left(\frac{1.35}{1.05} \right) = \lambda_z(1.29) = 0.6610.$$

Thus $Y_z^1 < Y_z^0$: output falls by 28.5%. Protection eliminates the incentive to innovate, and the resulting loss of competitiveness more than offsets the direct protective effect.

Appendix D.2 Structural Derivation of Elasticity Term

We start from the finite decomposition in Equation (1). To obtain a crisp form for the structural elasticity, we take a first-order approximation around the pre-protection equilibrium. This gives

$$\frac{\Delta \log Y_{iz}}{\Delta \log(1 + TP_{iz})} \approx \left. \frac{\partial \log Y_{iz}}{\partial \log(1 + TP_{iz})} \right|_0 + \left. \frac{\partial \log Y_{iz}}{\partial \log T_z^{US}} \right|_0 \cdot \frac{\Delta \log T_z^{US}}{\Delta I_{iz}^*} \cdot \frac{\Delta I_{iz}^*}{\Delta \log(1 + TP_{iz})}.$$

The derivation proceeds in three steps: (i) showing that the substitution and technology elasticities are equal, (ii) computing the technology step $\Delta \log T_z^{US} / \Delta I_{iz}^*$, and (iii) deriving the common elasticity under Bertrand-CES competition.

Step 1: Equality of Elasticities

US output is $Y_{iz} = E_{iz} \lambda_{iz}(r_{iz})$, where relative competitiveness is

$$r_{iz} = \frac{\alpha^{I_{iz}} (1 + TP_{iz})}{D_z}.$$

and $D_z = T_z^{UK} / T_z^{US}$. Equivalently,

$$\log r_{iz} = I_{iz} \log \alpha + \log(1 + TP_{iz}) + \log T_z^{US} - \log T_z^{UK}.$$

Since $\log Y_{iz} = \log E_i \chi_{iz} + \log \lambda_{iz}(r_{iz})$, by the chain rule:

$$\frac{\partial \log Y_{iz}}{\partial \log(1 + TP_{iz})} = \frac{d \log \lambda_{iz}}{d \log r_{iz}} \cdot \underbrace{\frac{\partial \log r_{iz}}{\partial \log(1 + TP_{iz})}}_{=1} = \frac{d \log \lambda_{iz}}{d \log r_{iz}}.$$

$$\frac{\partial \log Y_{iz}}{\partial \log T_z^{US}} = \frac{d \log \lambda_{iz}}{d \log r_{iz}} \cdot \underbrace{\frac{\partial \log r_{iz}}{\partial \log T_z^{US}}}_{=1} = \frac{d \log \lambda_{iz}}{d \log r_{iz}}.$$

Step 2: Technology Step

When the US firm innovates, its technology improves from T_z^{US} to αT_z^{US}. Therefore:

$$\frac{\Delta \log T_z^{US}}{\Delta I_{iz}^*} = \log(\alpha T_z^{US}) - \log(T_z^{US}) = \log \alpha.$$

Step 3: Derivation of $d \log \lambda_{iz} / d \log r_{iz}$

Under Bertrand-CES competition, the US and UK prices are:

$$p_z^{US} = \mu^{US}(\lambda_{iz}) \cdot c_z^{US}, \qquad p_z^{UK} = \mu^{UK}(\lambda_{iz}) \cdot c_z^{UK}.$$

where

$$\mu^{US}(\lambda_{iz}) = \frac{\varepsilon_z^{US}}{\varepsilon_z^{US} - 1} = \frac{\sigma - (\sigma - 1)\lambda_{iz}}{(\sigma - 1)(1 - \lambda_{iz})}$$

and

$$\mu^{UK}(\lambda_{iz}) = \frac{\varepsilon_z^{UK}}{\varepsilon_z^{UK} - 1} = \frac{1 + (\sigma - 1)\lambda_{iz}}{(\sigma - 1)\lambda_{iz}}.$$

The CES expenditure share satisfies:

$$\log \frac{\lambda_{iz}}{1 - \lambda_{iz}} = (\sigma - 1) \left[\log r_{iz} + \log \mu^{UK}(\lambda_{iz}) - \log \mu^{US}(\lambda_{iz}) \right].$$

Implicitly differentiating with respect to $\log r_{iz}$:

$$\frac{1}{\lambda_{iz}(1 - \lambda_{iz})} \frac{d\lambda_{iz}}{d \log r_{iz}} = (\sigma - 1) \left[1 + \left(\frac{d \log \mu^{UK}}{d\lambda_{iz}} - \frac{d \log \mu^{US}}{d\lambda_{iz}} \right) \frac{d\lambda_{iz}}{d \log r_{iz}} \right].$$

Computing the markup elasticities:

$$\frac{d \log \mu^{US}}{d\lambda_{iz}} = \frac{1}{\varepsilon_z^{US}(1 - \lambda_{iz})}, \qquad \frac{d \log \mu^{UK}}{d\lambda_{iz}} = \frac{-1}{\lambda_{iz} \varepsilon_z^{UK}}.$$

Substituting and solving for $d\lambda_{iz} / d \log r_{iz}$:

$$\frac{d \log \lambda_{iz}}{d \log r_{iz}} = \frac{(\sigma - 1)(1 - \lambda_{iz})}{1 + \frac{(\sigma - 1)(1 - \lambda_{iz})}{\varepsilon_z^{UK}} + \frac{(\sigma - 1)\lambda_{iz}}{\varepsilon_z^{US}}}.$$

Substituting $\varepsilon_z^{US} = \sigma - (\sigma-1)\lambda_{iz}$ and $\varepsilon_z^{UK} = 1 + (\sigma-1)\lambda_{iz}$, and letting $a = (\sigma-1)\lambda_{iz}$ and $b = (\sigma-1)(1-\lambda_{iz})$ so that $a + b = \sigma - 1$, the denominator becomes:

$$1 + \frac{b}{1+a} + \frac{a}{1+b} = \frac{(1+a)(1+b) + b(1+b) + a(1+a)}{(1+a)(1+b)} = \frac{1 + 2(a+b) + a^2 + b^2 + ab}{(1+a)(1+b)}.$$

Using $a^2 + b^2 = (a+b)^2 - 2ab$:

$$= \frac{(1+a+b)^2 - ab}{(1+a)(1+b)} = \frac{\sigma^2 - (\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}{(1+a)(1+b)}.$$

And

$$(1+a)(1+b) = 1 + a + b + ab = \sigma + (\sigma-1)^2\lambda_{iz}(1-\lambda_{iz}).$$

Therefore:

$$\frac{d\log\lambda_{iz}}{d\log r_{iz}} = (\sigma-1)(1-\lambda_{iz}) \cdot \frac{\sigma + (\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}{\sigma^2 - (\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}.$$

Combining

Substituting Steps 1–3 into the first-order approximation above gives:

$$\begin{aligned}\frac{\Delta\log Y_{iz}}{\Delta\log(1+TP_{iz})} &\approx \frac{d\log\lambda_{iz}}{d\log r_{iz}} + \frac{d\log\lambda_{iz}}{d\log r_{iz}} \cdot \log\alpha \cdot \frac{\Delta I_{iz}^*}{\Delta\log(1+TP_{iz})} \\ &= \frac{d\log\lambda_{iz}}{d\log r_{iz}} \cdot \left(1 + \log\alpha \cdot \frac{\Delta I_{iz}^*}{\Delta\log(1+TP_{iz})}\right).\end{aligned}$$

Using the expression for $d\log\lambda_{iz}/d\log r_{iz}$ derived above:

$$\frac{\Delta\log Y_{iz}}{\Delta\log(1+TP_{iz})} \approx (\sigma-1)(1-\lambda_{iz}) \cdot \frac{\sigma + (\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})}{\sigma^2 - (\sigma-1)^2\lambda_{iz}(1-\lambda_{iz})} \cdot \left(1 + \log\alpha \cdot \frac{\Delta I_{iz}^*}{\Delta\log(1+TP_{iz})}\right),$$

which yields Equation (8). ■

Appendix D.3 Comparative Statics Results

Appendix D.3.1 Proof of Proposition 1

Proof. We have

$$G_{iz}(D_z, TP_{iz}^1) = G_{iz}\left(\frac{1+TP_{iz}^0}{1+TP_{iz}^1} D_z,\ TP_{iz}^0\right).$$

Above-threshold result. Since $r \to 0$ as $D_z \to \infty$, there exists $\bar{D}$ such that both $(1 + TP_{iz}^0)/D_z$ and $(1 + TP_{iz}^1)/D_z$ lie in $(0, \bar{r}/\alpha)$ for all $D_z \geq \bar{D}$, where $\bar{r}$ is given by Lemma A1 below. Writing $u = \log r$ and $\psi(u) = \Pi^{op}(e^u)$, the gain from innovation is

$$G_{iz} = \psi(u + \log\alpha) - \psi(u).$$

By Lemma A1, ψ is strictly convex on this region, so $\psi'(u+\log\alpha) > \psi'(u)$ and hence

$$\frac{\partial G_{iz}}{\partial u} = \psi'(u+\log\alpha) - \psi'(u) > 0.$$

Thus G_{iz} is strictly increasing in r, and since r is increasing in TP_{iz},

$$G_{iz}(D_z, TP_{iz}^1) \geq G_{iz}(D_z, TP_{iz}^0) \qquad \text{for all } D_z \geq \bar{D}.$$

Because $I_{iz}^* = \mathbf{1}\{G_{iz} \geq F_z\}$ is nondecreasing in G_{iz}, it follows that $I_{iz}^*(D_z, TP_{iz}^1) \geq I_{iz}^*(D_z, TP_{iz}^0)$, and therefore

$$\Delta I_{iz}^* \geq 0.$$

Let $D_{iz}^{I,*}$ denote the smallest $\bar{D}$ for which this holds on $[\bar{D}, \infty)$.

Below-threshold result. Since Π^{op} is strictly increasing and $\alpha > 1$, $G_{iz} > 0$ for all D_z. As $D_z \to 0$, relative competitiveness diverges, $\lambda \to 1$ and $\Pi^{op} \to E_{iz}$ at both r and αr, so $G_{iz} \to 0$: the US firm captures the market with or without innovating. As $D_z \to \infty$, $\lambda \to 0$ and $\Pi^{op} \to 0$ at both arguments, so again $G_{iz} \to 0$: the firm is uncompetitive either way. A continuous, strictly positive function vanishing at both ends of $(0, \infty)$ attains an interior maximum. Hence there is a region of r on which G_{iz} is strictly decreasing, and therefore strictly decreasing in TP_{iz}:

$$G_{iz}(D_z, TP_{iz}^1) < G_{iz}(D_z, TP_{iz}^0).$$

For any such D_z, any fixed cost satisfying

$$G_{iz}(D_z, TP_{iz}^1) < F_z \leq G_{iz}(D_z, TP_{iz}^0)$$

yields $I_{iz}^*(D_z, TP_{iz}^0) = 1$ and $I_{iz}^*(D_z, TP_{iz}^1) = 0$, so

$$\Delta I_{iz}^* = -1.$$

Appendix D.1 exhibits parameter values for which this holds. ■

Lemma A1. There exists $\bar{r} > 0$ such that Π^{op} is strictly convex in $\log r$ on $(0, \bar{r})$.

Proof. Write $\psi(u) \equiv \Pi^{op}(e^u)$ and let $\varepsilon(u)$ denote the elasticity of operating profit with respect to competitiveness,

$$\varepsilon(u) \equiv \left.\frac{d\log\Pi^{op}}{d\log r}\right|_{r=e^u} = \underbrace{\frac{\sigma}{\sigma-(\sigma-1)\lambda}}_{d\log\Pi^{op}/d\log\lambda} \cdot \underbrace{(\sigma-1)(1-\lambda)\,\frac{\sigma+(\sigma-1)^2\lambda(1-\lambda)}{\sigma^2-(\sigma-1)^2\lambda(1-\lambda)}}_{d\log\lambda/d\log r},$$

where the second factor is derived in Appendix D.2. Then $\psi' = \varepsilon\psi$ and

$$\psi'' = \left(\varepsilon' + \varepsilon^2\right)\psi.$$

As $u \to -\infty$ we have $\lambda \to 0$, so

$$\varepsilon \;\longrightarrow\; \frac{\sigma}{\sigma} \cdot (\sigma - 1) \cdot \frac{\sigma}{\sigma^2} = \frac{\sigma - 1}{\sigma} \equiv \nu > 0.$$

Moreover ε is a smooth function of λ with bounded derivative near $\lambda = 0$, while $d\lambda/du = \lambda \cdot (d \log \lambda / d \log r) = O(\lambda) \to 0$; hence $\varepsilon' \to 0$. Therefore

$$\frac{\psi''}{\psi} \;\longrightarrow\; \nu^2 > 0,$$

and since $\psi > 0$, there exists $\bar{r} > 0$ such that $\psi'' > 0$ on $(0, \bar{r})$. ■

Appendix D.3.2 Proof of Proposition 2

Proof. *Above-threshold result.* Let $I^0_{iz} = I^*_{iz}(D_z, TP^0_{iz})$ and $I^1_{iz} = I^*_{iz}(D_z, TP^1_{iz})$. For $D_z \geq D^{I,*}_{iz}$, Proposition 1 gives $I^1_{iz} \geq I^0_{iz}$. Since $TP^1_{iz} > TP^0_{iz}$ and $\alpha > 1$,

$$\frac{\alpha^{I^1_{iz}}(1 + TP^1_{iz})}{D_z} > \frac{\alpha^{I^0_{iz}}(1 + TP^0_{iz})}{D_z}.$$

Because the Bertrand-CES expenditure share $\lambda(\cdot)$ is strictly increasing in relative US competitiveness,

$$Y^1_{iz} = E_{iz}\lambda\left(\frac{\alpha^{I^1_{iz}}(1 + TP^1_{iz})}{D_z}\right) > E_{iz}\lambda\left(\frac{\alpha^{I^0_{iz}}(1 + TP^0_{iz})}{D_z}\right) = Y^0_{iz},$$

so $\Delta \log Y_{iz} > 0$. Since this holds for all $D_z \geq D^{I,*}_{iz}$, the smallest threshold above which output rises satisfies $D^{Y,*}_{iz} \leq D^{I,*}_{iz}$.

Below-threshold result. Suppose the parameters are such that $\Delta I^*_{iz} = -1$, which Proposition 1 shows is possible, and suppose in addition that

$$\alpha > \frac{1 + TP^1_{iz}}{1 + TP^0_{iz}},$$

so that $\alpha(1 + TP^0_{iz}) > 1 + TP^1_{iz}$: the loss of the innovation step more than offsets the direct increase in protection. The firm innovates at TP^0_{iz} but not at TP^1_{iz}, so, since $\lambda(\cdot)$ is strictly increasing,

$$Y^0_{iz} = E_z\lambda\left(\frac{\alpha(1 + TP^0_{iz})}{D_z}\right) > E_z\lambda\left(\frac{1 + TP^1_{iz}}{D_z}\right) = Y^1_{iz},$$

and therefore $\Delta \log Y_{iz} < 0$. ■

Appendix D.3.3 Proof of Corollary 1

Proof. By single-peakedness, the set of products that innovate at the initial protection level is an interval:

$$\{D_z : G_{iz}(D_z, TP_{iz}^0) \geq F_z\} = [D_L, D_H].$$

Let

$$k = \frac{1 + TP_{iz}^1}{1 + TP_{iz}^0} > 1.$$

Since D_z and TP_{iz} enter only through $(1 + TP_{iz})/D_z$, the corresponding set at the higher protection level is

$$\{D_z : G_{iz}(D_z, TP_{iz}^1) \geq F_z\} = [kD_L, kD_H].$$

Protection strictly reduces innovation exactly when the firm innovates at TP_{iz}^0 but not at TP_{iz}^1. Since $kD_H > D_H$, this occurs precisely when

$$D_L \leq D_z \leq D_H \qquad \text{and} \qquad D_z < kD_L.$$

Therefore,

$$\{D_z : \Delta I_{iz}^* = -1\} = \begin{cases} [D_L, kD_L), & \text{if } kD_L \leq D_H, \\ [D_L, D_H], & \text{if } kD_L > D_H. \end{cases}$$

Outside this region, $\Delta I_{iz}^* \geq 0$.

Define the upper boundary of the region where innovation falls as

$$D_{iz}^{I,*} = \min\{kD_L, D_H\}.$$

When the innovation intervals overlap, including when they touch, this boundary is excluded from the region where innovation falls. When they are disjoint, the boundary is included.

It remains to apply the parity condition:

$$G_{iz}(1, TP_{iz}^0) \geq F_z > G_{iz}(1, TP_{iz}^1).$$

This condition implies that the firm innovates at the initial protection level but not at the higher protection level when $D_z = 1$. Hence,

$$D_L \leq 1 \leq D_H, \qquad 1 < kD_L.$$

Thus, on the domain $D_z \geq 1$,

$$\Delta I_{iz}^* < 0 \quad \text{for } 1 \leq D_z < D_{iz}^{I,*}, \qquad \Delta I_{iz}^* \geq 0 \quad \text{for } D_z > D_{iz}^{I,*}.$$

At the threshold itself,

$$\Delta I_{iz}^*|_{D_z = D_{iz}^{I,*}} = \begin{cases} 0, & \text{if } kD_L \leq D_H, \\ -1, & \text{if } kD_L > D_H. \end{cases}$$

■

Appendix D.3.4 Proof of Corollary 2

Proof. Let

$$k = \frac{1 + TP_{iz}^{1}}{1 + TP_{iz}^{0}} > 1.$$

By Corollary 1, on the domain $D_z \geq 1$, $\Delta I_{iz}^{*} = -1$ for $D_z < D_{iz}^{I,*}$, and $\Delta I_{iz}^{*} \geq 0$ for $D_z > D_{iz}^{I,*}$.

For $D_z < D_{iz}^{I,*}$, the below-threshold argument in the proof of Proposition 2 applies. Since $\alpha > k$, the loss of innovation more than offsets the direct increase in protection, so

$$\Delta \log Y_{iz} < 0.$$

For $D_z > D_{iz}^{I,*}$, innovation does not fall. The increase in protection therefore strictly increases domestic competitiveness and output, so

$$\Delta \log Y_{iz} > 0.$$

At $D_z = D_{iz}^{I,*}$, there are two cases. If $kD_L \leq D_H$, Corollary 1 implies $\Delta I_{iz}^{*} = 0$, so protection strictly raises output. If $kD_L > D_H$, Corollary 1 implies $\Delta I_{iz}^{*} = -1$; since $\alpha > k$, protection strictly lowers output.

Thus, the output effect is strictly negative below $D_{iz}^{I,*}$ and strictly positive above it. The innovation and output thresholds, interpreted as the upper boundaries of their respective negative-response regions, therefore coincide:

$$D_{iz}^{Y,*} = D_{iz}^{I,*}.$$

The output response at the common threshold is positive when $kD_L \leq D_H$ and negative when $kD_L > D_H$. ■

Appendix D.4 Full General Equilibrium Model

In Section 7, for ease of exposition, I focused on the local market problem to avoid introducing additional notation. Here, I fully specify the local and British firms' decisions in both markets, together with the UK equilibrium conditions. Superscripts identify firm origin, while subscripts identify the destination market and product.

Appendix D.4.1 Model Setup

I build on the environment in Section 3, allowing wages and expenditures to adjust. For each product z, submarkets are indexed by $\psi_z \in [0, 1]$. In each submarket, the local firm and a UK firm compete à la Bertrand in both the local and British markets. The local firm chooses whether to innovate before prices are set. Each submarket is measure zero in the economy, so individual firms take aggregate wages and expenditures as given.

Preferences have three nests: products, submarkets within products, and varieties within submarkets. Preferences across products are Cobb–Douglas:

$$E_{iz} = \chi_{iz} E_i, \qquad E_{UK,z} = \chi_{UK,z} E_{UK}, \qquad \sum_z \chi_{iz} = \sum_z \chi_{UK,z} = 1. \tag{19}$$

Within each product, preferences are Cobb–Douglas across submarkets:

$$\log Q_{iz} = \int_0^1 \log Q_{iz}(\psi_z)\, d\psi_z, \qquad \log Q_{UK,z} = \int_0^1 \log Q_{UK,z}(\psi_z)\, d\psi_z.$$

Here $Q_{iz}(\psi_z)$ and $Q_{UK,z}(\psi_z)$ denote consumption in submarket ψ_z in the local and British markets, respectively. The corresponding product-level price indices satisfy

$$\log P_{iz} = \int_0^1 \log P_{iz}(\psi_z)\, d\psi_z, \qquad \log P_{UK,z} = \int_0^1 \log P_{UK,z}(\psi_z)\, d\psi_z.$$

Within each submarket, the local and British varieties are CES substitutes with elasticity $\sigma > 1$. The submarket price indices are

$$P_{iz}(\psi_z) = \left[p_{iz}^i(\psi_z)^{1-\sigma} + p_{iz}^{UK}(\psi_z)^{1-\sigma}\right]^{\frac{1}{1-\sigma}},$$

$$P_{UK,z}(\psi_z) = \left[p_{UK,z}^i(\psi_z)^{1-\sigma} + p_{UK,z}^{UK}(\psi_z)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}.$$

Here $p_{dz}^o(\psi_z)$ denotes the price charged by the firm from location o in destination d for submarket ψ_z of product z, where $o, d \in \{i, UK\}$. Prices include any import tariff.

Aggregate price indices are

$$P_i = \prod_z P_{iz}^{\chi_{iz}}, \qquad P_{UK} = \prod_z P_{UK,z}^{\chi_{UK,z}}.$$

Submarket Firm Problem. Let E_{i,ψ_z} and E_{UK,ψ_z} denote expenditure in submarket ψ_z in the local and British markets, respectively. The local firm chooses whether to innovate:

$$I_{iz}(\psi_z) \in \{0, 1\}.$$

Its production marginal costs without and with innovation are

$$c_{iz}(0) = \frac{w_i}{T_z^{US}}, \qquad c_{iz}(1) = \frac{w_i}{\alpha T_z^{US}}, \qquad \alpha > 1.$$

The British firm's production marginal cost is

$$c_{UK,z} = \frac{w_{UK}}{T_z^{UK}},$$

where w_i and w_{UK} are the local and British wages.

Domestic market. The local and British firms' market shares in county i's market are

$$\lambda_{iz}^{i}(\psi_z) = \left[\frac{p_{iz}^{i}(\psi_z)}{P_{iz}(\psi_z)}\right]^{1-\sigma}, \qquad \lambda_{iz}^{UK}(\psi_z) = 1 - \lambda_{iz}^{i}(\psi_z).$$

Their residual demand elasticities are

$$\varepsilon_{iz}^{i}(\psi_z) = \sigma - (\sigma-1)\lambda_{iz}^{i}(\psi_z),$$
$$\varepsilon_{iz}^{UK}(\psi_z) = \sigma - (\sigma-1)\lambda_{iz}^{UK}(\psi_z).$$

Optimal prices satisfy

$$\begin{aligned} p_{iz}^{i}(\psi_z) &= \frac{\varepsilon_{iz}^{i}(\psi_z)}{\varepsilon_{iz}^{i}(\psi_z)-1} c_{iz}\big(I_{iz}(\psi_z)\big), \\ p_{iz}^{UK}(\psi_z) &= \frac{\varepsilon_{iz}^{UK}(\psi_z)}{\varepsilon_{iz}^{UK}(\psi_z)-1}\left[(1+t_z)c_{UK,z} + \tau_{iz}\right]. \end{aligned} \tag{20}$$

Here t_z is the US import tariff and τ_{iz} is the transportation-cost.

British market. The local and British firms' market shares in the UK are

$$\lambda_{UK,z}^{i}(\psi_z) = \left[\frac{p_{UK,z}^{i}(\psi_z)}{P_{UK,z}(\psi_z)}\right]^{1-\sigma}, \qquad \lambda_{UK,z}^{UK}(\psi_z) = 1 - \lambda_{UK,z}^{i}(\psi_z).$$

Their residual demand elasticities are

$$\varepsilon_{UK,z}^{i}(\psi_z) = \sigma - (\sigma-1)\lambda_{UK,z}^{i}(\psi_z),$$
$$\varepsilon_{UK,z}^{UK}(\psi_z) = \sigma - (\sigma-1)\lambda_{UK,z}^{UK}(\psi_z).$$

For simplicity, I assume the UK imposes no tariff. Let τ_{iz}^{X} denote the local firm's monetary transportation cost per unit exported to the UK. Optimal prices satisfy

$$\begin{aligned} p_{UK,z}^{i}(\psi_z) &= \frac{\varepsilon_{UK,z}^{i}(\psi_z)}{\varepsilon_{UK,z}^{i}(\psi_z)-1}\left[c_{iz}\big(I_{iz}(\psi_z)\big) + \tau_{iz}^{X}\right], \\ p_{UK,z}^{UK}(\psi_z) &= \frac{\varepsilon_{UK,z}^{UK}(\psi_z)}{\varepsilon_{UK,z}^{UK}(\psi_z)-1} c_{UK,z}. \end{aligned} \tag{21}$$

Operating profits and innovation. The local firm's operating profits from domestic and British sales are, respectively,

$$\Pi_{iz}^{i}(\psi_z) = \frac{E_{i,\psi_z}\lambda_{iz}^{i}(\psi_z)}{\varepsilon_{iz}^{i}(\psi_z)}, \qquad \Pi_{UK,z}^{i}(\psi_z) = \frac{E_{UK,\psi_z}\lambda_{UK,z}^{i}(\psi_z)}{\varepsilon_{UK,z}^{i}(\psi_z)}.$$

The British firm's operating profits from US and British sales are

$$\Pi_{iz}^{UK}(\psi_z) = \frac{E_{i,\psi_z}\lambda_{iz}^{UK}(\psi_z)}{(1+t_z)\varepsilon_{iz}^{UK}(\psi_z)}, \qquad \Pi_{UK,z}^{UK}(\psi_z) = \frac{E_{UK,\psi_z}\lambda_{UK,z}^{UK}(\psi_z)}{\varepsilon_{UK,z}^{UK}(\psi_z)}.$$

Total operating profits are therefore

$$\Pi_{iz}(\psi_z) = \Pi_{iz}^{i}(\psi_z) + \Pi_{UK,z}^{i}(\psi_z),$$

$$\Pi_{UK,z}(\psi_z) = \Pi_{iz}^{UK}(\psi_z) + \Pi_{UK,z}^{UK}(\psi_z).$$

Innovation lowers the local firm's production cost in both markets. The gain from innovation is

$$G_{iz}(\psi_z) = \Pi_{iz}(\psi_z; I=1) - \Pi_{iz}(\psi_z; I=0).$$

The local firm innovates if this gain is at least as large as the fixed cost:

$$I_{iz}(\psi_z) = \mathbf{1}\{G_{iz}(\psi_z) \geq F\}. \tag{22}$$

Aggregation Across Submarkets. Within each product, submarkets share common technological and innovation primitives and face identical market conditions in each destination. I maintain the main-text restrictions on innovation incentives and restrict attention to pure symmetric equilibria:

$$I_{iz}(\psi_z) = I_{iz} \qquad \forall \psi_z \in [0,1]. \tag{23}$$

Given common primitives and this common innovation choice, firms face the same pricing problem across submarkets in each destination. Equilibrium prices and market shares are therefore constant across submarkets, allowing sales to aggregate as follows:

$$\begin{aligned} Y_{iz} &= \int_0^1 \left[\lambda_{iz}^{i} E_{i,\psi_z} + \lambda_{UK,z}^{i} E_{UK,\psi_z}\right] d\psi_z \\ &= \lambda_{iz}^{i} E_{iz} + \lambda_{UK,z}^{i} E_{UK,z}. \end{aligned}$$

British firms' sales, net of US tariffs, aggregate as

$$\begin{aligned} Y_{UK,z} &= \int_0^1 \left[\frac{\lambda_{iz}^{UK} E_{i,\psi_z}}{1+t_z} + \lambda_{UK,z}^{UK} E_{UK,\psi_z}\right] d\psi_z \\ &= \frac{\lambda_{iz}^{UK} E_{iz}}{1+t_z} + \lambda_{UK,z}^{UK} E_{UK,z}. \end{aligned}$$

Operating profits aggregate as

$$\Pi_{iz} = \int_0^1 \Pi_{iz}(\psi_z)\, d\psi_z, \qquad \Pi_{UK,z} = \int_0^1 \Pi_{UK,z}(\psi_z)\, d\psi_z,$$

and

$$\Pi_{iz}^X = \int_0^1 \Pi_{iz}^X(\psi_z)\, d\psi_z.$$

Appendix D.4.2 General Equilibrium

First, goods-market clearing requires

$$\begin{aligned} Y_{iz} &= E_{iz}\lambda_{iz}^i + E_{UK,z}\lambda_{UK,z}^i, \\ Y_{UK,z} &= \frac{E_{iz}\lambda_{iz}^{UK}}{1+t_z} + E_{UK,z}\lambda_{UK,z}^{UK}. \end{aligned} \tag{24}$$

Second, aggregate income equals labor income and operating profits net of innovation costs:

$$\begin{aligned} Y_i &= \sum_z Y_{iz} = w_i L_i + \Pi_i - F\sum_z I_{iz}, \\ Y_{UK} &= \sum_z Y_{UK,z} = w_{UK} L_{UK} + \Pi_{UK}. \end{aligned} \tag{25}$$

Here L_i and L_{UK} are labor endowments, and aggregate operating profits are

$$\Pi_i = \sum_z \Pi_{iz}, \qquad \Pi_{UK} = \sum_z \Pi_{UK,z}.$$

Third, aggregate expenditure satisfies

$$E_i = Y_i + TR_i, \qquad E_{UK} = Y_{UK}, \tag{26}$$

where tariff revenue is rebated to the local household:

$$TR_i = \sum_z \frac{t_z}{1+t_z} E_{iz}\lambda_{iz}^{UK}. \tag{27}$$

These conditions imply balanced trade at producer prices:

$$\sum_z E_{UK,z}\lambda_{UK,z}^i = \sum_z \frac{E_{iz}\lambda_{iz}^{UK}}{1+t_z}.$$

Given primitives

$$\left\{\sigma, \alpha, F, L_i, L_{UK}, \left\{T_z^{US}, T_z^{UK}, \chi_{iz}, \chi_{UK,z}, t_z, \tau_{iz}, \tau_{iz}^X\right\}_z\right\},$$

an equilibrium is a vector of prices, innovation choices, wages, and expenditures

$$\left\{\left\{p_{iz}^i, p_{iz}^{UK}, p_{UK,z}^i, p_{UK,z}^{UK}, I_{iz}\right\}_z, w_i, w_{UK}, E_i, E_{UK}\right\}$$

such that expenditure allocations satisfy Equation (19), firms set prices according to Equations (20) and (21), innovation choices satisfy Equation (22), submarkets satisfy the symme-

try restriction in Equation (23), and the equilibrium conditions in Equations (24), (25), (26), and (27) hold. Market shares, elasticities, price indices, sales, and profits are determined by the definitions above.

Appendix D.5 Welfare Appendix

Appendix D.5.1 Numerical Welfare Examples

Consider a one-product economy with one US county and the UK, balanced trade, and no tariffs. Let $w_i = 1$ be the numeraire and take

$$\sigma = 8, \qquad \alpha = 1.2, \qquad F = 0.052, \qquad L_i = 1, \qquad L_{UK} = 2, \qquad T^{US} = 1.$$

Import transportation costs fall from $\tau_i = 0.3$ to $\tau_i = 0$, while export transportation costs remain fixed at $\tau_i^X = 0.5$. The welfare change is

$$\Delta \log W_i = -\frac{1}{\sigma - 1} \Delta \log \lambda_i + \Delta I_i \log \alpha + \Delta \log \left(L_i - F I_i + \Pi_i^X \right).$$

Close to frontier. Let $D = T^{UK}/T^{US} = 1.2$. The innovation gains satisfy

$$G_i^{0,\text{pre}} \approx 0.102702 > G_i^{1,\text{pre}} \approx 0.081841 > F,$$
$$G_i^{0,\text{post}} \approx 0.084101 > G_i^{1,\text{post}} \approx 0.058276 > F.$$

Hence, the unique pure symmetric equilibrium features continued innovation:

$$I_i^{\text{pre}} = I_i^{\text{post}} = 1.$$

Therefore,

$$\begin{aligned} \Delta \log W_i &= -\frac{1}{7} \log \left(\frac{0.455261}{0.609021} \right) + \log \left(\frac{1 - 0.052 + 0.096453}{1 - 0.052 + 0.069955} \right) \\ &\approx 0.067267 > 0. \end{aligned}$$

Welfare rises from 1.147320 to 1.227151, a gain of 6.9581%.

Farther from frontier. Now let $D = 2.5$, holding all other parameters fixed. The innovation gains satisfy

$$G_i^{0,\text{pre}} \approx 0.074390 > G_i^{1,\text{pre}} \approx 0.055761 > F,$$

whereas after liberalization,

$$F > G_i^{0,\text{post}} \approx 0.051390 > G_i^{1,\text{post}} \approx 0.036389.$$

Thus, the unique pure symmetric equilibrium switches from innovation to no innovation:

$$I_i^{\text{pre}} = 1, \qquad I_i^{\text{post}} = 0.$$

The ACR term is

$$-\frac{1}{7}\log\left(\frac{0.254128}{0.479060}\right) \approx 0.090570.$$

The innovation effect is

$$-\log(1.2) + \log\left(\frac{1+0.123993}{1-0.052+0.083042}\right) \approx -0.096004.$$

Therefore,

$$\Delta \log W_i \approx 0.090570 - 0.096004 = -0.005434 < 0.$$

Welfare falls from 1.202608 to 1.196090, a loss of 0.5420%. Holding innovation fixed at $I_i = 1$, the same liberalization would instead raise welfare by 10.5141%.

Appendix D.5.2 Derivation of One-Product Welfare Formula

Welfare is

$$W_i = \frac{E_i}{P_i}.$$

Aggregate income is

$$Y_i = w_i L_i + \Pi_i - F I_i,$$

where Π_i is total operating profit from domestic and UK sales.

We set w_i as the numeraire, so $w_i = 1$. In addition, L_i is fixed.

Let λ_i denote the share of expenditure spent on the domestic variety. CES demand implies

$$\lambda_i = \left(\frac{p_i}{P_i}\right)^{1-\sigma}.$$

Taking logs and rearranging,

$$-\log P_i = -\log p_i + \frac{1}{1-\sigma}\log \lambda_i.$$

Domestic Bertrand pricing implies

$$p_i = \mu_i c_i(I_i),$$

where

$$\mu_i = \frac{\varepsilon_i}{\varepsilon_i - 1}, \qquad \varepsilon_i = \sigma - (\sigma - 1)\lambda_i,$$

and

$$c_i(I_i) = \frac{1}{\alpha^{I_i} T^{US}}.$$

Therefore,

$$-\log P_i = -\log \mu_i + I_i \log \alpha + \log T^{US} + \frac{1}{1-\sigma}\log \lambda_i.$$

Total operating profits are

$$\Pi_i = \frac{\lambda_i E_i}{\varepsilon_i} + \Pi_i^X,$$

where Π_i^X denotes operating profits from UK sales.

Under balanced trade and zero tariffs, $E_i = Y_i = L_i + \Pi_i - FI_i$, so

$$E_i = L_i - FI_i + \Pi_i^X + \frac{\lambda_i E_i}{\varepsilon_i}.$$

Rearranging,

$$E_i = \frac{L_i - FI_i + \Pi_i^X}{1 - \frac{\lambda_i}{\varepsilon_i}}.$$

Combining terms,

$$\log W_i = \log(L_i - FI_i + \Pi_i^X) - \log\left(1 - \frac{\lambda_i}{\varepsilon_i}\right) - \log \mu_i + I_i \log \alpha + \frac{1}{1-\sigma} \log \lambda_i + \log T^{US}.$$

The domestic markup and domestic-profit-income terms cancel because

$$\mu_i \left(1 - \frac{\lambda_i}{\varepsilon_i}\right) = \frac{\varepsilon_i}{\varepsilon_i - 1}\left(1 - \frac{\lambda_i}{\varepsilon_i}\right) = \frac{\varepsilon_i - \lambda_i}{\varepsilon_i - 1}.$$

Using $\varepsilon_i = \sigma - (\sigma - 1)\lambda_i$,

$$\varepsilon_i - \lambda_i = \sigma(1 - \lambda_i), \qquad \varepsilon_i - 1 = (\sigma - 1)(1 - \lambda_i).$$

Thus

$$\mu_i \left(1 - \frac{\lambda_i}{\varepsilon_i}\right) = \frac{\sigma}{\sigma - 1},$$

which is constant. Since T^{US} is fixed, differencing the level expression between two equilibria gives the exact welfare change

$$\Delta \log W_i = \frac{1}{1-\sigma} \Delta \log \lambda_i + \Delta I_i \log \alpha + \Delta \log(L_i - FI_i + \Pi_i^X).$$

Appendix D.5.3 Derivation of Multiproduct Welfare Formula

With Cobb–Douglas preferences across products,

$$P_i = \prod_z P_{iz}^{\chi_{iz}},$$

so

$$\Delta \log P_i = \sum_z \chi_{iz} \Delta \log P_{iz}.$$

Within each product,

$$\lambda_{iz} = \left(\frac{p_{iz}}{P_{iz}}\right)^{1-\sigma}.$$

Therefore,

$$\Delta \log P_{iz} = \Delta \log p_{iz} - \frac{1}{1-\sigma}\Delta \log \lambda_{iz}.$$

We set w_i as the numeraire, so $w_i = 1$. Domestic Bertrand pricing gives

$$p_{iz} = \mu_{iz} c_{iz}(I_{iz}), \qquad c_{iz}(I_{iz}) = \frac{1}{\alpha^{I_{iz}} T_z^{US}}.$$

Since T_z^{US} is fixed,

$$\Delta \log p_{iz} = \Delta \log \mu_{iz} - \Delta I_{iz} \log \alpha.$$

Thus

$$\Delta \log P_{iz} = \Delta \log \mu_{iz} - \Delta I_{iz} \log \alpha - \frac{1}{1-\sigma}\Delta \log \lambda_{iz}.$$

Welfare is $W_i = E_i/P_i$, so

$$\Delta \log W_i = \Delta \log E_i - \Delta \log P_i.$$

Aggregate income is

$$Y_i = L_i + \sum_z \Pi_{iz} - F \sum_z I_{iz},$$

where Π_{iz} denotes total operating profits from domestic and UK sales.

Under balanced trade and zero tariffs, $E_i = Y_i$. Separating domestic and export profits, we can write

$$E_i = \left(L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}^X\right)\left(1 + \frac{\sum_z (\Pi_{iz} - \Pi_{iz}^X)}{L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}^X}\right),$$

so

$$\begin{aligned}\Delta \log E_i = {} & \Delta \log \left(L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}^X\right) \\ & + \Delta \log \left(1 + \frac{\sum_z (\Pi_{iz} - \Pi_{iz}^X)}{L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}^X}\right).\end{aligned}$$

Substituting the price-index expression gives

$$\Delta \log W_i = \underbrace{\sum_z \chi_{iz} \frac{1}{1-\sigma} \Delta \log \lambda_{iz}}_{\text{ACR term}} + \underbrace{\left[\sum_z \chi_{iz} \Delta I_{iz} \log \alpha + \Delta \log \left(L_i - F \sum_z I_{iz} + \sum_z \Pi_{iz}^X \right) \right]}_{\text{innovation effect}}$$
$$\underbrace{- \sum_z \chi_{iz} \Delta \log \mu_{iz}}_{\text{procompetitive effect}} + \underbrace{\Delta \log \left(1 + \frac{\sum_z (\Pi_{iz} - \Pi_{iz}^X)}{L_i - F \sum_z I_{iz} + \sum_z \Pi_{iz}^X} \right)}_{\text{rent effect}}.$$

Appendix D.5.4 Tariff-Revenue Extension

Single Product Case:

Tariff revenue enters county i's expenditure and can change following shocks to transportation costs or tariffs. We set w_i as the numeraire, so $w_i = 1$. Revenue is collected nationally and rebated to county i in share ρ_i:

$$TR = \sum_j \sum_z \frac{t_z}{1+t_z} (1 - \lambda_{jz}) E_{jz}, \qquad TR_i = \rho_i TR.$$

The county budget constraint in Appendix D.5.2 becomes

$$E_i = L_i - FI_i + \Pi_i^X + \rho_i TR + \frac{\lambda_i E_i}{\varepsilon_i},$$

so

$$E_i = \frac{L_i - FI_i + \Pi_i^X + \rho_i TR}{1 - \frac{\lambda_i}{\varepsilon_i}}.$$

Combining expenditure with the price-index expression gives

$$\log W_i = \log \left(L_i - FI_i + \Pi_i^X + \rho_i TR \right) - \log \left(1 - \frac{\lambda_i}{\varepsilon_i} \right) - \log \mu_i + I_i \log \alpha + \frac{1}{1-\sigma} \log \lambda_i + \log T^{US}.$$

The domestic markup and domestic-profit-income terms cancel exactly as before. Writing

$$L_i - FI_i + \Pi_i^X + \rho_i TR = \left(L_i - FI_i + \Pi_i^X \right) \left(1 + \frac{\rho_i TR}{L_i - FI_i + \Pi_i^X} \right)$$

and differencing between two equilibria, the exact one-product welfare formula becomes

$$\Delta \log W_i = \frac{1}{1-\sigma} \Delta \log \lambda_i + \Delta I_i \log \alpha + \Delta \log \left(L_i - FI_i + \Pi_i^X \right) + \Delta \log \left(1 + \frac{\rho_i TR}{L_i - FI_i + \Pi_i^X} \right).$$

The final term captures the tariff-revenue effect.

Multiproduct Case:

In the multiproduct case, the county budget constraint becomes

$$E_i = L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz} + \rho_i TR,$$

which can be written as

$$E_i = \left(L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}^X\right)\left(1 + \frac{\sum_z(\Pi_{iz} - \Pi_{iz}^X)}{L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}^X}\right) \times \left(1 + \frac{\rho_i TR}{L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}}\right).$$

Thus,

$$\begin{aligned}\Delta \log E_i = {} & \Delta \log \left(L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}^X\right) \\ & + \Delta \log \left(1 + \frac{\sum_z(\Pi_{iz} - \Pi_{iz}^X)}{L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}^X}\right) \\ & + \Delta \log \left(1 + \frac{\rho_i TR}{L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}}\right).\end{aligned}$$

Accounting for tariff revenue therefore adds the following term to the multiproduct welfare formula:

$$\Delta \log \left(1 + \frac{\rho_i TR}{L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}}\right).$$

The resulting expression is

$$\begin{aligned}\Delta \log W_i = {} & \underbrace{\sum_z \chi_{iz} \frac{1}{1-\sigma} \Delta \log \lambda_{iz}}_{\text{ACR term}} + \underbrace{\left[\sum_z \chi_{iz} \Delta I_{iz} \log \alpha + \Delta \log \left(L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}^X\right)\right]}_{\text{innovation effect}} \\ & \underbrace{- \sum_z \chi_{iz} \Delta \log \mu_{iz}}_{\text{procompetitive effect}} + \underbrace{\Delta \log \left(1 + \frac{\sum_z(\Pi_{iz} - \Pi_{iz}^X)}{L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}^X}\right)}_{\text{rent effect}} \\ & + \underbrace{\Delta \log \left(1 + \frac{\rho_i TR}{L_i - F\sum_z I_{iz} + \sum_z \Pi_{iz}}\right)}_{\text{tariff-revenue effect}},\end{aligned}$$

where TR is defined above.

Appendix D.6 Monopolistic Competition and Kimball Demand Model

The model in Section 3 assumes a CES demand structure with Bertrand oligopoly. We can instead generate a declining residual demand elasticity by departing from CES demand. In this section I show that a model of monopolistic competition with Kimball demand, in

the parameterization of Klenow and Willis (2016), delivers the same comparative statics. Here the declining residual demand elasticity arises from demand saturation rather than from strategic interaction between large firms.

Appendix D.6.1 Environment

Fix county i and product z. Expenditure is $E_{iz} = \chi_{iz} E_i > 0$, where preferences across products are Cobb–Douglas and $\sum_z \chi_{iz} = 1$. Expenditure and wages are held fixed in partial equilibrium. Normalize US and UK wages to one.

The market contains fixed continua of US and UK firms,

$$\Psi^{US} = [0, N_{US}], \qquad \Psi^{UK} = (N_{US}, 1], \qquad N_{US} + N_{UK} = 1, \quad N_{US}, N_{UK} > 0.$$

Each firm supplies one horizontally differentiated variety. Preferences treat variety labels symmetrically, without an additional country-of-origin term. Firms are homogeneous within country before innovation. There is no entry, exit, or fixed operating cost, but firms may optimally sell zero. Domestic firms serve their local market; British pricing is specific to the destination market. Competition is monopolistic competition.

Technology and protection. Initial technologies are $T_z^{US} > 0$ and $T_z^{UK} > 0$, with

$$D_z := \frac{T_z^{UK}}{T_z^{US}}, \qquad c_z^{UK} = \frac{1}{T_z^{UK}}.$$

A US firm ψ chooses $I_{iz}(\psi) \in \{0, 1\}$. Innovation costs $F > 0$ and raises productivity by a factor $\alpha > 1$, so

$$c_{iz}^{US}(\psi) = \frac{1}{T_z^{US} \alpha^{I_{iz}(\psi)}}, \qquad c_{iz}^{US,a} = \frac{1}{T_z^{US} \alpha^a}, \quad a \in \{0, 1\}.$$

British technology is fixed. Let t_z be the ad valorem tariff and $\tau_{iz} \geq 0$ the per-unit freight cost. As in the baseline model,

$$\begin{aligned} TP_{iz} &:= t_z + \frac{\tau_{iz}}{c_z^{UK}}, \\ \widetilde{c}_{iz}^{UK} &:= (1 + t_z) c_z^{UK} + \tau_{iz} = (1 + TP_{iz}) c_z^{UK}. \end{aligned}$$

The delivered foreign cost relative to a non-innovating US firm's cost is

$$R_{iz} := \frac{\widetilde{c}_{iz}^{UK}}{c_{iz}^{US,0}} = \frac{1 + TP_{iz}}{D_z}.$$

Protection raises R_{iz}, while greater distance to the frontier lowers it. The distance comparisons focus on $D_z \geq 1$. Comparative statics in D_z and TP_{iz} hold other primitives fixed; one can vary D_z by varying T_z^{US} while holding British technology fixed.

Timing. First, US firms simultaneously choose innovation. Second, all firms choose prices, observing the resulting technology distribution. Consumers then purchase varieties. Firms are atomistic and take aggregate demand conditions as given in both their pricing and their individual innovation deviations.

Demand. Let Q_{iz} denote the product consumption composite, reserving Y_{iz} for domestic output measured as revenue. For positive composite consumption, the Kimball aggregator satisfies

$$\int_0^1 \Upsilon\left(\frac{q_{iz}(\psi)}{Q_{iz}}\right) d\psi = 1, \qquad x_{iz}(\psi) := \frac{q_{iz}(\psi)}{Q_{iz}}.$$

Within products, preferences follow the Klenow–Willis specification

$$f(x) := \Upsilon'(x) = \frac{\sigma-1}{\sigma}\exp\left(\frac{1-x^{\theta/\sigma}}{\theta}\right), \quad \sigma > 1, \quad \theta > 0,$$
$$\Upsilon(x) = 1 + \int_1^x f(u)\, du.$$

Thus $\Upsilon(1) = 1$, $f > 0$, and $\Upsilon'' < 0$ for $x > 0$. In general $\Upsilon(0) \neq 0$; this term is retained in the aggregation conditions below.

Write $p_{iz}(\psi)$ for a consumer's delivered price and define

$$P_{iz} := \frac{E_{iz}}{Q_{iz}}, \qquad \Lambda_{iz} := \frac{P_{iz}}{\Omega_{iz}},$$
$$\Omega_{iz} := \int_0^1 f(x_{iz}(\psi)) x_{iz}(\psi)\, d\psi.$$

For a consumed variety, expenditure minimization gives

$$p_{iz}(\psi) = \Lambda_{iz} f(x_{iz}(\psi)), \qquad q_{iz}(\psi) = Q_{iz} x_{iz}(\psi).$$

Demand is zero at a price at or above the choke price $\Lambda_{iz} f(0)$. An individual firm treats Λ_{iz} and Q_{iz} as given; equilibrium determines both.

The perceived demand elasticity and optimal markup for an active firm are

$$\varepsilon_{iz}(\psi) = -\frac{f(x_{iz}(\psi))}{x_{iz}(\psi) f'(x_{iz}(\psi))} = \sigma[x_{iz}(\psi)]^{-\theta/\sigma},$$
$$\mu_{iz}(\psi) = \frac{\varepsilon_{iz}(\psi)}{\varepsilon_{iz}(\psi) - 1}.$$

The elasticity decreases with relative quantity. Define

$$\widehat{x} := \sigma^{\sigma/\theta}, \qquad \varepsilon(\widehat{x}) = 1, \qquad f(0) = \frac{\sigma-1}{\sigma} e^{1/\theta} < \infty.$$

Appendix D.6.2 Pricing and Innovation

Pricing. For an effective marginal cost c, the pricing problem is

$$Q_{iz} \max_{x \geq 0} \{\Lambda_{iz} f(x) - c\} x.$$

The effective cost is $c_{iz}^{US,a}$ for a US firm and $\widetilde{c}_{iz}^{UK}$ for a British firm.[46]

Define

$$g(x) := f(x) \left(1 - \frac{1}{\varepsilon(x)}\right).$$

For $c < \Lambda_{iz} f(0)$, the unique optimal relative quantity lies in $(0, \widehat{x})$ and solves

$$c = \Lambda_{iz} g(x), \qquad p = \frac{\varepsilon(x)}{\varepsilon(x) - 1} c.$$

For $c \geq \Lambda_{iz} f(0)$, optimal sales are zero. Posted prices of inactive varieties are immaterial to allocations.

Aggregation conditional on innovation. Let

$$s_{iz} := \frac{1}{N_{US}} \int_{\Psi^{US}} I_{iz}(\psi)\, d\psi.$$

Use group labels $a \in \{0, 1, K\}$ for US non-innovators, US innovators, and British firms, respectively. Their masses are

$$m_{iz}^0 = N_{US}(1 - s_{iz}), \qquad m_{iz}^1 = N_{US} s_{iz}, \qquad m_{iz}^K = N_{UK},$$

and their effective costs are $c_{iz}^0 = c_{iz}^{US,0}$, $c_{iz}^1 = c_{iz}^{US,1}$, and $c_{iz}^K = \widetilde{c}_{iz}^{UK}$. Conditional on s_{iz}, let x_{iz}^a be each type's optimal relative quantity. The aggregate demand shifter solves

$$\sum_{a \in \{0,1,K\}} m_{iz}^a \Upsilon(x_{iz}^a) = 1.$$

This equation determines a unique Λ_{iz}. To see this, each type's optimal relative quantity is continuous and nondecreasing in Λ_{iz}, and strictly increasing whenever it is positive. The left-hand side tends to $\Upsilon(0) < 1$ as $\Lambda_{iz} \downarrow 0$ and to $\Upsilon(\widehat{x}) > 1$ as $\Lambda_{iz} \to \infty$; it is strictly increasing once a type with positive mass is active. Active prices, quantities, and operating profits are consequently unique conditional on s_{iz}. Writing $h(x) := x f(x)$, the remaining aggregates are

$$\Omega_{iz} = \sum_a m_{iz}^a h(x_{iz}^a), \qquad Q_{iz} = \frac{E_{iz}}{\Lambda_{iz} \Omega_{iz}}.$$

[46] If a British firm's factory-gate price is $p_{iz}^{UK,f}(\psi)$, its delivered price is $p_{iz}^{UK}(\psi) = (1 + t_z) p_{iz}^{UK,f}(\psi) + \tau_{iz}$. Its actual operating profit is therefore $[p_{iz}^{UK}(\psi) - \widetilde{c}_{iz}^{UK}] q_{iz}(\psi)/(1 + t_z)$. The factor $1/(1 + t_z)$ does not change the pricing choice.

Types with zero mass are still assigned their hypothetical optimal quantities when computing the return to an individual innovation deviation.

Innovation. For $a \in \{0, 1\}$, a US firm's operating profit is

$$\Pi_{iz}^{op,a}(s_{iz}, D_z, TP_{iz}) = \frac{E_{iz}}{\Omega_{iz}} \frac{h(x_{iz}^a)}{\varepsilon(x_{iz}^a)}.$$

The right-hand side is defined as zero when $x_{iz}^a = 0$. Its private gain from innovation is

$$\begin{aligned} G_{iz}(s_{iz}, D_z, TP_{iz}) := \Pi_{iz}^{op,1}(s_{iz}, D_z, TP_{iz}) \\ - \Pi_{iz}^{op,0}(s_{iz}, D_z, TP_{iz}). \end{aligned}$$

The two profits are evaluated at the same aggregate demand conditions: a single atomistic firm's deviation does not change s_{iz} or the aggregates. The firm innovates if $G_{iz} > F$, does not innovate if $G_{iz} < F$, and either choice is optimal at equality.

An equilibrium innovation share satisfies

$$\begin{cases} G_{iz}(0, D_z, TP_{iz}) \leq F, & s_{iz}^* = 0, \\ G_{iz}(s_{iz}^*, D_z, TP_{iz}) = F, & 0 < s_{iz}^* < 1, \\ G_{iz}(1, D_z, TP_{iz}) \geq F, & s_{iz}^* = 1. \end{cases}$$

An interior equilibrium permits some otherwise identical firms to innovate and others not to innovate. It does not impose a common binary adoption decision on all US firms.

Appendix D.6.3 Equilibrium and Assumptions

Assumption A1: British varieties can supply the market on the elastic portion of demand. Assume

$$\boxed{N_{UK}\Upsilon(\widehat{x}) + N_{US}\Upsilon(0) > 1.}$$

Then there is a unique $\overline{x}^{UK} < \widehat{x}$ satisfying

$$N_{UK}\Upsilon(\overline{x}^{UK}) + N_{US}\Upsilon(0) = 1.$$

Define

$$\overline{\kappa}^{UK} := g(\overline{x}^{UK}) > 0.$$

Thus British varieties alone can supply the composite while British demand remains strictly elastic. British markups remain bounded as US productivity becomes arbitrarily low.

Assumption A2: strategic substitutes in aggregate innovation. For each (D_z, TP_{iz}) in the domain of the comparison, $G_{iz}(s, D_z, TP_{iz})$ is nonincreasing in $s \in [0, 1]$ and strictly

decreasing wherever it is positive. Where differentiable, a sufficient formulation is

$$\frac{\partial G_{iz}(s, D_z, TP_{iz})}{\partial s} < 0 \quad \text{whenever } G_{iz}(s, D_z, TP_{iz}) > 0.$$

Zero-gain plateaus are allowed because $F > 0$. This is an additional restriction where needed, not a general implication of Kimball demand at all technology gaps.

Under A2 the aggregate innovation share is unique. Continuity gives existence: either an adoption corner satisfies the equilibrium inequalities or G_{iz} crosses F in the interior. Monotonicity gives at most one crossing because $F > 0$. Individual innovation profiles need not be unique when $0 < s^*_{iz} < 1$: different subsets of identical firms can innovate while generating the same aggregates. Thus the equilibrium is symmetric in prices within each technology type, but a common binary innovation decision is not imposed on all domestic firms.

A concrete sufficient condition for A2 is that US non-innovators cannot produce and

$$c^{US,1}_{iz} \geq \widetilde{c}^{UK}_{iz}, \quad \text{equivalently} \quad D_z \geq \alpha(1 + TP_{iz}).$$

Under A1, a sufficient bound for non-innovators to be inactive at every candidate adoption share is

$$D_z \geq (1 + TP_{iz})\frac{f(0)}{\overline{\kappa}^{UK}}.$$

The proof below establishes that these two bounds imply A2. Thus uniqueness follows directly in this region; uniqueness at smaller distances requires an additional argument or restriction.

Equilibrium definition. Given primitives, an equilibrium comprises an innovation profile $\{I_{iz}(\psi)\}_{\psi \in \Psi^{US}}$, its aggregate share s^*_{iz}, prices, quantities, and $(Q_{iz}, P_{iz}, \Lambda_{iz}, \Omega_{iz})$ such that consumers minimize expenditure, firms price optimally, the aggregation and expenditure identities hold, and each US firm's innovation decision is optimal. A1 and A2 give the version with a unique aggregate equilibrium on the stated domain.

Outcomes. The domestic expenditure share and output measure are

$$\lambda_{iz} = \frac{N_{US}(1 - s^*_{iz})h(x^0_{iz}) + N_{US}s^*_{iz}h(x^1_{iz})}{\Omega_{iz}}, \qquad \boxed{Y_{iz} = \lambda_{iz}E_{iz}}.$$

Aggregate innovation and innovation expenditure are

$$I_{iz} := \int_{\Psi^{US}} I_{iz}(\psi)\, d\psi = N_{US}s^*_{iz}, \qquad FI_{iz}.$$

Here $I_{iz}(\psi)$ is binary, while I_{iz} is the aggregate mass of innovators. The output measure Y_{iz} is domestic revenue, as in Section 3, rather than the sum of physical quantities.[47] Comparisons that allow zero domestic production are stated in levels.

[47] The latter is $N_{US}Q_{iz}[(1 - s^*_{iz})x^0_{iz} + s^*_{iz}x^1_{iz}]$.

Appendix D.6.4 Comparative Statics

Fix a county-product market (i, z) and two protection levels $TP_{iz}^1 > TP_{iz}^0 \geq 0$. Hold $E_{iz}, F, \alpha, N_{US}, N_{UK}, \sigma, \theta$ fixed. Write

$$I_{iz}^*(D_z, TP_{iz}) = N_{US} s_{iz}^*(D_z, TP_{iz}),$$
$$Y_{iz}^*(D_z, TP_{iz}) = E_{iz} \lambda_{iz}^*(D_z, TP_{iz}).$$

Under A1, use the positive constant $\overline{\kappa}^{UK}$ defined above to construct the finite, sufficient distance bound

$$\boxed{\widetilde{D}_{iz} := (1 + TP_{iz}^1) \frac{f(0)}{\overline{\kappa}^{UK}}.} \tag{28}$$

For the unique-equilibrium statements below, assume A2 for every candidate adoption share, every $D_z \geq \widetilde{D}_{iz}$, and every protection level in $[TP_{iz}^0, TP_{iz}^1]$. Requiring A2 at smaller distances is unnecessary for these threshold statements. Continuity of G gives existence through an adoption corner or an interior root; A2 makes the aggregate equilibrium unique because $F > 0$.

Proposition K1 (innovation). For every $D_z \geq \widetilde{D}_{iz}$,

$$\boxed{I_{iz}^*(D_z, TP_{iz}^1) \geq I_{iz}^*(D_z, TP_{iz}^0).} \tag{29}$$

Thus one admissible innovation threshold is $D_{iz}^{I,*} = \widetilde{D}_{iz}$. Innovation can decrease below this sufficient threshold, including in comparisons with a unique aggregate equilibrium, as established by the construction below.

Proposition K2 (domestic output). For every $D_z \geq \widetilde{D}_{iz}$,

$$\boxed{Y_{iz}^*(D_z, TP_{iz}^1) \geq Y_{iz}^*(D_z, TP_{iz}^0).} \tag{30}$$

Thus one admissible output threshold is $D_{iz}^{Y,*} = \widetilde{D}_{iz}$. Output can decrease below this bound, as the same construction below establishes. The common bound is sufficient and need not be the smallest threshold for either outcome. When both domestic outputs are positive, (30) also implies $\Delta \log Y_{iz}^* \geq 0$. Stating the result in levels accommodates shutdown. If $TP_{iz}^0 > 0$, dividing the inequalities by $\Delta \log TP_{iz} > 0$ gives the corresponding signs in the notation of Section 3. Neither proposition asserts a unique sign crossing or a negative response at every distance below the bound.

The mechanism in the far-distance region is that non-innovators cannot profitably serve the market. The return to innovation is therefore the innovator's entire operating profit, which increases with protection. Strategic substitutes translate this increase in the private return into a weak increase in aggregate adoption. Both the direct substitution effect and the induced adoption response then increase domestic revenue.

Appendix D.6.5 Proofs of Propositions K1 and K2

Suppress (i, z), and write $n = N_{US}$, $k = N_{UK}$ and $R = (1 + TP)/D$. Define the normalized demand shifter $\ell = \Lambda/c^{US,0}$. Normalized effective costs are then $(1, 1/\alpha, R)$; nominal expenditure E is unchanged. Set

$$b = \frac{\theta}{\sigma}, \quad w(x) = 1 - \frac{1}{\varepsilon(x)}, \quad h(x) = xf(x), \quad v(x) = \frac{h(x)}{\varepsilon(x)}, \quad g(x) = f(x)w(x).$$

For an active type, define

$$\begin{aligned} \eta(x) &:= \left.\frac{\partial \log x}{\partial \log \ell}\right|_{\text{normalized cost}} = \frac{\varepsilon(x)(\varepsilon(x) - 1)}{(1 + b)\varepsilon(x) - 1}, \\ \beta(x) &:= \eta(x)w(x) = \frac{(\varepsilon(x) - 1)^2}{(1 + b)\varepsilon(x) - 1}. \end{aligned} \tag{31}$$

In particular,

$$0 < \beta(x) < \varepsilon(x) - 1, \qquad h'(x) = f(x)w(x) > 0, \qquad \frac{\partial \log v(x)}{\partial \log \ell} = \varepsilon(x) - 1. \tag{32}$$

All partial derivatives involving an individual type's quantity treat its normalized cost as fixed, unless its cost change is explicitly included.

Lemma K1: A1 bounds the choke price. For every candidate adoption share, aggregation and monotonicity of Υ imply

$$k\Upsilon(x_K) + n\Upsilon(0) \leq 1.$$

Hence $x_K \leq \overline{x}^{UK} < \widehat{x}$. If British firms are active, $R = \ell g(x_K)$ and the strict decrease of g on its active branch give

$$\ell \leq \frac{R}{\overline{\kappa}^{UK}}. \tag{33}$$

If Britain is inactive, its choke inequality gives $\ell \leq R/f(0) < R/\overline{\kappa}^{UK}$, so (33) still holds.

For $D \geq \widetilde{D}$ and any $TP \in [TP^0, TP^1]$,

$$\ell f(0) \leq \frac{1 + TP}{D} \frac{f(0)}{\overline{\kappa}^{UK}} \leq 1.$$

Thus $x_0 = 0$ at every candidate adoption share throughout the policy comparison: only innovators can produce. If instead $D \geq \alpha\widetilde{D}$, the same calculation gives $\ell f(0) \leq 1/\alpha$, so even innovators cannot produce. In that extreme region $G = 0$, and $F > 0$ implies the unique equilibrium $s^* = I^* = Y^* = 0$.

Lemma K2: protection raises the return to innovation when non-innovators cannot produce. In the region $x_0 = 0$,

$$H := \Omega = ns\, h(x_1) + kh(x_K), \qquad G = \frac{Ev(x_1)}{H}.$$

When both producing types are active, define their revenue shares $\lambda_1 = ns\, h(x_1)/H$ and $\lambda_K = kh(x_K)/H$, and write

$$M = \lambda_1\eta_1 + \lambda_K\eta_K, \qquad A = \lambda_1\beta_1 + \lambda_K\beta_K, \qquad \zeta = \log R.$$

Differentiating aggregation at fixed s gives

$$B_\zeta := \frac{\partial \log \ell}{\partial \zeta} = \frac{\lambda_K\eta_K}{M}. \tag{34}$$

Because British normalized cost changes with R, $\partial \log H/\partial\zeta = AB_\zeta - \lambda_K\beta_K$. Using (32),

$$\begin{aligned} \frac{\partial \log G}{\partial \zeta} &= B_\zeta[\varepsilon_1 - 1 - A + Mw_K] \\ &= B_\zeta[\varepsilon_1 - 1 + \lambda_1\eta_1(w_K - w_1)]. \end{aligned} \tag{35}$$

The expression in brackets is bounded below by $\varepsilon_1 - 1 - \lambda_1\beta_1 > 0$, by (32). Therefore $G_R > 0$ when potential US innovators and British firms are active. The formula also applies at $s = 0$ with the potential innovator assigned zero revenue weight. When potential innovators are inactive, $G = 0$; when Britain is inactive, changes in its cost have no local effect. Continuity at these production boundaries gives $G_R \geq 0$ throughout the only-innovators-can-produce region.

Proof of Proposition K1. Fix $D \geq \widetilde{D}$, and let $R_1 > R_0$ denote the two relative costs. Lemma K1 puts every candidate adoption share in the region covered by Lemma K2, so $G(s, R_1) \geq G(s, R_0)$ for every s.

Write $s_a = s^*(R_a)$ for $a = 0, 1$. Suppose, to obtain a contradiction, that $s_1 < s_0$. Since $s_0 > 0$, equilibrium requires $G(s_0, R_0) \geq F$. Monotonicity in protection and A2 imply

$$G(s_1, R_1) > G(s_0, R_1) \geq G(s_0, R_0) \geq F.$$

But $s_1 < 1$ requires $G(s_1, R_1) \leq F$, a contradiction. Thus $s_1 \geq s_0$, including adoption corners. Multiplication by n proves (29).

Lemma K3: output increases with protection and with adoption when non-innovators cannot produce. Here $Y = E\lambda_1$. At fixed s, (34) implies that an increase in R weakly increases x_1 and weakly decreases x_K. Since $h' > 0$ on the active branch, $Y_R \geq 0$, strictly when both origins produce.

For the adoption effect, let

$$L = \Upsilon(x_1) - \Upsilon(0), \qquad J = \frac{h(x_1)}{L}.$$

Since $dh/d\Upsilon = w$ and w is strictly decreasing,

$$J = \frac{\int_0^{x_1} f(u)w(u)\,du}{\int_0^{x_1} f(u)\,du} > w_1. \tag{36}$$

Differentiation of aggregation and the domestic revenue share gives

$$\begin{aligned} B_s &:= \frac{\partial \log \ell}{\partial s} = -\frac{nL}{HM} < 0, \\ \frac{\partial \lambda_1}{\partial s} &= \frac{\lambda_1 \lambda_K}{s}\left[1 - \frac{\lambda_1(\beta_1 - \beta_K)}{JM}\right]. \end{aligned} \tag{37}$$

The bracket is positive because

$$JM - \lambda_1(\beta_1 - \beta_K) = \lambda_1\eta_1(J - w_1) + \lambda_K\eta_K J + \lambda_1\beta_K > 0.$$

Thus $Y_s > 0$ when $s > 0$ and both origins produce. The cases $s = 0$, zero US production, and zero British production follow by continuity or direct calculation, giving $Y_s \geq 0$ throughout this region.

Proof of Proposition K2. By Proposition K1, $s_1 \geq s_0$. Lemma K3 implies

$$\begin{aligned} Y(s_1, R_1) - Y(s_0, R_0) = &[Y(s_1, R_1) - Y(s_0, R_1)] \\ &+ [Y(s_0, R_1) - Y(s_0, R_0)] \geq 0. \end{aligned}$$

The first term is the innovation effect and the second is the substitution effect. This proves (30). Monotonicity of Y in s here is a property of the only-innovators-can-produce region, not an assumed global property.

Uniqueness from a concrete cost condition. If $x_0 = 0$ and $R \leq 1/\alpha$, then $x_1 \leq x_K$ and $\varepsilon_1 \geq \varepsilon_K$ wherever US innovators can produce. By (32), $A < \varepsilon_1 - 1$. Consequently,

$$\frac{\partial \log G}{\partial s} = (\varepsilon_1 - 1 - A)B_s - \frac{nh(x_1)}{H} < 0 \quad \text{where } G > 0. \tag{38}$$

This establishes A2 throughout that region. Under A1 alone, a sufficient bound for both this condition and $x_0 = 0$ at all policies is

$$D \geq \max\{\widetilde{D}, \alpha(1 + TP^1)\}. \tag{39}$$

Thus A1 alone delivers a sufficiently-far region with a unique aggregate equilibrium and the threshold comparative statics, using this possibly larger distance bound. It does not

ensure uniqueness at every smaller distance.

Appendix D.6.6 Strictly Positive Effects with Active Production

Write $\rho = \overline{\kappa}^{UK}/f(0)$. We have $0 < \rho < 1$. Choose

$$\frac{\rho}{\alpha} < R^\circ < \min\{\rho, 1/\alpha\}. \tag{40}$$

The interval is nonempty because $\alpha > 1$ and $\rho < 1$. At this relative cost, non-innovators are inactive for every s, and innovators are no cheaper than British firms. Therefore (38) gives A2. At $s = 0$, Britain supplies the market alone, $\ell = R^\circ/\overline{\kappa}^{UK}$, and the first inequality in (40) implies that a potential US innovator can produce. Thus $G(0, R^\circ) > 0$ and $G(1, R^\circ) < G(0, R^\circ)$.

Choose the fixed cost in the nonempty interval

$$G(1, R^\circ) < F < G(0, R^\circ). \tag{41}$$

There is a unique interior adoption equilibrium, and both origins produce. Equations (35), (37), and (38) give

$$\frac{ds^*}{dR} = -\frac{G_R}{G_s} > 0, \qquad \frac{dY^*}{dR} = Y_R + Y_s \frac{ds^*}{dR} > 0. \tag{42}$$

All inequalities are strict and persist in a neighborhood. A sufficiently small finite protection increase therefore strictly raises both innovation and output over an open interval of distances, with positive US production under both policies. Because $R^\circ < \rho$, the comparison can be chosen inside $D \geq \widetilde{D}$. This construction does not rely on eventual zero output.

Appendix D.6.7 Negative Effects Below the Threshold

This construction establishes that protection can strictly reduce both innovation and output below the sufficient threshold. A2 holds across all candidate adoption shares throughout the policy interval used in the construction, so both outcomes are evaluated at unique aggregate equilibria.

We first record derivative identities valid when both US technologies and British firms can produce. Let

$$H := \sum_a m_a h(x_a), \qquad \lambda_a := \frac{m_a h(x_a)}{H},$$
$$M := \sum_a \lambda_a \eta_a, \qquad A := \sum_a \lambda_a \beta_a, \qquad L := \Upsilon(x_1) - \Upsilon(x_0).$$

$$J := \frac{h(x_1) - h(x_0)}{L}, \qquad \xi := \frac{h(x_1) - h(x_0)}{v(x_1) - v(x_0)},$$
$$Z := \xi - 1 - A + MJ, \qquad W := \xi - 1 - A + Mw_K.$$

Direct differentiation of aggregation and operating profits gives

$$\begin{aligned} B_s &= -\frac{nL}{HM} < 0, \qquad B_\zeta = \frac{\lambda_K \eta_K}{M} > 0, \\ G_s &= GB_s Z, \qquad G_\zeta = GB_\zeta W. \end{aligned} \tag{43}$$

Along an interior innovation equilibrium with $Z > 0$,

$$\frac{dY^*}{d\zeta} = E\lambda_K \left[\beta_K + \frac{\lambda_K \eta_K (J - w_K)(\xi - 1 - \beta_K)}{Z} \right]. \tag{44}$$

Equation (44) follows by substituting $s_\zeta^* = -B_\zeta W/(B_s Z)$ into $Y_\zeta + Y_s s_\zeta^*$, using $Y_s = E\lambda_K B_s(\xi - 1 - \beta_K - Z)$ and $Y_\zeta = E\lambda_K[(A - \beta_K)B_\zeta + (1 - \lambda_K)\beta_K]$.

Set $\theta = \sigma$ and choose

$$1 < \sigma < \frac{3 + \sqrt{5}}{2}.$$

Choose $n \in (0, 1)$ sufficiently small that A1 holds, and let $k = 1 - n$. Such a choice exists because $\widehat{x} = \sigma > 1$ and $\Upsilon(\sigma) > \Upsilon(1) = 1$. In this family,

$$\eta(x) = \frac{\sigma(\sigma - x)}{x(2\sigma - x)}.$$

This function is strictly decreasing on $(0, \sigma)$ and equals one at

$$x^\dagger := \frac{\sigma(3 - \sqrt{5})}{2} \in (0, 1).$$

Begin at the limit $\alpha = 1$. For any $x_U \in (0, 1)$, let x_K solve

$$n\Upsilon(x_U) + k\Upsilon(x_K) = 1.$$

A1 implies $1 < x_K < \overline{x}^{UK} < \sigma$, so both origins are on the elastic portion of demand. Define

$$R^\circ := \frac{g(x_K)}{g(x_U)}, \qquad \lambda_U := \frac{nh(x_U)}{nh(x_U) + kh(x_K)}, \qquad \lambda_K := 1 - \lambda_U,$$
$$d := w(x_U) - w(x_K) > 0, \qquad C := \lambda_K \eta_K d > 0.$$

At $x_U = x^\dagger$, $\eta_U - 1 = 0$ and $C > 0$. By continuity, choosing $x_U > x^\dagger$ sufficiently close to $x^\dagger$ gives

$$\gamma := \eta_U - 1 < 0, \qquad \gamma + C > 0. \tag{45}$$

Since $x_K > x_U$ and g is decreasing, $R^\circ < 1$. Moreover, $g(x_K) > \overline{\kappa}^{UK}$ and $g(x_U) < f(0)$

imply $R^\circ > \overline{\kappa}^{UK}/f(0)$. Thus, at zero baseline protection,

$$1 < D^\circ := \frac{1}{R^\circ} < \frac{f(0)}{\overline{\kappa}^{UK}},$$

which lies below the sufficient threshold for any positive protection increase from zero.

Now take $\alpha = e^\delta$ with $\delta > 0$ sufficiently small. As $\delta \downarrow 0$, the two domestic types converge to x_U, uniformly over $s \in [0,1]$. Since $dh/d\Upsilon = w$ and $h'/v' = \eta$,

$$J \longrightarrow w_U, \qquad \xi \longrightarrow \eta_U.$$

Consequently,

$$Z \longrightarrow \gamma + C > 0, \qquad W \longrightarrow \gamma - \lambda_U \eta_U d < 0. \tag{46}$$

The pricing system is smooth in a neighborhood of this configuration, with all types strictly active. The limits are therefore uniform over the compact set of candidate shares. For all sufficiently small positive δ, (43) implies $G_s < 0$ and $G_\zeta < 0$ for every s. A2 thus holds throughout a neighborhood of the proposed policy comparison.

Choose $F = G(1/2, R^\circ) > 0$. The unique baseline equilibrium is $s^* = 1/2$. Equations (43) and (46) imply $ds^*/d\zeta < 0$. Moreover, (44) yields

$$\lim_{\delta \downarrow 0} \frac{dY^*}{d\zeta} = E\lambda_K \frac{\gamma(\beta_K + C)}{\gamma + C} < 0. \tag{47}$$

Thus, for sufficiently small positive δ, a sufficiently small finite protection increase strictly reduces both aggregate innovation and output. The equilibrium is unique at both policies and throughout the intervening policy interval, and the strict inequalities persist over an open interval of distances below the sufficient bound. This establishes the possibility clauses in Propositions K1 and K2.

Appendix D.6.8 Equilibrium Selection

Lemmas K1–K3 also imply the threshold results under either a consistently smallest or a consistently largest equilibrium selection. Indeed,

$$s_{\min}(R) = \inf\big(\{s \in [0,1] : G(s,R) \le F\} \cup \{1\}\big),$$
$$s_{\max}(R) = \sup\big(\{s \in [0,1] : G(s,R) \ge F\} \cup \{0\}\big).$$

Since $G(s,R)$ is nondecreasing in R in this region, both selections are nondecreasing in R. Lemma K3 then orders output as well. Switching selection rules between policies is not covered. At $D \ge \alpha\widetilde{D}$, the unique zero-innovation equilibrium needs no selection rule.

Appendix D.7 Continuous Model For Comparative Statics

This appendix presents a continuous-innovation version of the model in 3. Here the US firm chooses a continuous innovation effort rather than making a discrete upgrade decision.

Appendix D.7.1 Setup

A US firm and a UK firm compete in the US market for product z à la Bertrand. US expenditure on product z is E_z, which is fixed in this partial-equilibrium model. Demand is CES with elasticity $\sigma > 1$. If the US and UK prices are p_z^{US} and p_z^{UK}, the US expenditure share is

$$\lambda_z = \frac{(p_z^{US})^{1-\sigma}}{(p_z^{US})^{1-\sigma} + (p_z^{UK})^{1-\sigma}}.$$

Let US and UK technologies be T_z^{US} and T_z^{UK}. Define distance to the frontier as

$$D_z \equiv \frac{T_z^{UK}}{T_z^{US}}.$$

The tariff raises the delivered marginal cost of the UK firm in the US market:

$$c_z^{UK} = \frac{1 + t_z}{T_z^{UK}}.$$

The US firm chooses innovation effort $n_z \geq 0$. Innovation raises US technology from T_z^{US} to

$$T_z^{US} \exp(n_z^{\alpha}), \qquad 1 < \alpha < 2.$$

Thus, the US unit cost is

$$c_z^{US}(n_z) = \frac{1}{T_z^{US} \exp(n_z^{\alpha})}.$$

Define relative US competitiveness as

$$r_z(n_z, t_z) \equiv \frac{c_z^{UK}}{c_z^{US}(n_z)} = \frac{(1 + t_z)\exp(n_z^{\alpha})}{D_z}.$$

Appendix D.7.2 Pricing

Under Bertrand competition with CES demand, the perceived own-price elasticity of the US firm is

$$\varepsilon_z^{US} = \sigma - (\sigma - 1)\lambda_z.$$

The perceived own-price elasticity of the UK firm is

$$\varepsilon_z^{UK} = 1 + (\sigma - 1)\lambda_z.$$

Therefore, optimal prices satisfy

$$p_z^{US} = \frac{\sigma - (\sigma - 1)\lambda_z}{(\sigma - 1)(1 - \lambda_z)} c_z^{US},$$

and

$$p_z^{UK} = \frac{1 + (\sigma - 1)\lambda_z}{(\sigma - 1)\lambda_z} c_z^{UK}.$$

Combining the CES share equation with the Bertrand pricing equations, the US expenditure share $\lambda_z(r_z)$ is implicitly defined by

$$\frac{\lambda_z(r_z)}{1-\lambda_z(r_z)} = \left[\frac{[1+(\sigma-1)\lambda_z(r_z)][1-\lambda_z(r_z)]}{\lambda_z(r_z)[\sigma-(\sigma-1)\lambda_z(r_z)]} r_z\right]^{\sigma-1}.$$

Operating profit is

$$\Pi_z(r_z) = E_z \frac{\lambda_z(r_z)}{\sigma-(\sigma-1)\lambda_z(r_z)}.$$

Appendix D.7.3 Innovation Problem

Innovation effort is costly. Let

$$C_z(n_z) = \kappa_{0z} n_z + \frac{\kappa_{1z}}{2} n_z^2, \qquad \kappa_{0z} > 0, \qquad \kappa_{1z} > 0.$$

The firm chooses innovation effort to solve

$$n_z^* \in \arg\max_{n_z \geq 0} \left\{ \Pi_z\left(\frac{(1+t_z)\exp(n_z^\alpha)}{D_z}\right) - \kappa_{0z} n_z - \frac{\kappa_{1z}}{2} n_z^2 \right\}.$$

For an interior optimum, the first-order condition is

$$\frac{\partial \Pi_z(r_z^*)}{\partial \log r_z} \alpha (n_z^*)^{\alpha-1} = \kappa_{0z} + \kappa_{1z} n_z^*,$$

where

$$r_z^* = \frac{(1+t_z)\exp((n_z^*)^\alpha)}{D_z}.$$

Let

$$\Omega_z \equiv \kappa_{1z} - \frac{\partial \Pi_z(r_z^*)}{\partial \log r_z} \alpha(\alpha-1)(n_z^*)^{\alpha-2} - \frac{\partial^2 \Pi_z(r_z^*)}{\partial (\log r_z)^2} \alpha^2 (n_z^*)^{2\alpha-2}.$$

Throughout this appendix, I focus on regular interior optima, meaning optima with $n_z^* > 0$ and $\Omega_z > 0$.[48]

Appendix D.7.4 Comparative Statics

Differentiating the first-order condition with respect to $\log(1+t_z)$ gives

$$\frac{dn_z^*}{d\log(1+t_z)} = \frac{\frac{\partial^2 \Pi_z(r_z^*)}{\partial (\log r_z)^2} \alpha (n_z^*)^{\alpha-1}}{\Omega_z}.$$

[48] A sufficient condition for the optimum to be interior is that there exists some $\tilde{n}_z > 0$ such that

$$\Pi_z\left(\frac{(1+t_z)\exp(\tilde{n}_z^\alpha)}{D_z}\right) - \kappa_{0z}\tilde{n}_z - \frac{\kappa_{1z}}{2}\tilde{n}_z^2 > \Pi_z\left(\frac{1+t_z}{D_z}\right).$$

Differentiating the first-order condition with respect to $\log D_z$ gives

$$\frac{dn_z^*}{d\log D_z} = -\frac{\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2}\alpha(n_z^*)^{\alpha-1}}{\Omega_z}.$$

US sales are

$$Y_z = E_z\lambda_z\left(\frac{(1+t_z)\exp((n_z^*)^\alpha)}{D_z}\right).$$

Therefore,

$$\frac{d\log Y_z}{d\log(1+t_z)} = \frac{\partial \log \lambda_z(r_z^*)}{\partial \log r_z}\left[1+\alpha(n_z^*)^{\alpha-1}\frac{dn_z^*}{d\log(1+t_z)}\right].$$

Substituting the innovation response and using the first-order condition gives

$$\frac{d\log Y_z}{d\log(1+t_z)} = \frac{\frac{\partial \log \lambda_z(r_z^*)}{\partial \log r_z}}{\Omega_z}\left[\kappa_{1z}(2-\alpha) - \frac{\kappa_{0z}(\alpha-1)}{n_z^*}\right].$$

Define

$$\bar{n}_z \equiv \frac{\kappa_{0z}(\alpha-1)}{\kappa_{1z}(2-\alpha)}.$$

Appendix D.7.5 Main Propositions

I now state the main comparative statics results. The first proposition characterizes the innovation response. The second characterizes the output response. The third gives the local monotonicity result in distance to the frontier.

Proposition A1: There exists a threshold D_z^I such that for all $D_z \geq D_z^I$, protection weakly increases innovation $\left(\frac{dn_z^*}{d\log(1+t_z)} \geq 0\right)$. In addition, for some parameter values, there can exist products with $D_z < D_z^I$ for which protection reduces innovation $\left(\frac{dn_z^*}{d\log(1+t_z)} < 0\right)$.

Suppose there exists a unique value $\bar{r}_z$ such that

$$\frac{\partial^2 \Pi_z(\bar{r}_z)}{\partial(\log r_z)^2} = 0,$$

with

$$\frac{\partial^2 \Pi_z(r_z)}{\partial(\log r_z)^2} > 0 \quad \text{for} \quad r_z < \bar{r}_z,$$

and

$$\frac{\partial^2 \Pi_z(r_z)}{\partial(\log r_z)^2} < 0 \quad \text{for} \quad r_z > \bar{r}_z.$$

Consider a regular interval of distance-to-frontier values over which $r_z^*(D_z)$ is strictly decreasing in D_z and crosses $\bar{r}_z$ once. Then there exists a unique threshold D_z^I such that protection weakly increases innovation for $D_z \geq D_z^I$ and reduces innovation for $D_z < D_z^I$.

Proof. From the comparative statics above,

$$\frac{dn_z^*}{d\log(1+t_z)} = \frac{\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2}\alpha(n_z^*)^{\alpha-1}}{\Omega_z}.$$

Since the optimum is regular,

$$\Omega_z > 0.$$

Since $\alpha > 1$ and $n_z^* > 0$,

$$\alpha(n_z^*)^{\alpha-1} > 0.$$

Therefore the sign of the innovation response is the sign of

$$\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2}.$$

By assumption, this term is positive when $r_z^* < \bar{r}_z$ and negative when $r_z^* > \bar{r}_z$. Since $r_z^*(D_z)$ is strictly decreasing in D_z and crosses $\bar{r}_z$ once, there exists a unique D_z^I such that

$$r_z^*(D_z^I) = \bar{r}_z.$$

For $D_z > D_z^I$, $r_z^*(D_z) < \bar{r}_z$, so

$$\frac{dn_z^*}{d\log(1+t_z)} \geq 0.$$

For $D_z < D_z^I$, $r_z^*(D_z) > \bar{r}_z$, so

$$\frac{dn_z^*}{d\log(1+t_z)} < 0.$$

At $D_z = D_z^I$, the innovation response is zero. Thus protection weakly increases innovation above the threshold and reduces innovation below it. ■

Proposition A2: There exists a threshold D_z^Y such that for all $D_z \geq D_z^Y$, protection raises output $\left(\frac{d\log Y_z}{d\log(1+t_z)} \geq 0\right)$. In addition, for some parameter values, there can exist products with $D_z < D_z^Y$ for which protection reduces output $\left(\frac{d\log Y_z}{d\log(1+t_z)} < 0\right)$.

Consider a regular interval $[D_L, D_H]$ over which

$$\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2} < 0.$$

Suppose

$$n_z^*(D_L) < \bar{n}_z < n_z^*(D_H).$$

Then there exists a unique threshold $D_z^Y \in (D_L, D_H)$ such that protection reduces output for $D_z < D_z^Y$ and raises output for $D_z > D_z^Y$.

Proof. Over this interval,

$$\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2} < 0.$$

Therefore,

$$\frac{dn_z^*}{d\log D_z} = -\frac{\frac{\partial^2 \Pi_z(r_z^*)}{\partial(\log r_z)^2}\alpha(n_z^*)^{\alpha-1}}{\Omega_z} > 0.$$

Thus $n_z^*(D_z)$ is strictly increasing in distance to frontier over $[D_L, D_H]$.

The output elasticity is

$$\frac{d\log Y_z}{d\log(1+t_z)} = \frac{\frac{\partial \log \lambda_z(r_z^*)}{\partial \log r_z}}{\Omega_z}\left[\kappa_{1z}(2-\alpha) - \frac{\kappa_{0z}(\alpha-1)}{n_z^*}\right].$$

Because

$$\frac{\partial \log \lambda_z(r_z^*)}{\partial \log r_z} > 0 \qquad \text{and} \qquad \Omega_z > 0,$$

the sign of the output response is the sign of

$$\kappa_{1z}(2-\alpha) - \frac{\kappa_{0z}(\alpha-1)}{n_z^*}.$$

This expression is negative when

$$n_z^* < \bar{n}_z$$

and positive when

$$n_z^* > \bar{n}_z.$$

Since $n_z^*(D_z)$ is strictly increasing and

$$n_z^*(D_L) < \bar{n}_z < n_z^*(D_H),$$

there exists a unique $D_z^Y \in (D_L, D_H)$ such that

$$n_z^*(D_z^Y) = \bar{n}_z.$$

Hence protection reduces output below D_z^Y and raises output above D_z^Y. ■

Proposition A3: There exists a range such that the effect of protection on output is monotonic in distance to frontier.

Under the conditions of Proposition A2, at the threshold $D_z = D_z^Y$,

$$\frac{\partial}{\partial \log D_z}\left[\frac{d\log Y_z}{d\log(1+t_z)}\right] > 0.$$

Therefore, in a neighborhood of D_z^Y, the effect of protection on output is increasing in distance to the frontier.

Proof. At $D_z = D_z^Y$,

$$n_z^*(D_z^Y) = \bar{n}_z.$$

Therefore,

$$\kappa_{1z}(2-\alpha) - \frac{\kappa_{0z}(\alpha-1)}{n_z^*} = 0.$$

Differentiating the output elasticity with respect to $\log D_z$ and evaluating at $D_z = D_z^Y$ gives

$$\frac{\partial}{\partial \log D_z}\left[\frac{d\log Y_z}{d\log(1+t_z)}\right] = \frac{\frac{\partial \log \lambda_z(r_z^*)}{\partial \log r_z}}{\Omega_z}\frac{\kappa_{0z}(\alpha-1)}{(n_z^*)^2}\frac{dn_z^*}{d\log D_z}.$$

Each term on the right-hand side is positive. Therefore,

$$\frac{\partial}{\partial \log D_z}\left[\frac{d\log Y_z}{d\log(1+t_z)}\right] > 0.$$

By continuity, the same inequality holds in a neighborhood of D_z^Y. ■